

Technical training.
Product information.

F30H Complete Vehicle



BMW Service

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General information

Symbols used

The following symbol/schematic diagram is used in this document to facilitate better comprehension or to draw attention to very important information:



Contains important safety information and information that needs to be observed strictly in order to guarantee the smooth operation of the system.

Information status and national-market versions

BMW Group vehicles meet the requirements of the highest safety and quality standards. Changes in requirements for environmental protection, customer benefits and design render necessary continuous development of systems and components. Consequently, there may be discrepancies between the contents of this document and the vehicles available in the training course.

This document basically relates to the European version of left-hand drive vehicles. Some operating elements or components are arranged differently in right-hand drive vehicles than shown in the graphics in this document. Further deviations may arise as a result of the equipment specification in specific markets or countries.

Additional sources of information

Further information on the individual topics can be found in the following:

- Owner's Handbook
- Integrated Service Technical Application.

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The information contained in this document forms an integral element of the technical training of the BMW Group and is intended for the trainer and participants in the seminar. Refer to the current respective information systems of the BMW Group for any changes/additions to the technical data.

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BV-72/Technical Training

F30H Complete Vehicle

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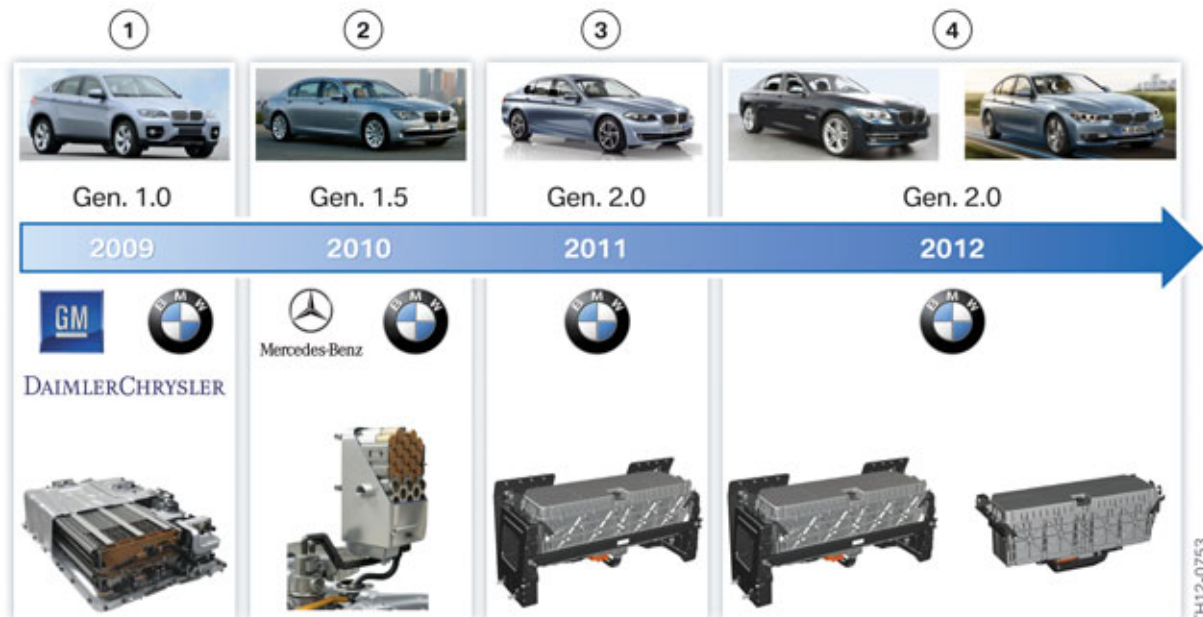
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1. Introduction

1.1. Positioning

Hybrid technology in BMW vehicles (to date) was only available in the upper class vehicle segments, as well as in a sports activity coupe. With the introduction of the BMW ActiveHybrid 3 (in 2012) the BMW Group introduces for the first time hybrid technology in the compact premium sedan segment. The ActiveHybrid 3 (development code F30H) uses "second generation" hybrid technology introduced in the F10H (as the F01H/F02H). The F30H is also a full hybrid vehicle and uses a lithium-ion battery as an energy storage device for the electrical powertrain. The drive system comprises a hybrid-specific 8-speed automatic transmission (GA8P70HZ) with integrated electrical machine and a 225 kW/300 HP 6-cylinder in-line engine (**N55B30M0**) with BMW TwinPower Turbo technology. The (forward-looking) intelligent energy management which is characteristic of the second generation hybrid technology is thus also installed in the F30H: It ensures that the operating strategy can be adapted not only to the present driving situation but also to an immediately imminent driving situation, thereby delivering a much more efficient drive and an enhanced reproducible driving experience.



Generations of BMW ActiveHybrid vehicles to date

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1. Introduction

Index	Explanation
1	The E72 was launched on to the market at the end of 2009 as the first BMW hybrid vehicle. The technology used in generation 1.0 was a product created from the cooperation venture between General Motors, DaimlerChrysler and BMW. A nickel metal hydride battery was used as an electric energy storage device.
2	The second hybrid car from BMW came onto the market in 2010 under the name ActiveHybrid 7 with the development code F04. It is a mild hybrid vehicle which uses generation 1.5 technology. This technology was developed together with Mercedes Benz. A highly efficient lithium-ion battery was used in the high-voltage electrical system.
3	The ActiveHybrid 5 was the third BMW hybrid vehicle and has been built since the end of 2011. The F10H is the first vehicle with second generation hybrid technology. A lithium-ion battery is also used as an electrical energy storage unit.
4	In 2012 the production of further BMW hybrid vehicles with Generation 2.0 technology was expanded with the introduction of the new BMW ActiveHybrid 7 (development codes F01H/F02H) which replaces the F04 and the BMW ActiveHybrid 3 (development code F30H) which comes on to the market same time.

In comparison to the F10H or F01H/F02H, the ActiveHybrid technology has been considered right from the start in the design of the F30H. As a result, the hybrid components could be optimally integrated into the vehicle in order to maintain the F30 luggage compartment capacity.

The main advantage of the ActiveHybrid technology used in the F30H is the increase in driving power (compared to the conventional BMW 335i sedan) while, at the same time, lowering the fuel consumption. For example the powertrain of the ActiveHybrid 3 provides a maximum system power of 250 kW/340 HP, which accelerates the vehicle from 0 to 60 mph in 5.2 seconds. The F30H consumes only 5.9 l of gasoline per 100 km ("unofficial" 38 mpg) in the EU test cycle – this is approximately 18% less than the conventional F30 sedan (335i with automatic transmission).

All vehicles in which second generation hybrid technology is used have similar features, components and functions. These are described in full and in detail in the "ST1203 F10H Complete Vehicle" training material. In this F30H training manual only the features specific to the ActiveHybrid 3 are discussed as well as the technical differences to the F10H.



The second generation hybrid technology is described in detail in the "ST1203 F10H Complete Vehicle" training material available on TIS and ICP. This product information manual only describes the F30H.

These are the specific features of the F30H:

- Identifying features and technical data
- Modified electrical machine electronics (mainly the housing)
- Installation location of the high-voltage battery with effects on the removal and installation
- Changes to the hybrid brake system

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1. Introduction

- Bus systems (basic vehicle electrical system of the F30 instead of the one in the F10)
- 12 V voltage supply
- Displays in the instrument cluster and in the central information display.

1.2. Identifying features

1.2.1. Exterior trim

Similar to the familiar BMW ActiveHybrid vehicles, the F30H can also be distinguished from a conventional F30 by some features.

The F10H can be differentiated from the conventional F10 by two main components of the exterior trim: the longitudinal bar of the kidney grills and the exhaust system tailpipes. For the F30H these are not definitive identifying features as the customer can order the F30H in the F30 in three different trim Lines (Sport, Luxury and Modern) or with the M Sports package. If the customer orders a F30H without one of the Lines, the exhaust tailpipes of the exhaust system are designed in "chrome matt". with regard to the wheels and tires the customer of an ActiveHybrid 3 without a Line or M Sports package receives the wheel rim "V-spoke 413" as standard equipment. If the customer configures his F30H with a Line or with the M Sports package he receives the wheel rim of the selected Line as standard equipment. All other wheel and tire combinations of the F30 335i can also be ordered for the F30H. What remains exclusive for the F30H however is the 18" wheel rim in the aerodynamic design ("Streamline 419") with 225/45 R18 tires. Irrespective of the Lines or the M Sports package there are still the following exclusive designer features in the exterior trim of the F30H:

- "ActiveHybrid 3" inscription: Located on the entrance sills, the C-pillars and on the trunk lid.
- Acoustic cover of the combustion engine: This component also contains a hybrid-specific inscription ("ActiveHybrid").

The paint color "Liquid Blue Metallic" can be ordered for the F30H but is also for the conventional F30 models. To give the F30H higher visibly and quality appearance, the exterior trims are designed in high gloss chrome or high gloss black as standard.

F30H Complete Vehicle

1. Introduction



External view of the ActiveHybrid 3

Index	Explanation
1	"ActiveHybrid 3" inscription on the trunk lid
2	"ActiveHybrid 3" inscription on the C-pillars
3	Exterior trims in aluminium satined
4	Exterior trims in high gloss black
5	Entry sills with "ActiveHybrid 3" writing
6	18" wheel rims ("Streamline 419")
7	Acoustic cover with "ActiveHybrid" writing

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1. Introduction

1.2.2. Interior



Interior of the ActiveHybrid 3

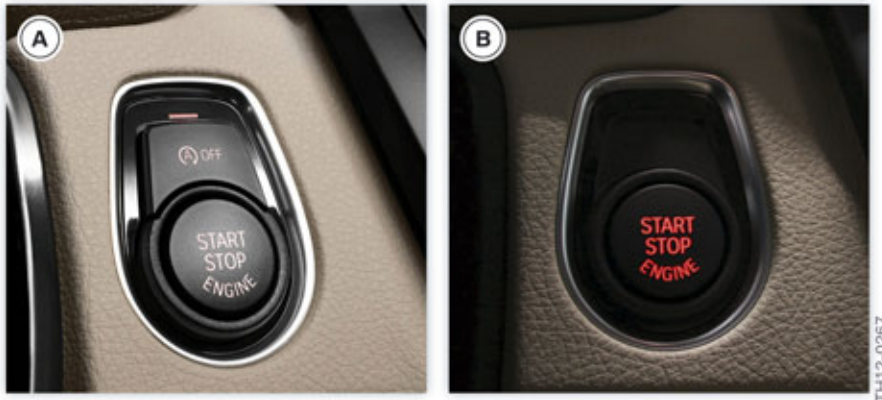
The interior equipment of the BMW ActiveHybrid 3 also has special features which distinguish it from a conventional F30. The "ActiveHybrid 3" inscription is located on the link around the electronic gear selector switch.

The hybrid-specific operating conditions and the state of charge of the high-voltage battery are displayed in the instrument cluster and in the Central Information Display. The hybrid-specific displays in the CID were visibly enhanced due to the introduction of the new HU-H - the content of which is comparable to that of the F10H. Both the displays in the CID and in the instrument cluster are activated when the ignition is turned on. These hybrid-specific displays are described in detail in the section entitled "Displays and Controls".

The START-STOP button of the F30H is also different from that of a conventional F30: In the F30H there is no button for deactivating the automatic engine start-stop function. The operating strategy of the F30H ensures that the combustion engine is always shut down appropriate to the position of the driving experience switch and the current driver's choice when it makes sense in the current driving situation. For more information on how the combustion engine can be switched on permanently refer to the "Displays and Controls" chapter of this training manual.

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1. Introduction



START-STOP button

Index	Explanation
A	START-STOP button in the F30 with integrated button for deactivation of the automatic engine start-stop function
B	START-STOP button in the F30H

Compared to other Generation 2 hybrid vehicles there is no distinctive difference (in usage) of the luggage compartment of the F30H compared to the conventional model. The reason for this is the installation location of the high-voltage battery unit. The high-voltage battery unit of the F30H is located under the luggage compartment floor panel. As a result, the luggage compartment capacity is much the same size as in the F30 and can even be flexibly extended through the through-loading system also available for the F30H.

F30H Complete Vehicle

1. Introduction



Luggage compartment of the ActiveHybrid 3

TH12-0258

Index	Explanation
1	Folded-down rear seat backrests of the through-loading system
2	Luggage compartment floor panel
3	High-voltage battery unit

1.3. Operating strategy

Similar to the F10H the operating strategy of the F30H is also characterized by innovative functions. This includes the intelligent energy management and the Coasting function, which was introduced with the F10H, and is also in the F30H.

Only the “Home” feature of the intelligent energy management function is available for the US and Canadian markets.

The coasting function means that the combustion engine is also shut down at higher speeds up to 100 mph, if it is not required for the powertrain. At the same time the decoupler clutch in the drivetrain is opened so that the vehicle rolls without engine braking effect. The electrical machine is still operated

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1. Introduction

(in this case) at moderate power as an alternator. Some kinetic energy is taken from the vehicle and the electrical energy produced is used primarily for the supply of the low-voltage vehicle electrical system. Excess energy can of course be used to charge the high-voltage battery.

To be able to use "coasting" the driver must have selected ECO PRO mode. If the driver takes his foot off the accelerator pedal, the function becomes active after a short while, the combustion engine switches off and the decoupler clutch is opened. "Coasting" can also be experienced in COMFORT mode if for example one is driving on a main road and the speed is reduced by gently pressing the brake pedal before reaching a city boundary. The advantage of increased efficiency through "coasting" is clearly visible: in this operating condition no fuel whatsoever is used.

1.4. Technical data

	Unit	BMW 335i	BMW ActiveHybrid 3
Engine and transmission			
Design		R6	R6
Number of valves per cylinder		4	4
Displacement	[cm ³]	2979	2979
Transmission		GA8HP45Z	GA8P70HZ
Powertrain		Rear-wheel	Rear-wheel
Final drive ratio		3.154	2.813
Maximum power, combustion engine	[kW (HP)] [rpm]		225 (300) 5800
Maximum torque, combustion engine	[Nm (lb-ft)]		400 (300)
Complete system power	[kW (HP)]		250 (340)
Overall system torque	[Nm (lb-ft)]		450 (330)
High-voltage battery type		-	Lithium-ion
Power of electrical machine	[kW / HP]	-	40/55
Maximum torque, electrical machine	[Nm (lb-ft)]	-	210 (184)
Vehicle performance			
Acceleration 0 to 60 mph	[s]	5.1	5.2
Maximum speed (limited)	[km/h (mph)]	250 (155)	250 (155)
Consumption and emissions			
Fuel consumption (EU cycle, urban driving)	[l/100 km]	10.2	5.3
Consumption EU cycle, non-urban driving	[l/100 km]	5.5	6.4
Consumption EU cycle, total	[l/100 km]	7.2	5.9

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1. Introduction

	Unit	BMW 335i	BMW ActiveHybrid 3
CO ₂ emissions	[g/km]	169	139
Dimensions and weights			
Length/width/height	[mm]	4636/1811/1429	
Wheelbase	[mm]	2810	
Turning circle	[m]	11.3	
Vehicle curb weight (DIN/EU)	[kg]	1520/1595	1655/1730
Payload in accordance with DIN	[kg]	550	530
Fuel tank capacity	[liters (gal)]	60 (15.8)	57 (15)
Luggage compartment capacity	[liters]	480	390

These above values are according to EU test cycle depending on the tire dimension. The values specified apply for the standard tires (225/50 R17). Through the possibility of electric driving fuel consumption in the city cycle is significantly lower than for the comparative model 335i. The fact that the fuel consumption for the subsequent overland cycle is slightly higher must be taken into account for this: the energy which was used for the electric driving in the city cycle must now be acquired again. This happens not only through brake energy regeneration but also through load point shifting of the combustion engine. In the overall cycle there is a significant advantage in the fuel consumption for the ActiveHybrid 3.

1.5. Equipment

The F30H and the F10H are available in a European and a US version, as well as a left-hand drive vehicle and a right-hand drive vehicle. Due to the optimized housing of the high-voltage battery unit in the F30H in comparison to the F10H, the customer can also order the through-loading system with ski bag in the F30H. The adaptive chassis and suspension or the variable sport steering are also available as optional equipment for the F30H.

The following optional equipment is not available in the F30H:

- xDrive all-wheel drive system
- Active cruise control with Stop & Go function (valid at the time of market introduction).

The following equipment is part of the standard equipment:

- Electromechanical power steering (EPS) with Servotronic function
- Automatic transmission
- 2-zone A/C system
- Preparation for Navigation Professional system (option 6UF).

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1. Introduction

The HU-H including the 8.8" display screen is installed in each F30H to display the hybrid-specific displays in better quality in the central information display. However, the navigation function is not included as standard equipment on the F30H. The customer can order this as optional equipment (option 609) and thus receives the forward-looking operating strategy for even more efficient fuel consumption.

The longer final drive ratio of the F30H in comparison to the conventional F30 335i also helps reduce the fuel consumption through the reduced engine speed. This advantage is not only noticeable in normal consumption, but also in everyday use at higher driving speed. The slightly lower drag coefficient of the F30H also has a positive effect on the fuel consumption in day-to-day usage.

F30H Complete Vehicle

2. Drivetrain

2.1. Design

The powertrain of the F30H is in principle identical to that of the F10H and comprises the following key components:

- **N55B30M0** engine
- Fuel supply systems
- Electrical machine
- **GA8P70HZ** Automatic transmission

The details of the combustion engine and the fuel supply system are specific for the F30H and differ from the basic vehicle F30 and/or the F10H. The changes are described in the following chapters. The electrical machine and the automatic transmission are identical to the components used in the F10H. For this reason only the key properties are summarized here.

2.2. N55B30M0 engine

Similar to the F10H, the N55B30M0 was also modified for use in the F30H.

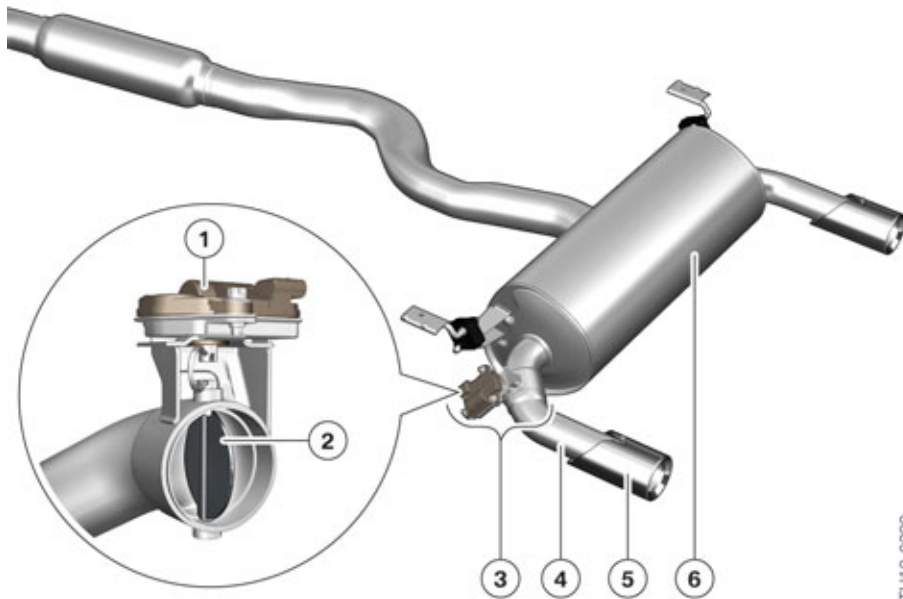
The **common features** with the engine used in the F10H are:

- Maximum power of 225 kW/300 HP and maximum torque of 400 Nm (300 lb-ft)
- As the only ancillary component the belt drive includes the starter motor generator (SGR) and two belt tensioners.
- The vacuum supply is guaranteed by a mechanical and an electric vacuum pump.
- The crankshaft bearings have a reinforced design due to the more frequent engine start processes.
- The exhaust system has an electric exhaust flap on the left exhaust tailpipe.

The operation is done via an electromechanical actuator like in the F30 with N55B30M0 engine (wired directly to the DME control unit). The operation is thus different than in the F10H, where the operation is done via vacuum and an electric changeover valve.

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2. Drivetrain



F30H exhaust flap

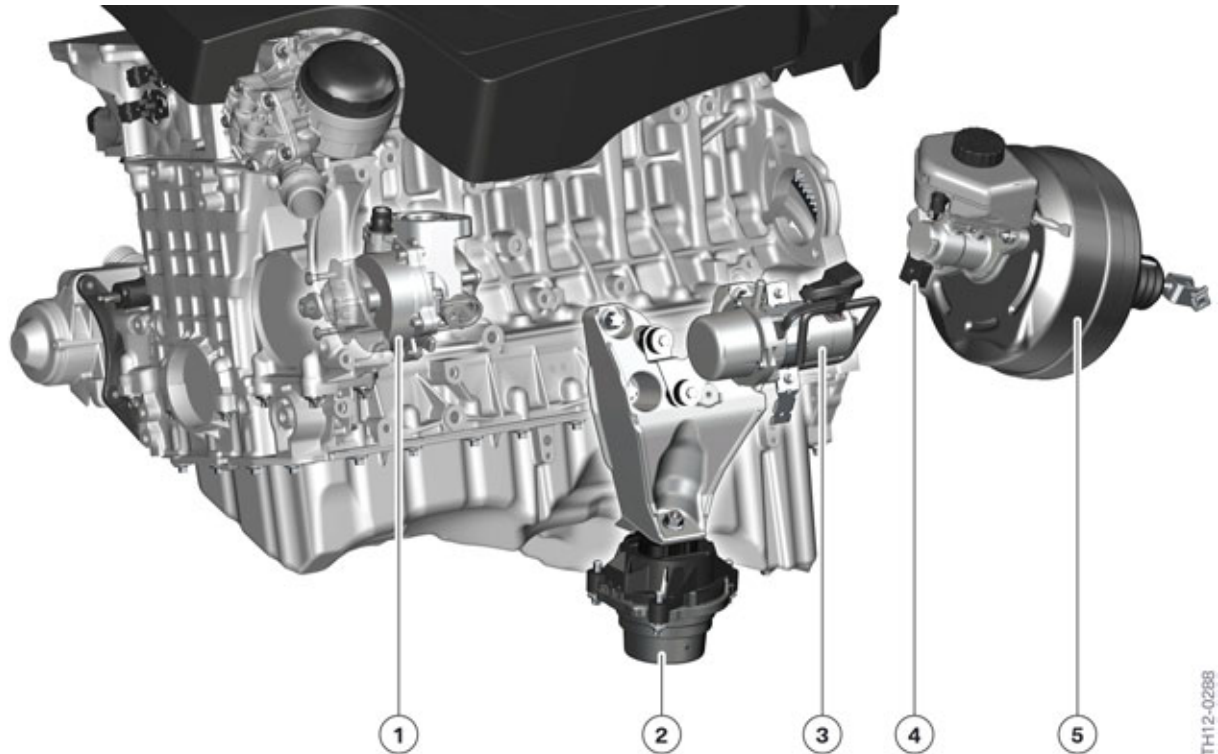
Index	Explanation
1	Electric motor with transmission
2	Exhaust flap
3	Electromechanical exhaust flap
4	Exhaust tailpipe
5	Cover over exhaust tailpipe (color coding dependent on Line ordered)
6	Rear silencer

Compared to the F10H there is **no** another vacuum actuator installed because the damping-controlled engine mounts are not used in the F30H. Tests have shown that in the customer-relevant operation the standard (**non** damping-controlled) engine mounts perform at a high comfort level for the engine start and shutdown processes.

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2. Drivetrain

The modified vacuum system of the F30H is shown in the following graphic.



F30H Vacuum system

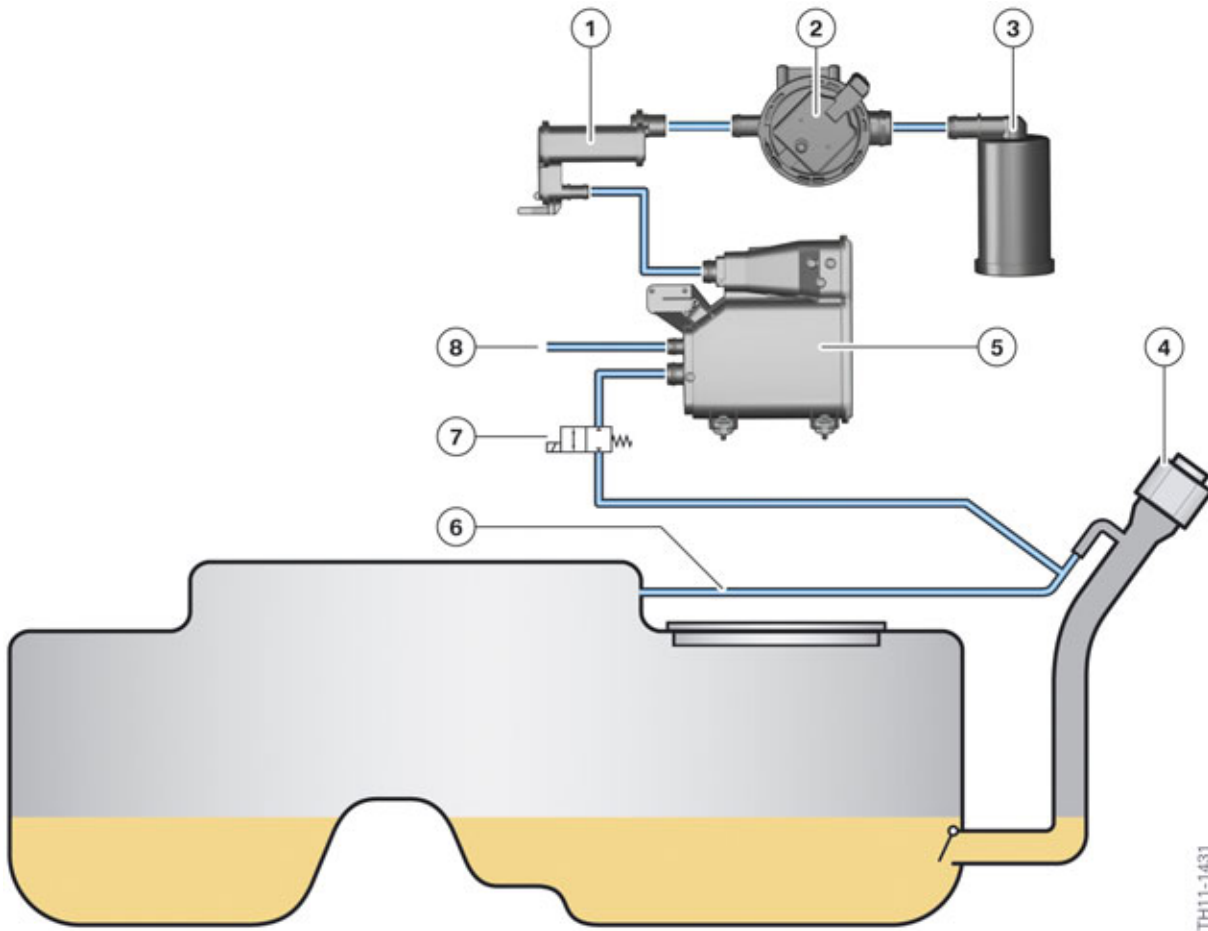
Index	Explanation
1	Mechanical vacuum pump
2	Engine mount (non damping-controlled)
3	Electric vacuum pump
4	Brake vacuum sensor
5	Brake servo

2.3. Fuel supply systems

The fuel supply system of the F30H is based on that of the conventional F30 335i. All versions of the F30H have a fuel tank capacity of 57 l (15 gal) in comparison to 60 l (15.8 gal) for the conventional 335i. Similar to the F10H, F30H vehicles in US version also have some special features in relation to the fuel tank ventilation. These technical measures are for Natural Vacuum Leak Detection and ensure that the legal regulations for hydrocarbon emissions are observed.

F30H Complete Vehicle

2. Drivetrain



Principal structure of the fuel supply system with NVLD (Natural Vacuum Leak Detection)

Index	Explanation
1	Additional carbon canister (activated carbon pressed in honeycomb shape)
2	Tank leak diagnosis (Natural Vacuum Leak Detection, NVLD)
3	Air cleaner (spinning strainer)
4	Fuel filler neck
5	Carbon canister
6	Fuel tank ventilation line
7	Shutoff valve for fuel tank ventilation
8	Purge air line to the combustion engine

The special features of the fuel tank ventilation for the current (US version) second generation hybrid cars are listed below.

F30H Complete Vehicle

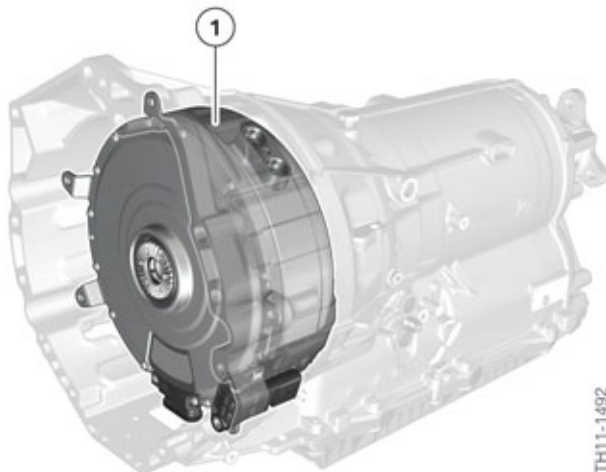
2. Drivetrain

- Pressure and temperature sensor in the fuel tank, connected electrically via local interconnect network bus to the DME control unit
- Shutoff valve for fuel tank ventilation, activation via hard-wired wiring to DME
- Tank leak diagnosis (Natural Vacuum Leak Detection NVLD), hard-wired to DME control unit
- Additional carbon canister with honeycomb structure.

2.4. Electrical machine

The electrical machine in the drivetrain of the F30H has the same technical features as those of the electrical machine used in the F10H. Together with the decoupler clutch it is integrated in the automatic transmission (GA8P70HZ) at the place where the hydraulic torque converter would be located in conventional vehicles.

This permanently excited synchronous machine can convert the electrical energy stored in the high-voltage battery to mechanical energy and thus drive the F30H. This ranges from supporting the combustion engine for example when overtaking to purely electric driving. In the opposite scenario, during brake energy regeneration, the electrical machine works as an alternator and converts the kinetic energy of the vehicle into electrical energy which can be stored in the high-voltage battery.



Installation location of the electrical machine

Index	Explanation
1	Electrical machine

The electrical machine is a high-voltage component.



High-voltage component warning sticker

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2. Drivetrain

Each high-voltage component has on its housing or casing an identifying label that enables service technicians and vehicle users to clearly identify the possible hazards that can result from the high electric voltages used.



Only service technicians who satisfy all the prerequisites are permitted to work on the designated high-voltage components. Technicians must have the suitable qualifications and comply with the electrical safety rules as well as follow the repair instructions and procedures.

The key technical data of the electrical machine is summarized below:

Supplier	ZF Sachs AG
Maximum torque (< 1 second)	210 Nm at 1300 rpm
Torque (continuous)	80 Nm to 900 rpm
Boost power (< 60 seconds)	38 kW from 1800 rpm
Maximum power (< 1 second)	40 kW at 3000 rpm
Efficiency	up to 91.8 %
Operating speed range	0 – 7500 rpm
Nominal flow temperature (cooling)	105 °C (221 °F)
Nominal volume flow (cooling)	6 l/min
Weight (with torsional vibration damper and clutch)	about 21 kg
Outer diameter of stator (with support)	285 mm
Inner diameter of stator	214 mm
Stator length	85 mm
Outer diameter of rotor	212 mm
Inner diameter of rotor (with support)	172 mm

2.4.1. Interfaces

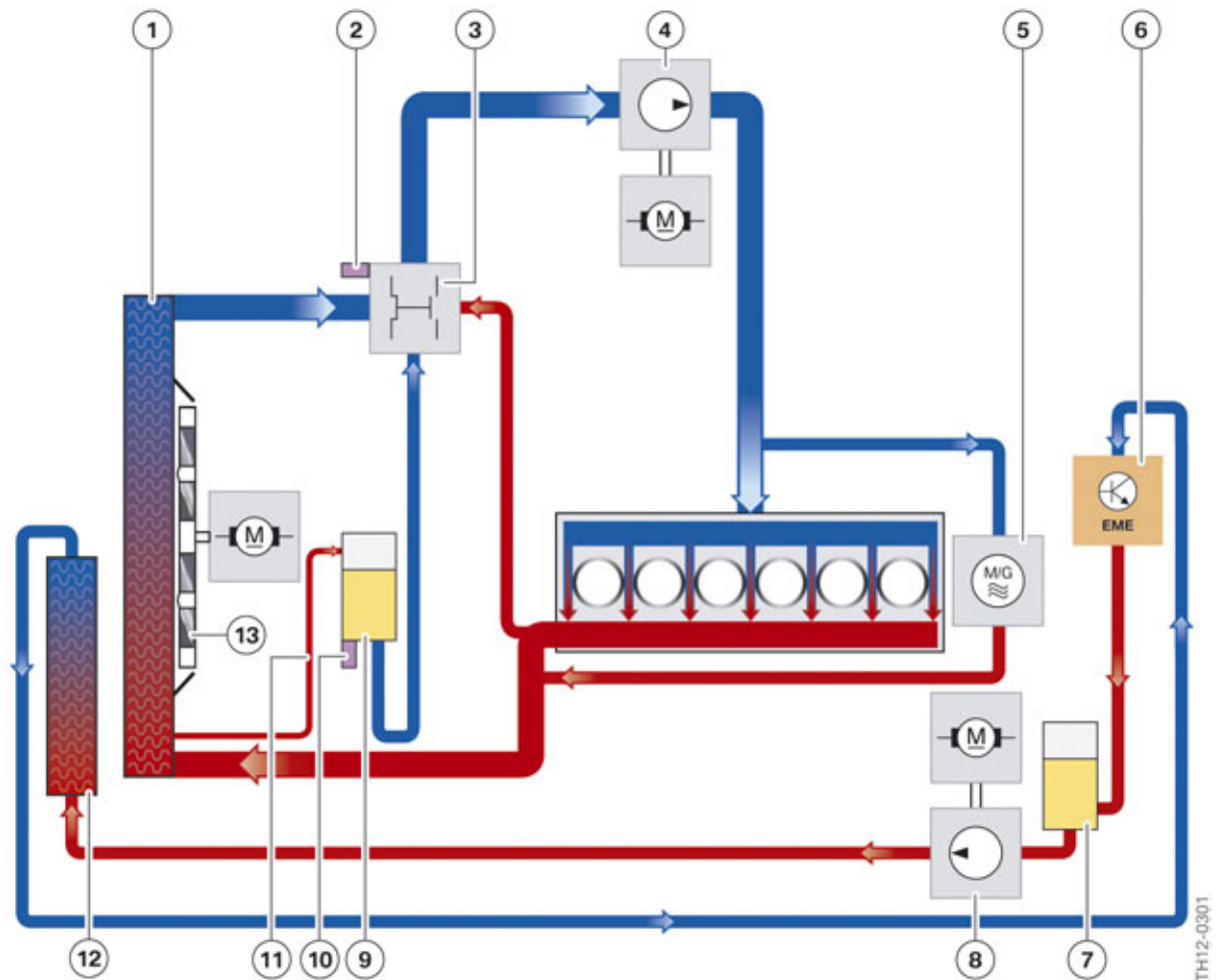
The electrical machine together with the decoupler clutch forms one unit. The mechanical interfaces of this unit comprise a positive connection to the torsional vibration damper in the forward travel direction. In the reverse travel direction the rotor is connected to the transmission input shaft via a positive hollow shaft. The housing of the electrical machine is connected mechanically using five screws to the transmission housing.

The electrical interfaces include; The three-phase high-voltage connection (which is accessible from the vehicle underside) A second low-voltage connection for the signals of the temperature sensor and the rotor position sensor (located on the left side of the housing of the electrical machine). The signal connector and the high-voltage connector must be removed before the transmission is removed.

The electrical machine is integrated in the cooling circuit of the combustion engine via two connections. The electrical machine is cooled via a coolant duct parallel to the combustion engine.

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2. Drivetrain



Cooling circuits F30H

Index	Explanation
1	Coolant/air heat exchanger in the cooling circuit of combustion engine and electrical machine
2	Heater coil
3	Map thermostat
4	Electric coolant pump in the cooling circuit of combustion engine and electrical machine
5	Electrical machine
6	Electrical machine electronics (EME)
7	Expansion tank in the cooling circuit of the EME
8	Electric coolant pump (50 W) in the cooling circuit of the EME
9	Expansion tank in the cooling circuit of combustion engine and electrical machine

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2. Drivetrain

Index	Explanation
10	Coolant level switch
11	Balancing line
12	Coolant/air heat exchanger in the cooling circuit of the EME
13	Electric fan

2.4.2. Service information

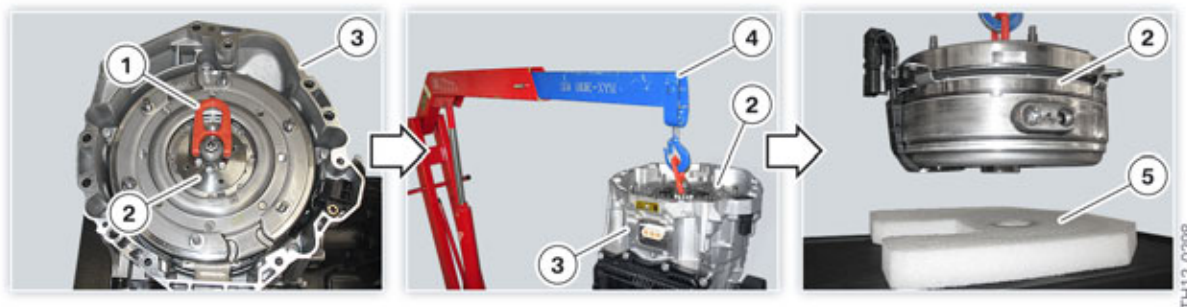
If the electrical machine has to be replaced, the following information must be observed:



Apply electrical safety rules before beginning work.



If the electrical machine must be replaced, the unit comprising the electrical machine, the decoupler clutch and the torsional vibration damper must be removed completely. The special tools intended for this must be used.



Special tools for the removal of the electrical machine

Index	Explanation
1	Special tool for removing the electrical machine with torsional vibration damper from the automatic transmission (special tool number 2 297 311)
2	Electrical machine with torsional vibration damper
3	Automatic transmission
4	Workshop crane
5	Special tool for setting down the electrical machine with torsional vibration damper (special tool number 2 297 312)



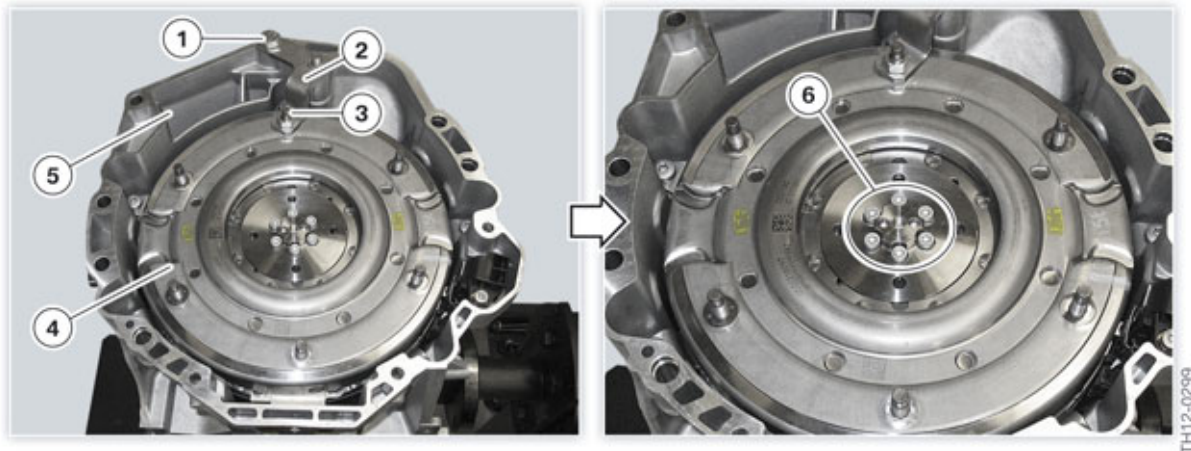
Upon replacement of the electrical machine or the electrical machine electronics, the rotor position sensor must be adjusted with help of the ISTA BMW diagnosis system.

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2. Drivetrain



If only the torsional vibration damper has to be replaced, this can be removed separately. There is also a special tool for this which must be used.



Special tool for removing the torsional vibration damper

Index	Explanation
1	Screw connection of the special tool at the automatic transmission
2	Special tool for disconnecting the torsional vibration damper (special tool number 2 297 313)
3	Screw connection of the special tool at the torsional vibration damper
4	Torsional vibration damper
5	Automatic transmission
6	Screw connection of the torsional vibration damper with electrical machine

2.5. GA8P70HZ automatic transmission

2.5.1. Overview

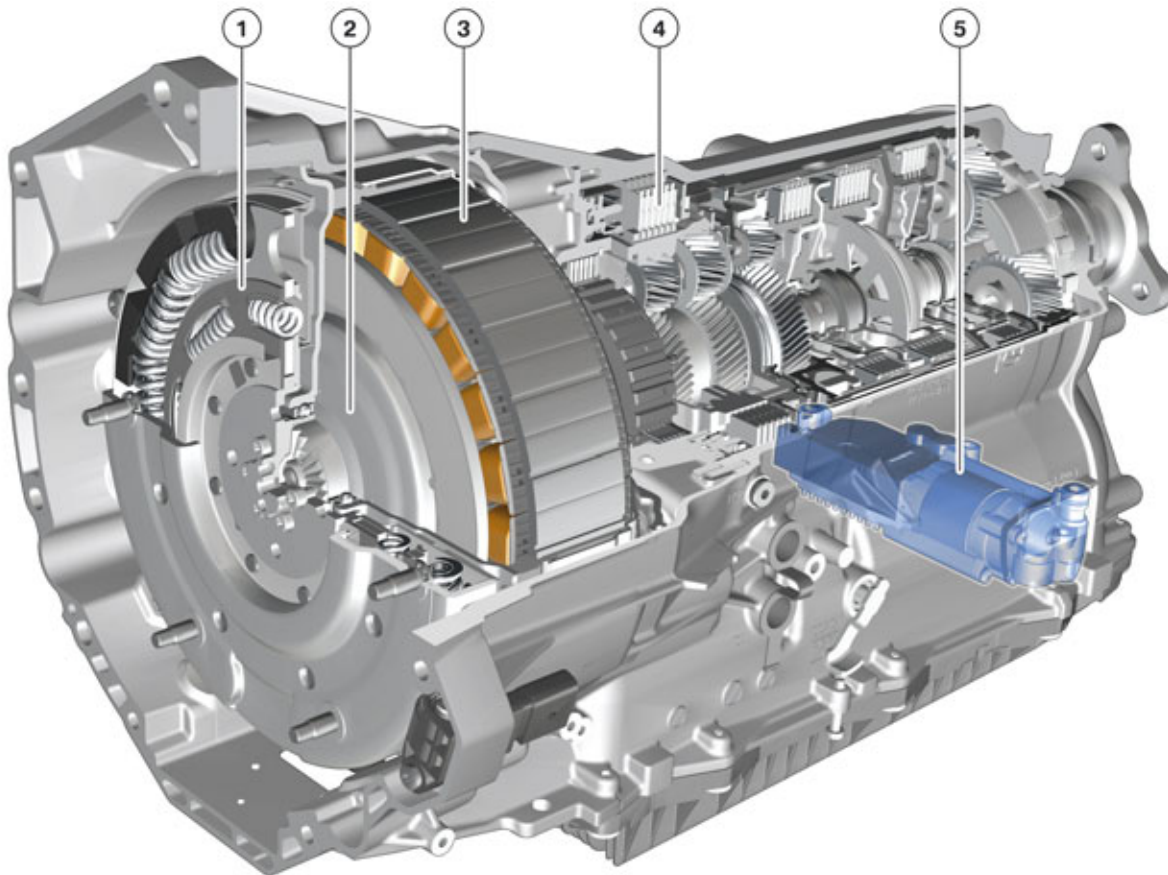
The GA8P70HZ automatic transmission of the F30H has the same structure as that of the automatic transmissions in the F10H and F01H/F02H. The following hybrid-specific components are integrated in the same location as that in the conventional GA8HP70Z transmission:

- Torsional vibration damper for balancing the irregular rotations of the combustion engine
- Decoupler clutch to be able to disconnect the combustion engine from the rest of the drivetrain and to enable purely electric driving and coasting
- Electrical machine which can work as an engine and alternator
- Additional electrical transmission oil pump to ensure the supply with transmission oil pressure in the case of a stationary transmission input shaft.

F30H Complete Vehicle

2. Drivetrain

As the GA8P70HZ does not contain a hydraulic torque converter, other components must assume the transmission of the torque when driving off and creeping. The electrical machine and the multi-disc brake B assume this task. For this purpose the multi-disc brake B (also called "integrated starting element") was modified in comparison to the conventional GA8HP70Z transmission: The number of discs and their diameter were increased so that the required torque can be transmitted when driving off with a slip for example.



TH11-1491

Hybrid-specific components in the automatic transmission

Index	Explanation
1	Torsional vibration damper
2	Decoupler
3	Electrical machine
4	Multi-disc brake B
5	Additional electrical transmission oil pump

The concept of the electrical machine integrated in the transmission makes the F30H a parallel hybrid: combustion engine and electrical machines can drive the vehicle parallel, i.e. at the same time. The installation of the decoupler clutch between the combustion engine and electrical machine allows purely electric driving without the combustion engine possible (which also makes the F30H is a full hybrid).

F30H Complete Vehicle

2. Drivetrain

2.5.2. Service information

Similar to the conventional 8-speed automatic transmission, there is also a mechanical emergency release from the underneath the vehicle, as well as an electronic emergency release, for the GA8P70HZ. The operations for performing the mechanical transmission emergency release are identical to those in the conventional 8-speed automatic transmissions. However, the technical process taking place in the transmission and the operations to be performed by the Service technicians for the electronic emergency release differ.

In conventional vehicles the starter motor rotates and also drives the mechanical transmission oil pump via the torque converter. With help of the built-up transmission oil pressure the parking lock can be released. In the F30H however, the decoupler clutch is open without transmission oil pressure. For this reason the transmission oil pressure cannot be generated for releasing the parking lock in the F30H via the rotating starter motor. Instead the transmission oil pressure can be built up via the additional electric transmission oil pump. Alternatively the electrical machine can drive the mechanical transmission oil pump and thus build up transmission oil pressure.

Conventional vehicle with GA8HP70Z	ActiveHybrid vehicle with GA8P70HZ
Starter motor rotates	Conventional starter motor rotates
Mechanical transmission oil pump is driven by the rotating starter motor and generates transmission oil pressure	Mechanical transmission oil pump via electrical machine or electric transmission oil pump generates transmission oil pressure
With help of the transmission oil pressure the parking lock is released	With help of the transmission oil pressure the parking lock is released

To perform the electronic emergency release a series of steps must be carried out.

In general the following applies:



Before performing an emergency release of the parking lock the vehicle must be secured against rolling away.

For conventional vehicles pressing the electronic gear selector switch twice to move one stage forward is included in the process for the electronic emergency release.

F30H Complete Vehicle

2. Drivetrain



TH12-0752

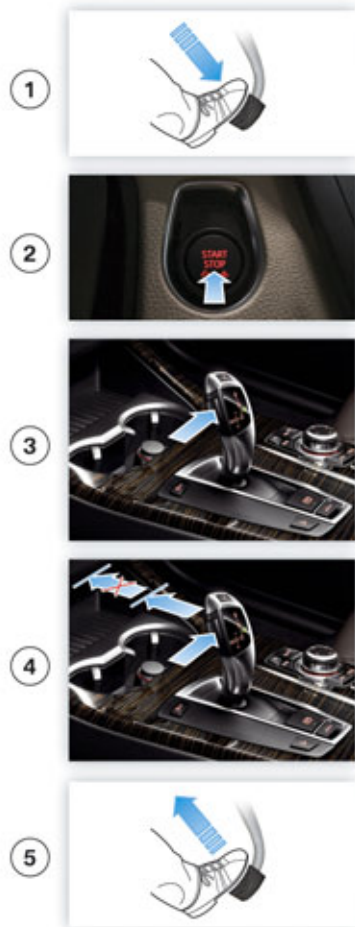
Electronic emergency release of the parking lock for conventional vehicles

Index	Explanation
1	Press brake pedal
2	Press START-STOP button
3	Press release button at electronic gear selector switch
4	Hold down release button and press the gear selector switch forward by one stage, hold there for approx. 2 seconds and then release again
5	(Briefly) press gear selector switch forward by one stage a second time if release button is still held down and release again
6	As soon as the neutral N gear is shown, the brake pedal can be released

F30H Complete Vehicle

2. Drivetrain

In the ActiveHybrid 3 the operation of the gear selector switch must only be effected forwards once and the neutral N gear is engaged directly. The following graphic highlights this difference.



TH12-0751

Electronic emergency release of the parking lock in the F30H

Index	Explanation
1	Press brake pedal
2	Press START-STOP button
3	Press release button at electronic gear selector switch
4	Hold down the release button and press the gear selector switch forward one stage for a short period and release again
5	As soon as the neutral N gear is shown, the brake pedal can be released

In the ActiveHybrid 3 the procedure for the electronic emergency release is thus simplified. As the vehicle still has an electric powertrain in addition to the combustion engine a further function is available to the customer and the Service technician. If the combustion engine does not start, the electrical powertrain is fully functional and with its help can thus move the vehicle at least a few

F30H Complete Vehicle

2. Drivetrain

meters. During the attempt to start the engine one can engage Drive (D) or Reverse (R) gear through the usual operation of the electronic gear selector switch. After the brake pedal is released the vehicle is then moved with help of the electric powertrain.



The Reverse (R) drive position is also engaged when the procedural steps for the electronic emergency release for a **conventional** vehicle are performed. If one has forgotten to secure the vehicle against rolling away, the vehicle may start moving unexpectedly.

F30H Complete Vehicle

3. Electrical Machine Electronics

3.1. Introduction

The electrical machine electronics in the F10H is very similar to the F30H. Although it has a slightly modified housing it assumes the same tasks as in the F10H. Here only the key properties are summarized and details of the differences to the F10H provided.



The electrical machine electronics is a high-voltage component! Therefore before working on the electrical machine electronics the electrical safety rules must be applied.



High-voltage component warning sticker

Each high-voltage component has on its housing or casing an identifying label that enables Service technicians and vehicle users to clearly identify the possible hazards that can result from the high electric voltages used.



For reasons of high-voltage safety the electrical machine electronics must not be opened or otherwise dismantled.

In the event of a fault in the electrical machine electronics it is always replaced as a complete unit.

After the electrical machine electronics has been replaced it must be put into operation with help of the ISTA BMW diagnosis system. Please observe the information in the repair instructions.

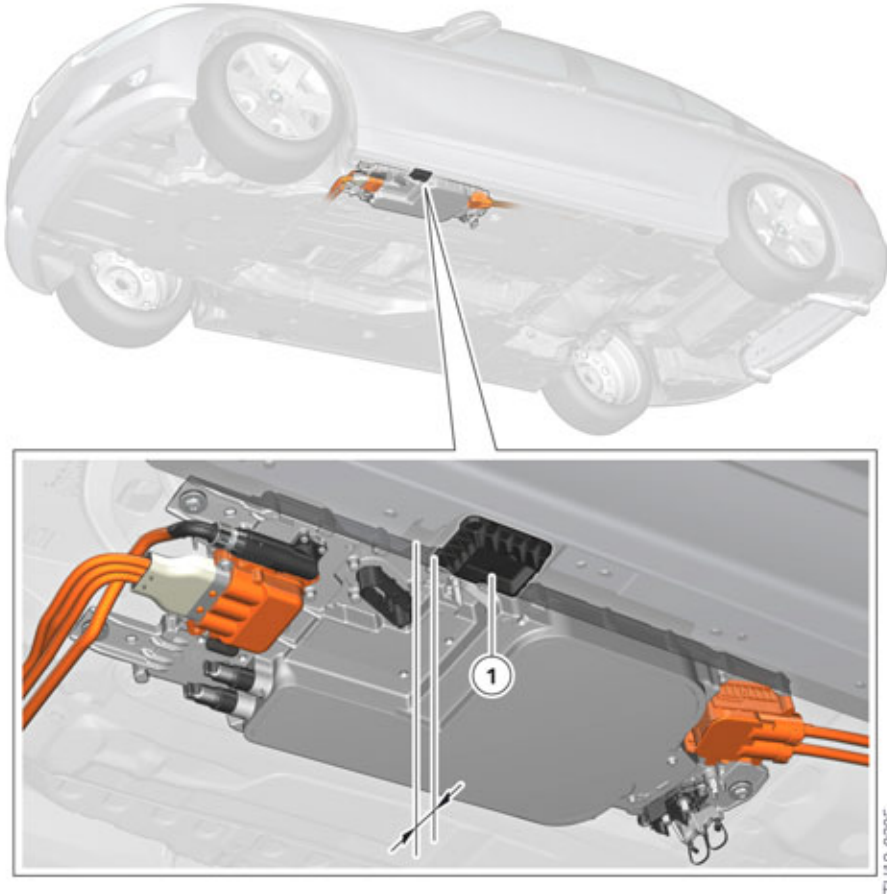
The housing of the electrical machine electronics also contains the following sub-components in addition to the control unit of the same name ("EME" for short):

- Bidirectional DC/AC converter for activating the electrical machine as a motor or alternator
- Unidirectional DC/DC converter for supplying the low-voltage vehicle electrical system with electrical energy
- Output stages for activating the electric vacuum pump, the cut-off relay between the two 12 V batteries, as well as the combined expansion and shutoff valve for the passenger compartment.

F30H Complete Vehicle

3. Electrical Machine Electronics

3.2. Installation location



Installation location of the electrical machine electronics

Index	Explanation
1	Jacking point

The electrical machine electronics of the F30H is installed at the vehicle underbody to the left, beside the automatic transmission as in the F10H. Due to the smaller vehicle dimensions of the F30H, its installation location touches slightly at the front left jacking point. This must be taken into consideration during service work, for which the vehicle must be raised.



Before a F30H is raised with help of a vehicle lift or a jack, the vehicle lift or the jack must be positioned precisely under the jacking point.

Otherwise there is a risk that the vehicle lift or the jack may be supported at the electrical machine electronics which will damage the component when the vehicle is raised!

To reach the connections of the electrical machine electronics, the corresponding underbody panelling must be removed.

F30H Complete Vehicle

3. Electrical Machine Electronics

3.3. Functions

The electrical machine electronics in the F30H completes the same functions as in the F10H.

The EME functions are summarized below:

- Activation of the electrical machine
- Control of start-up and shut-down of the high-voltage electrical system
- Control of the power of the DC/DC converter for supplying the 12 V vehicle electrical system
- Consumer coordination
- High-voltage electrical system diagnosis
- Activation of the cut-off relay between the two 12 V batteries
- High-voltage interlock loop
- Activation of the combined shutoff and expansion valve in the refrigerant circuit for cooling the passenger compartment
- Activating the electric vacuum pump
- Forward-looking operating strategy

3.4. Connections

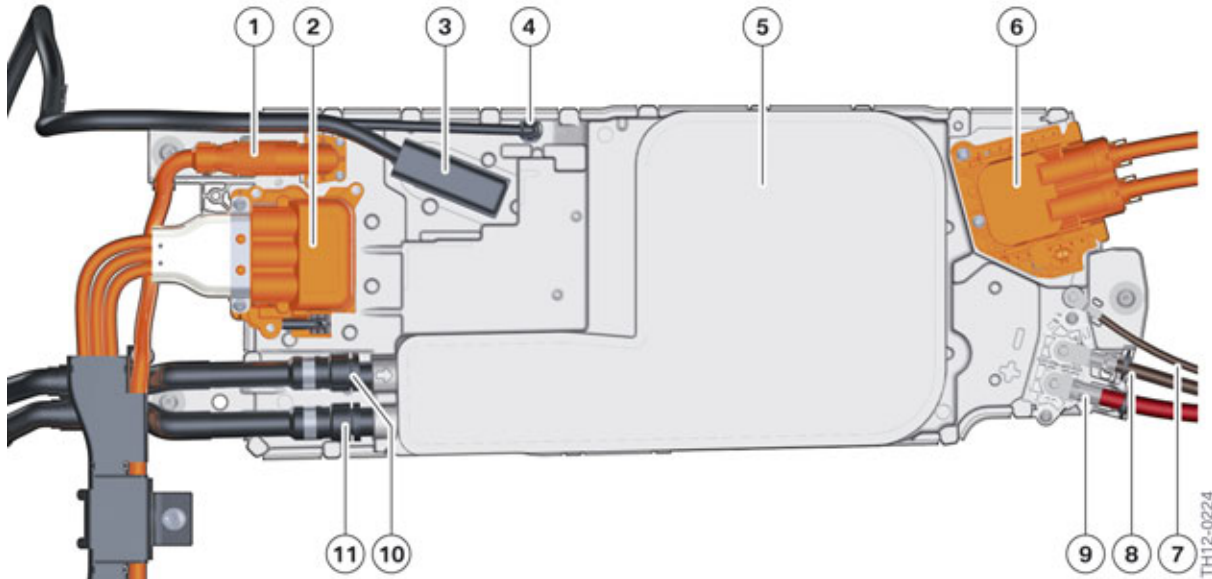
As with the F10H, the connections at the F30H EME can also be divided into five categories:

- Low-voltage connections
- High-voltage connections
- Connection for potential compensation lines
- Connections for coolant lines
- Connection for ventilation line

F30H Complete Vehicle

3. Electrical Machine Electronics

The following graphic shows the F30H EME connections.



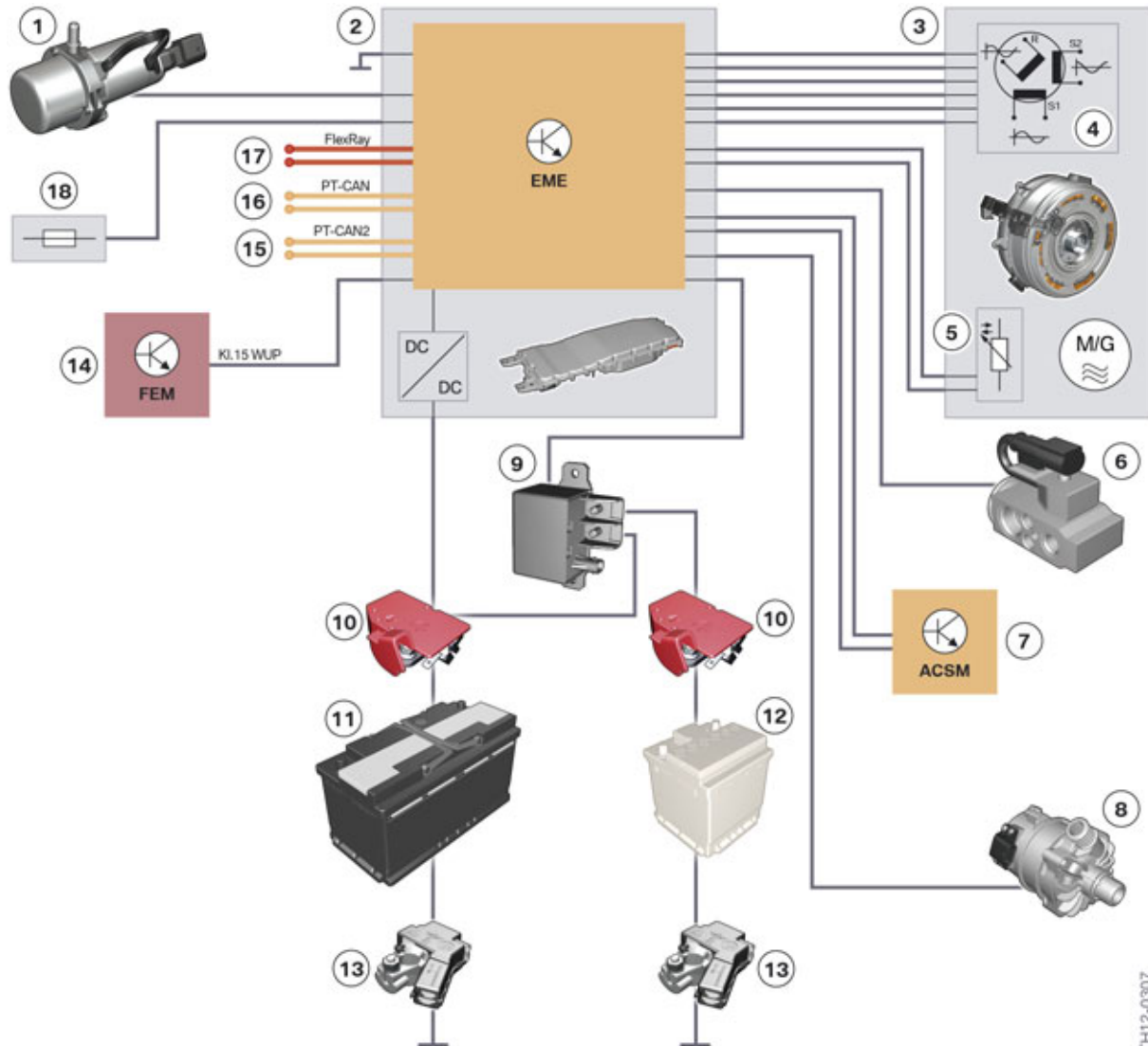
F30H connections, electrical machine electronics

Index	Explanation
1	Connector with high-voltage cable (to the electric A/C compressor)
2	Connector with three-phase high-voltage cable (to the electrical machine)
3	Signal connector
4	Tank ventilation line
5	Electrical machine electronics
6	Connector with two high-voltage cables (to the high-voltage battery unit)
7	Potential compensation line
8	Grounding cable
9	12 V connecting line
10	Coolant line, supply
11	Coolant line, return

F30H Complete Vehicle

3. Electrical Machine Electronics

3.4.1. Low-voltage connections



F30H low-voltage connections of the electrical machine electronics

Index	Explanation
1	Electric vacuum pump
2	Electrical machine electronics
3	Electrical machine
4	Rotor position sensor
5	Temperature sensor
6	Combined expansion and shutoff valve in the refrigerant circuit (for passenger compartment)
7	Crash Safety Module

F30H Complete Vehicle

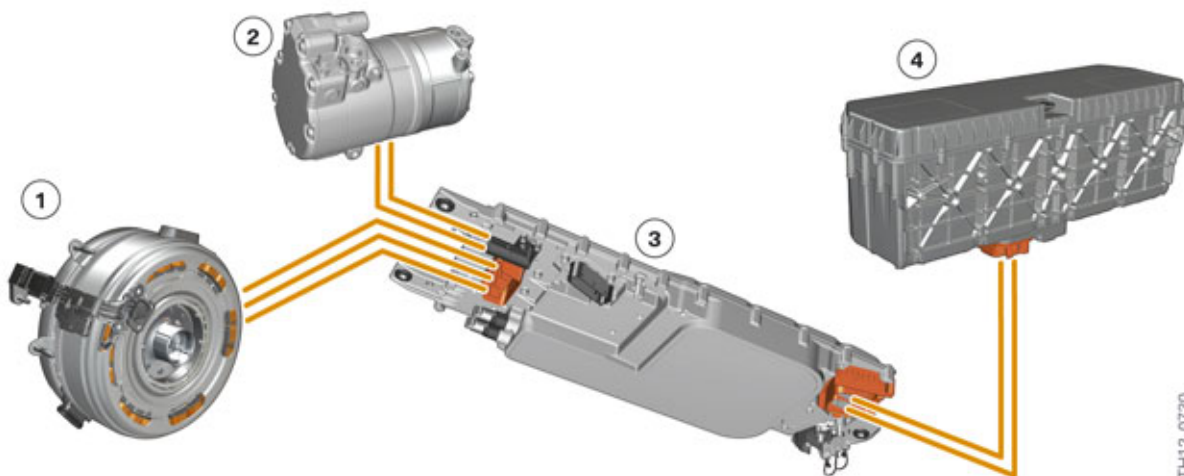
3. Electrical Machine Electronics

Index	Explanation
8	Electric coolant pump (EME cooling circuit)
9	Cut-off relay
10	Safety battery terminal
11	Vehicle battery (12 V)
12	Auxiliary battery (12 V)
13	Intelligent battery sensors (battery and auxiliary battery)
14	Front Electronic Module
15	Connection for PT-CAN2
16	Connection for PT-CAN
17	Connection for FlexRay
18	Fuse in the power distribution box, engine compartment

The electrical machine electronics is connected to the 12 V vehicle electrical system (terminals 30 and 31) via a positive battery cable and a grounding cable with large gauge wires. The DC/DC converter in the EME provides the entire 12 V vehicle electrical system with energy via this connection. These two lines are connected to the EME via a screw connection instead of plug connection.

All other lines and signals via the low-voltage connector have a relatively small current draw.

3.4.2. High-voltage connections



High-voltage connections of the electrical machine electronics (graphic shows housing of the electrical machine electronics in the F10H)

Index	Explanation
1	Electrical machine
2	Electric A/C compressor
3	Electrical machine electronics
4	High-voltage battery unit

F30H Complete Vehicle

3. Electrical Machine Electronics

There is a total of three high-voltage connections at the EME to establish contact between the lines and other high-voltage components:

- **Connection to the electrical machine:**
three-phase, AC voltage, 1 shielding for all 3 lines, high-voltage connector (screwed on)
- **Connection to the high-voltage battery:**
two-pin, direct current voltage, 1 shielding per line, flat high-voltage connector with mechanical locking, contact protection through cover over the contact blades and by bridge of high-voltage interlock loop
- **Connection to the electric A/C compressor:**
two-pin, direct current voltage, 1 shielding for both lines, round high-voltage connector with mechanical locking, contact protection through cover over the contact blades and by open circuit of the voltage supply for the electric A/C compressor control unit.

A certain sequence must be observed when breaking or establishing the contact connection both for flat and round high-voltage connectors. The individual steps are described below.

3.4.3. High-voltage cables

The high-voltage cables connect the high-voltage components and are labelled by orange cable sleeves.



Before removing the high-voltage cables the electrical safety rules must be applied.

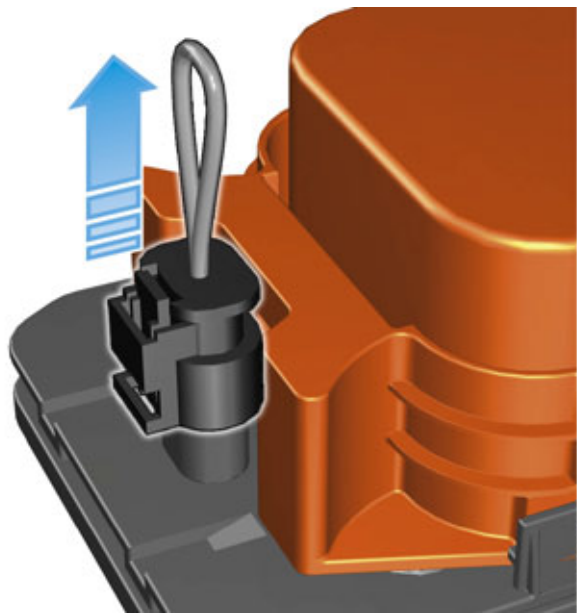
High-voltage cables must not be repaired. In the event of damage, the line must essentially be replaced!

Removing the flat high-voltage connector

The procedure described here applies to the corresponding high-voltage connector on the electrical machine electronics and to the high-voltage connector on the high-voltage battery unit.

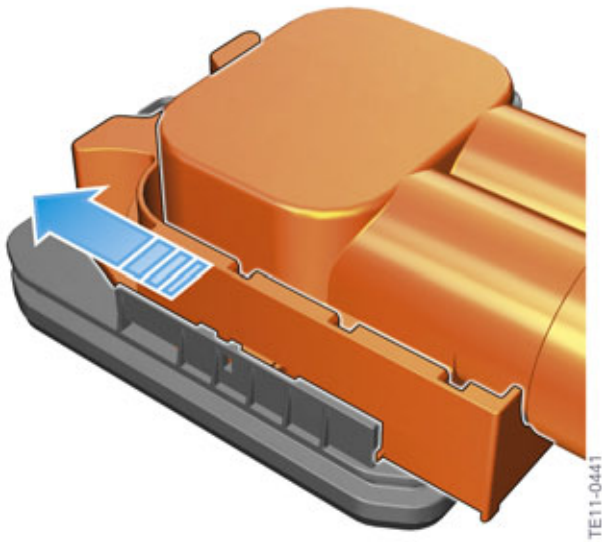
F30H Complete Vehicle

3. Electrical Machine Electronics



Removing bridge for high-voltage interlock loop

The bridge (when connected) closes the circuit of the high-voltage interlock loop. The battery management electronics continuously monitors the circuit of the high-voltage interlock loop and only when the circuit is closed is the high-voltage system active. If the circuit of the high-voltage interlock loop is interrupted by removing the bridge, the high-voltage system shuts down automatically. This is an additional safety precaution as the Service technician must have already switched off the high-voltage system before beginning work.

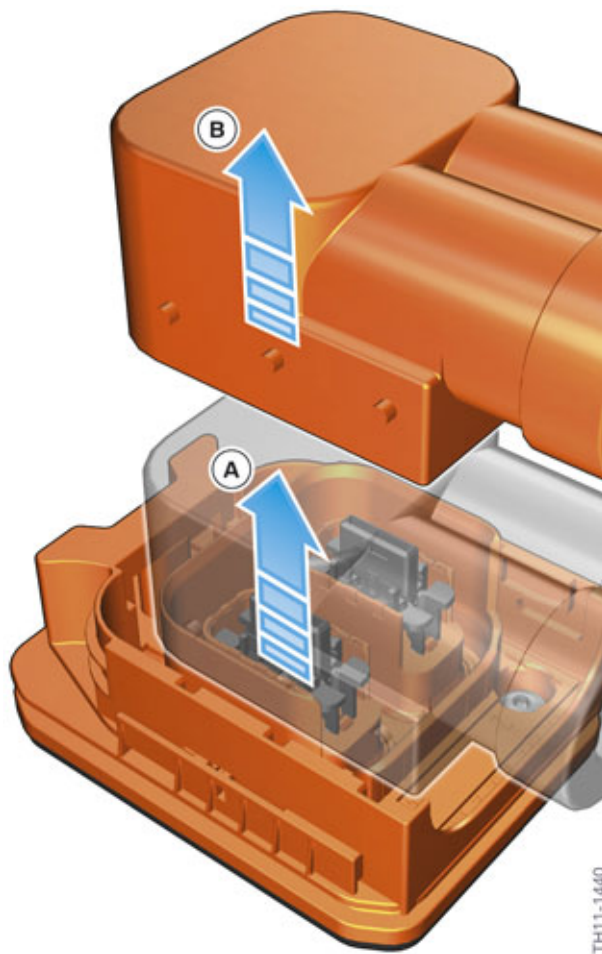


Removing the mechanical locking

Only after the bridge of the high-voltage interlock loop has been removed, can the mechanical locking be moved in the direction of the arrow. The mechanical locking is a component of the high-voltage connector on the high-voltage components. By moving the lock in the direction of the arrow the mechanical guide of the high-voltage connector on the high-voltage cable is released which permits the subsequent disconnection.

F30H Complete Vehicle

3. Electrical Machine Electronics



Removing the connector of the high-voltage cable

The connector of the high-voltage cable can now be removed in the direction of the arrow. After the connector has been pulled out a few millimeters (A), one encounters a higher counterforce. The connector must then be pulled out further in the same direction (B). Under no circumstances must the connector be pressed back into the socket on the high-voltage component after reaching position (A). This may damage the connector on the high-voltage components.



The high-voltage connector of the high-voltage cables must be pulled out at a right angle in two steps and in the same direction. Changing the direction of movement during removal is not permitted.

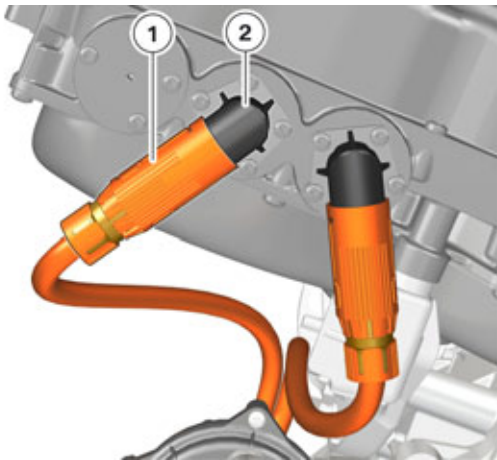
When re-inserting the high-voltage connector proceed in reverse order.

Removing the round high-voltage connector

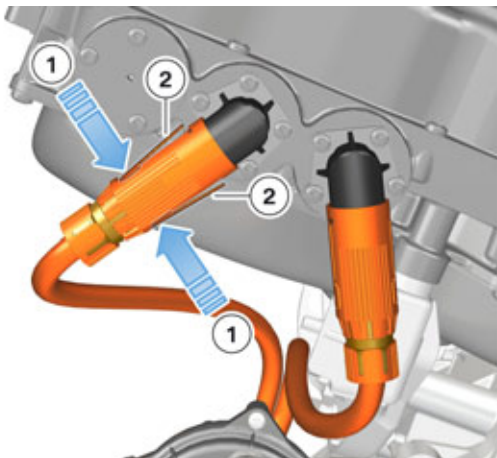
The procedure described here applies to the corresponding high-voltage connector on the electrical machine electronics and to the high-voltage connector on the electric A/C compressor. The graphics show the procedure at the electrical machine electronics of the E82E as an example, but the procedure can also be applied to the F30H.

F30H Complete Vehicle

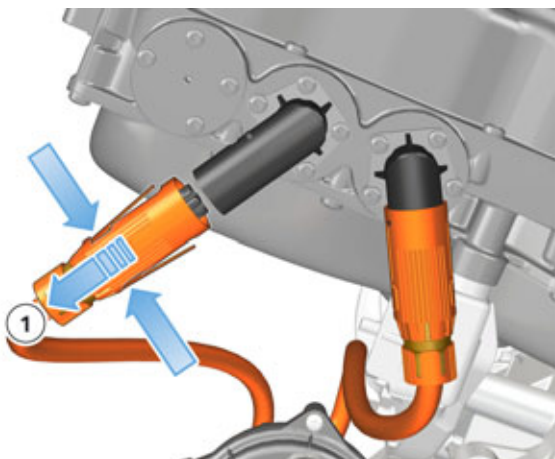
3. Electrical Machine Electronics



The connector of the high-voltage cable (1) is located at the high-voltage connection of the component (2) and is locked.



The two locking elements (2) must be pressed together in the direction of arrow (1). The mechanical lock of the connector at the connection of the high-voltage component is thus removed.



While the locking elements are further pushed together, the connector must be removed lengthways in the direction of arrow (1).

When reconnecting the high-voltage cable it is recommended to also push the locking elements together at the start. This ensures that the locking elements glide past the locks on the outside. If the locking elements are not pushed together at the start, there is a risk that they may slip inwards when

F30H Complete Vehicle

3. Electrical Machine Electronics

positioning and be damaged. At the end of the positioning ensure that the locking elements engage (clicking noise). In addition, the engaging of the locking elements should be checked by subsequent pulling on the connector.

The signal of the high-voltage interlock loop runs over the bridge in the high-voltage connector at the electrical machine electronics when the high-voltage cable is connected. If the circuit is interrupted this also causes the high-voltage system to shut down. This is an additional technical safety precaution as the Service technician should have already switched off the high-voltage system before beginning work. There is also a bridge in the round high-voltage connector at the electric A/C compressor. However the voltage supply line of the EKK control unit runs over it. **When the round high-voltage connector on the EKK is removed the circuit of the high-voltage interlock loop is thus NOT interrupted. Instead the current flow in the high-voltage cables is interrupted, because the EKK control unit can no longer draw any power. Therefore upon subsequent removal of the high-voltage connector no electric arcs can occur.**

3.4.4. Connection for potential compensation lines

The isolation monitoring determines whether the insulation resistance between active high-voltage components (e.g. high-voltage cables) and ground is above a required minimum value. The switching for the isolation monitoring is integrated in the battery management electronics (SME), and should not only detect isolation faults there, but also in the entire high-voltage electrical system. As the reference potential for the isolation monitoring is the ground potential, all conductive housing from high-voltage components must be grounded. In the case of the electrical machine electronics a separate potential compensation line is used for this.



The high-voltage system must not be operated if the potential compensation cables are not properly connected to the high-voltage components.



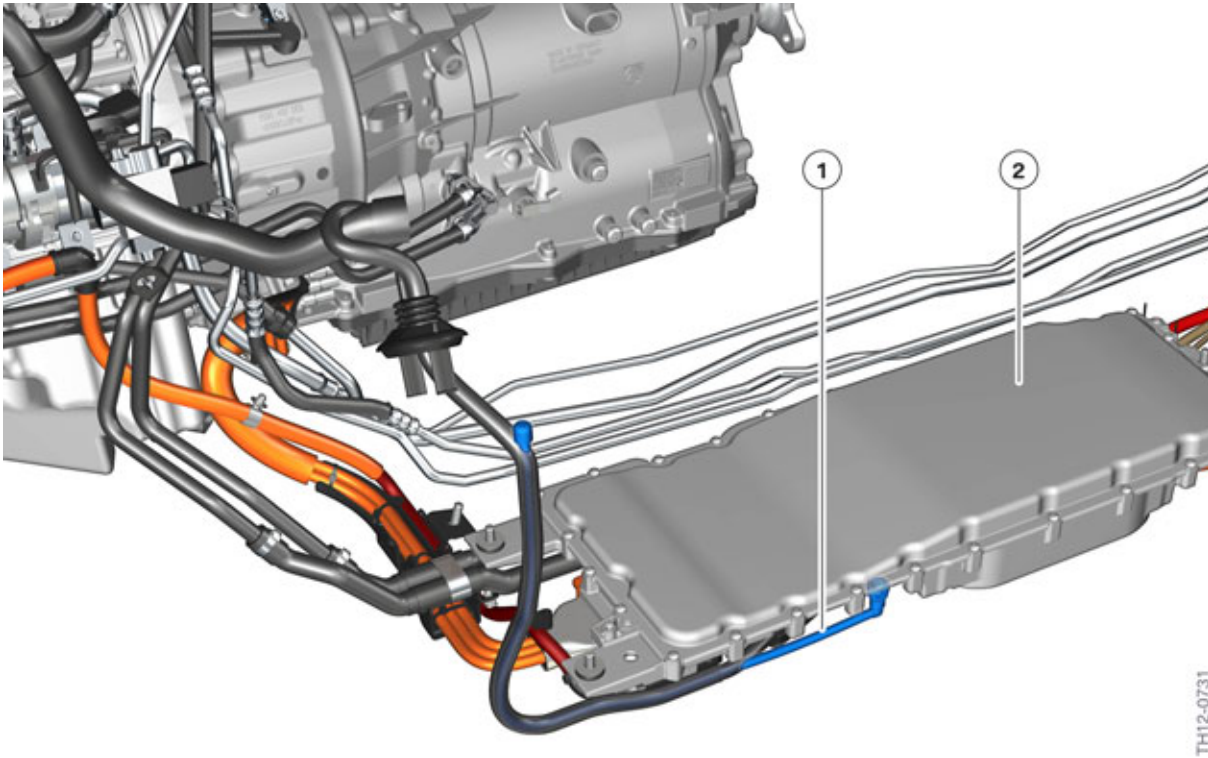
If in the event of a repair the high-voltage components or the body components are replaced, the following must be observed during assembly: The connection between the housing and the body must be properly re-established. The repair instructions must be strictly observed with regard to tightening torque and self-cutting screws.

3.4.5. Cooling/Ventilation

Beside the electrical connections, the electrical machine electronics in the F30H also has connections for coolant lines and for a tank ventilation line. A ventilation line is required to prevent water (resulting from temperature changes and thus possible condensation of air moisture) collecting inside the electrical machine electronics. The end of the ventilation line is situated high above the electrical machine electronics.

F30H Complete Vehicle

3. Electrical Machine Electronics



Tank ventilation line at the electrical machine electronics

Index	Explanation
1	Tank ventilation line
2	Electrical machine electronics

The incorporation of the EME in a separate cooling circuit ensures that the heat of the power electronics (occurring during operation) can be dissipated. The electrical machine electronics is cooled by its own cooling circuit.

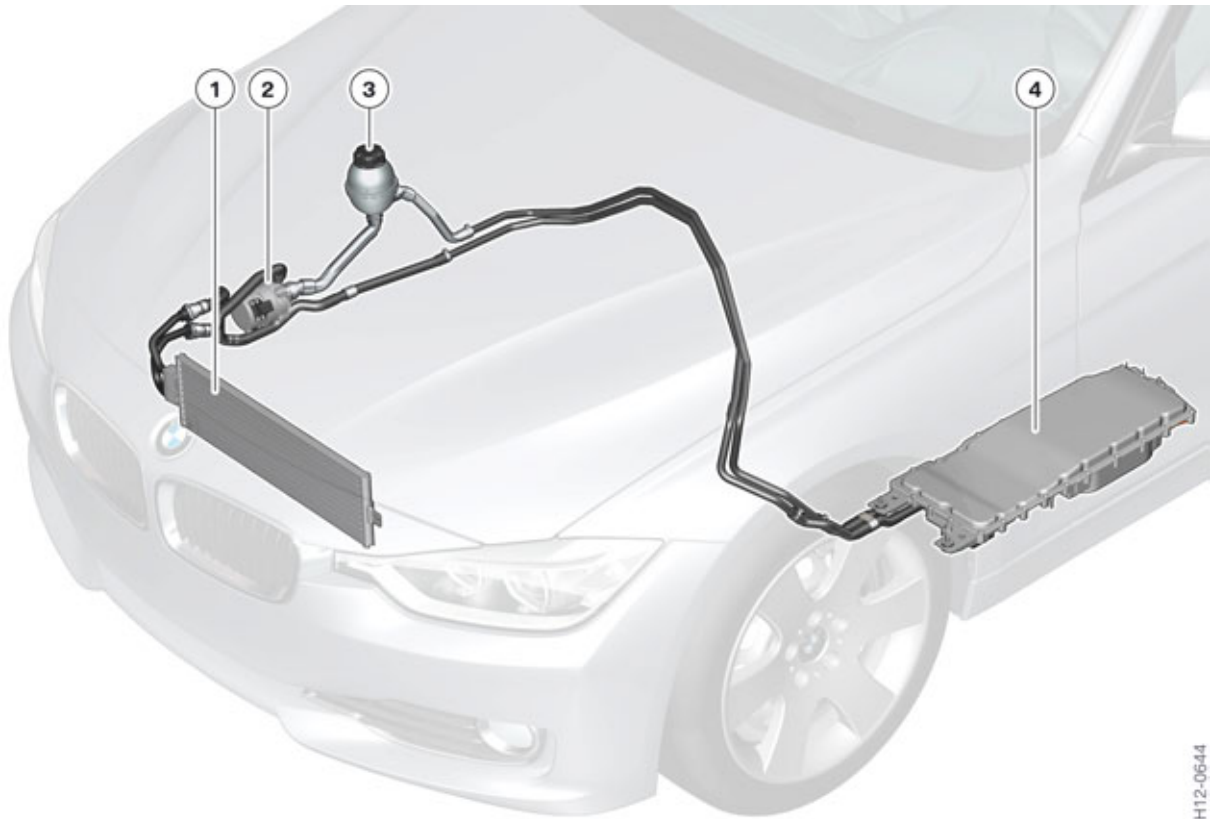
EME cooling circuit components:

- High-performance low-temperature cooler
- 50 W coolant pump
- Expansion tank
- Coolant lines.

The installation locations of these components, as well as a block diagram of all cooling circuits in the F30H, are shown in the following two graphics.

F30H Complete Vehicle

3. Electrical Machine Electronics



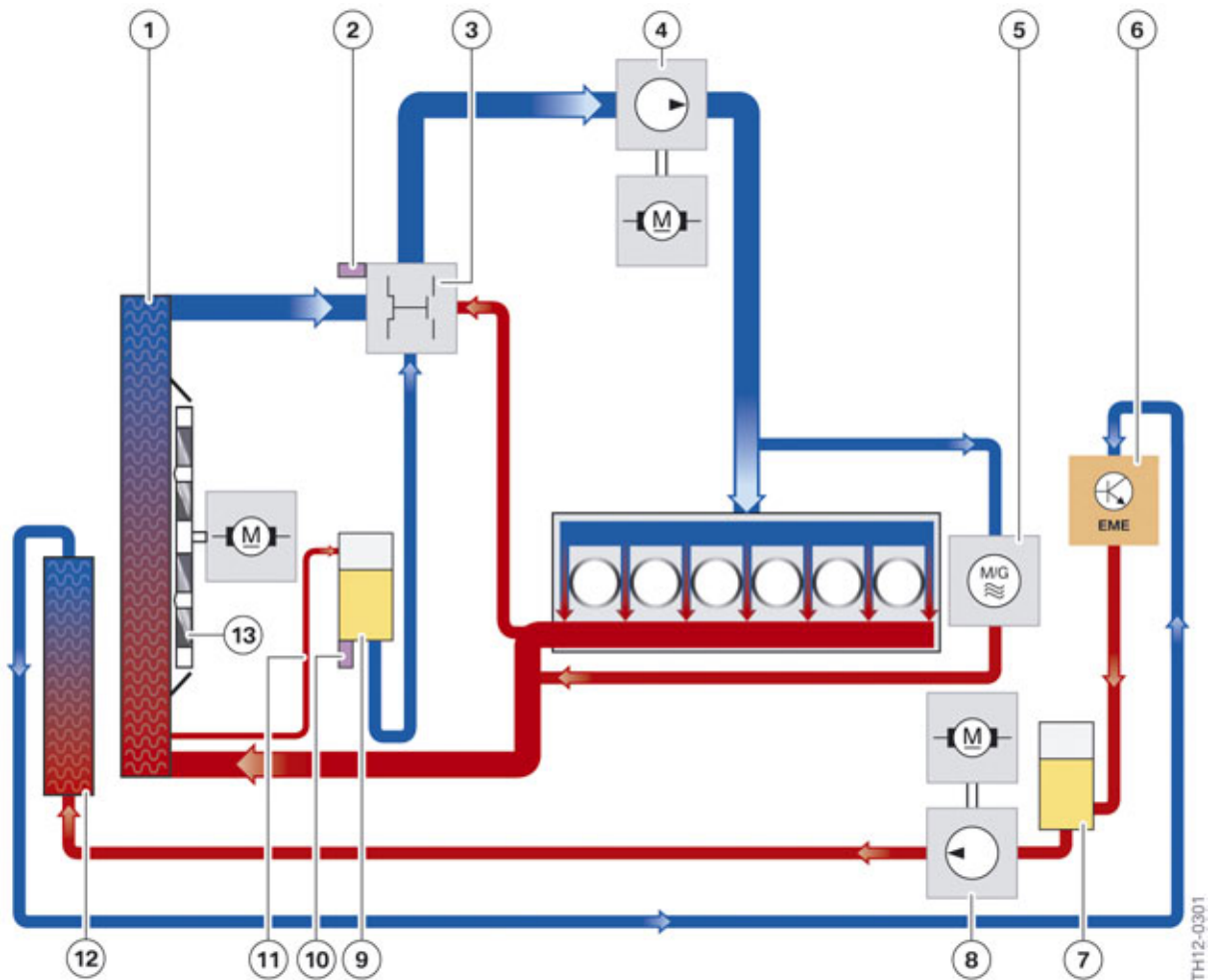
TH12-0644

Installation locations of the components in the cooling circuit of the electrical machine electronics

Index	Explanation
1	Coolant/air heat exchanger
2	Electric coolant pump
3	Expansion tank
4	Electrical machine electronics

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3. Electrical Machine Electronics



Cooling circuits F30H

Index	Explanation
1	Coolant/air heat exchanger in the cooling circuit of combustion engine and electrical machine
2	Heater coil
3	Map thermostat
4	Electric coolant pump in the cooling circuit of combustion engine and electrical machine
5	Electrical machine
6	Electrical machine electronics
7	Expansion tank in the EME cooling circuit
8	Electric coolant pump (50 W) in the EME cooling circuit
9	Expansion tank in the cooling circuit of combustion engine and electrical machine

F30H Complete Vehicle

3. Electrical Machine Electronics

Index	Explanation
10	Coolant level switch
11	Balancing line
12	Coolant/air heat exchanger in the EME cooling circuit
13	Electric fan

The coolant/air heat exchanger for the cooling circuit of the EME is integrated in the cooling module. Depending on the cooling requirement of the EME, the electric coolant pump and the electric fan are activated according to requirements thus contributing to a reduction in consumption levels.

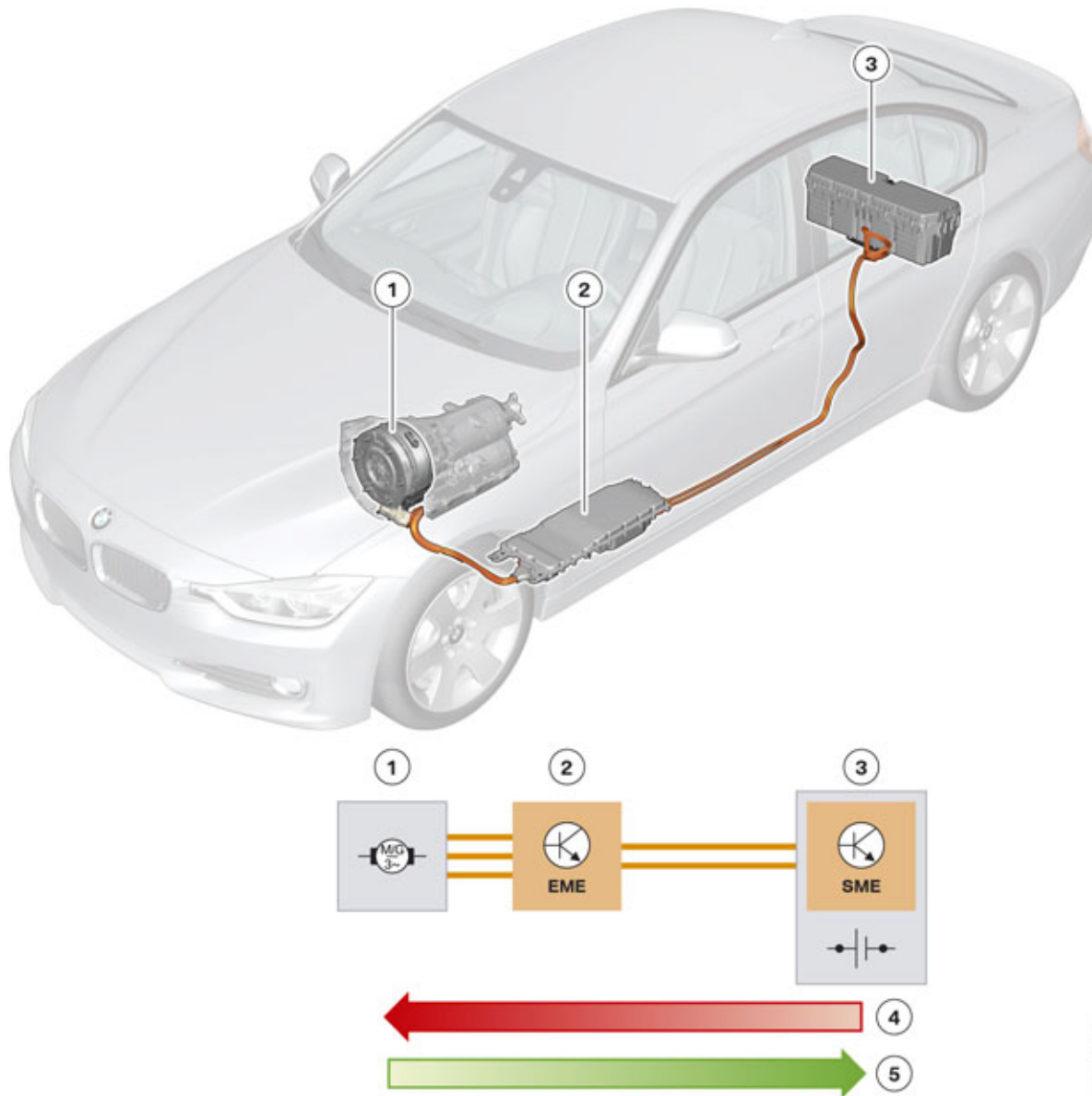
3.5. Energy flow

3.5.1. High-voltage battery – Electrical machine

The following graphic shows the energy flow between high-voltage battery and electrical machine.

F30H Complete Vehicle

3. Electrical Machine Electronics



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Energy flow between high-voltage battery and electrical machine

Index	Explanation
1	Electrical machine
2	Electrical machine electronics (EME)
3	High-voltage battery unit
4	Power electronics of the EME work as an inverter and convert high-voltage direct current into three-phase high-voltage alternating current
5	Power electronics of the EME work as a rectifier and convert three-phase high-voltage alternating current into high-voltage direct current

F30H Complete Vehicle

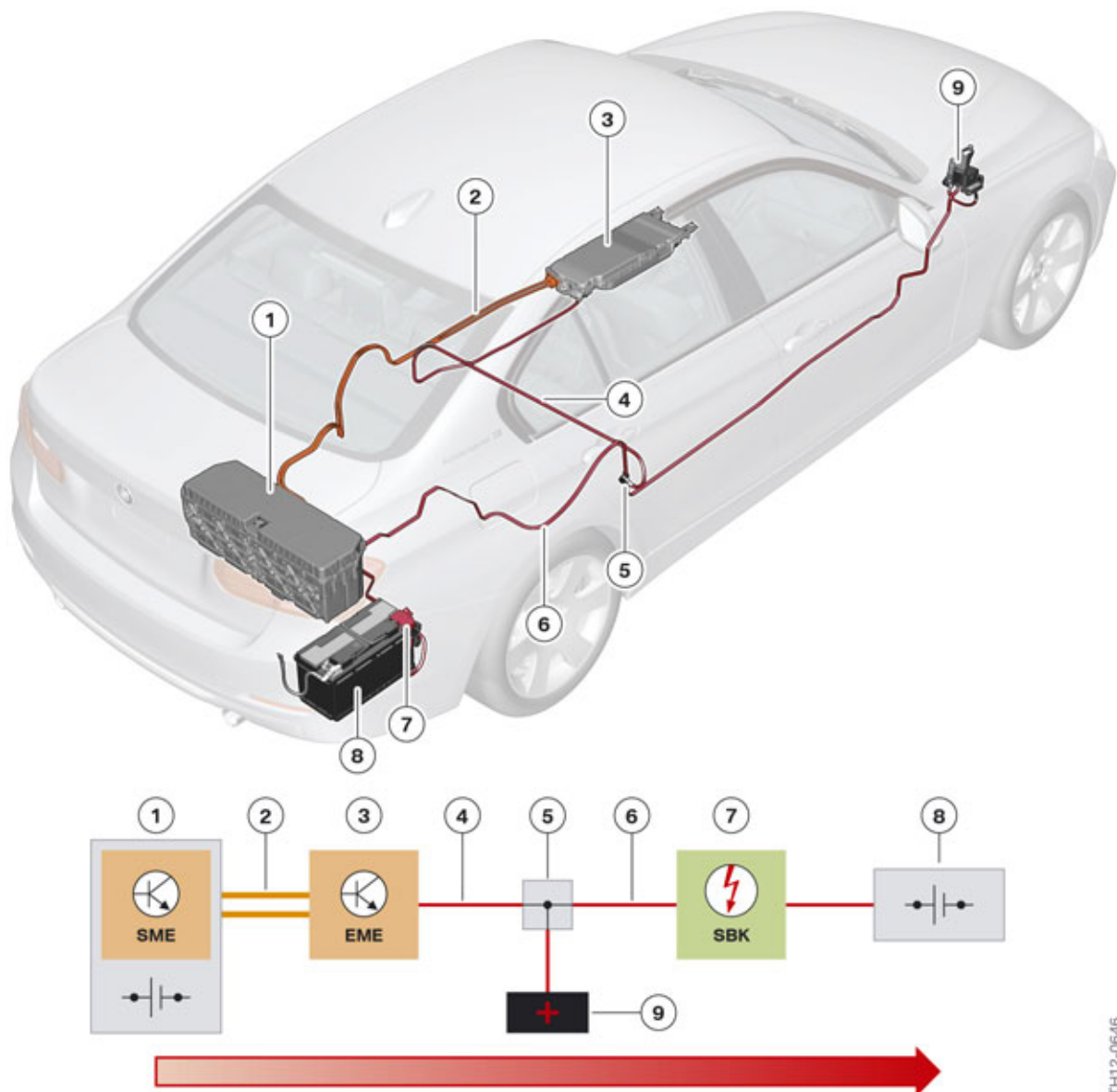
3. Electrical Machine Electronics

For activating and controlling the electrical machine a bidirectional inverter in the electrical machine electronics converts the direct current voltage from the high-voltage battery into three-phase alternating current for the electrical machine.

In alternator mode the high-voltage battery is charged via the inverter (energy recovery).

3.5.2. High-voltage – Low-voltage

The following graphic shows the energy flow between the low-voltage vehicle electrical system and the high-voltage electrical system in the F30H.



Energy flow between high-voltage battery and 12 V vehicle electrical system

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3. Electrical Machine Electronics

Index	Explanation
1	High-voltage battery unit
2	High-voltage cable
3	Electrical machine electronics (EME)
4	Positive battery cable from/to electrical machine electronics
5	Transfer support point, luggage compartment/under floor
6	Positive battery cable from/to 12 V battery
7	Safety battery terminal (SBK)
8	12 V battery
9	Positive battery connection point

In comparison to conventional vehicles, the 12 V vehicle electrical system is not supplied with energy via an alternator but via a DC/DC converter from the high-voltage electrical system.

Similar to the F10H, the DC/DC converter in the electrical machine electronics of the F30H is not a bidirectional but a unidirectional DC/DC converter. This means that only energy from the high-voltage electrical system can be transferred to the 12 V vehicle electrical system. Charging of the high-voltage battery from the 12 V vehicle electrical system therefore is not possible.

3.6. Technical data

Electrical machine electronics	
Supplier	Robert Bosch GmbH
Weight	10 kg (22 lbs)
Operating temperature range	–40 °C to +80 °C (–40 °F to 176 °F)
Power electronics	
Operating voltage range	Max. 385 V DC
Maximum current	150 A (continuous); 400 A (0.3 second)
DC/DC converter	
Nominal output voltage	14 V DC
Maximum output current	170 A (continuous)
Maximum output power	2.4 kW (continuous); 3 kW (peak value 100 ms)

F30H Complete Vehicle

4. High-voltage Battery Unit

4.1. Overview

The high-voltage battery unit of the F30H is identical to that in the F10H. It has the same dimensions, the same technical data and the same inner structure. The functions for operation and safety are also the same. For this reason only some of the key characteristic data is summarized below.

The high-voltage battery unit is the energy storage device for the electrical powertrain of the F30H. It is manufactured by BMW AG. The lithium-ion cells of the high-voltage battery unit are manufactured by A123 Systems. The high-voltage battery unit is a complete system that contains not only the cells themselves, but also the following additional components:

- Electronic control unit, battery management electronics (SME)
- Electromechanical switch contactors
- Connections for high-voltage cables
- Connections for signal lines
- Connections for refrigerant lines and drain line for condensation
- Cell supervision circuits (CSC)
- Vent hole.

In addition to the energy storage task, the high-voltage battery unit in the F30H also assumes essential tasks which contribute to the safety of the high-voltage system. These include for example the high-voltage interlock loop and isolation monitoring. The high-voltage safety connector (also called the "Service Disconnect") is not an integral part of the high-voltage battery unit in the F30H. Instead it is located under the luggage compartment floor behind the high-voltage battery unit.

The high-voltage battery unit of the F30H belongs to hazard class 9 (UN 3090) and cannot be transported via air freight.



The high-voltage battery unit is a complex, high-voltage component thus it is imperative that the handling and safety regulations are followed. In particular, lithium-ion batteries must not be overcharged or subjected to excessively high temperatures. Otherwise there is a risk of fire. The high-voltage battery unit must never be opened, otherwise there is a risk of fatal injury from high currents and voltages.



Work on live high-voltage components is strictly prohibited. Prior to every operation which involves a high-voltage component, it is essential to disconnect the high-voltage system from the voltage supply and to secure it against unauthorized return to service.

- 1 Switch off terminal 15
- 2 De-energize the high-voltage system by disconnecting the high-voltage safety connector.
- 3 Secure the high-voltage system against restart by fitting a pad lock at the high-voltage connector
- 4 Switch on terminal 15

F30H Complete Vehicle

4. High-voltage Battery Unit

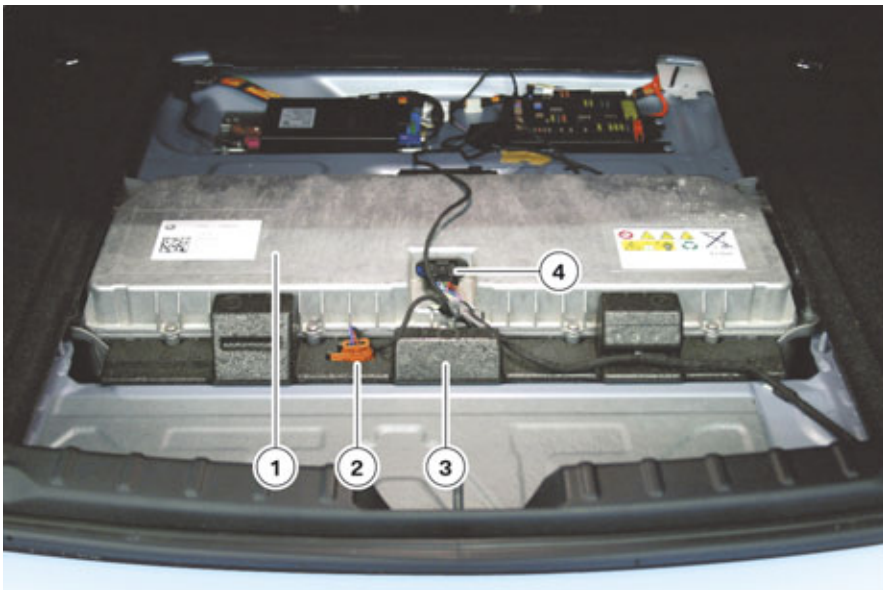
- 5 Wait until the Check Control message "High-voltage system switched off" appears in the instrument cluster – only then is the de-energized state properly determined in the Service workshop
- 6 Switch off terminal 15 and terminal R.

The vehicle-specific features of the high-voltage battery unit in the F30H (which distinguish it from the F10H) are described in the following subchapters.

4.2. Installation location and interfaces

The F30H high-voltage battery unit is in the luggage compartment, however in comparison to the F10H, it is installed under a cover, under the luggage compartment floor. This installation location has the following advantages which are relevant for the customer:

- Large usable luggage compartment capacity of 390 l (F01H has 360 l)
- Optional equipment offers "Through-loading system" for even further flexible use of luggage compartment.



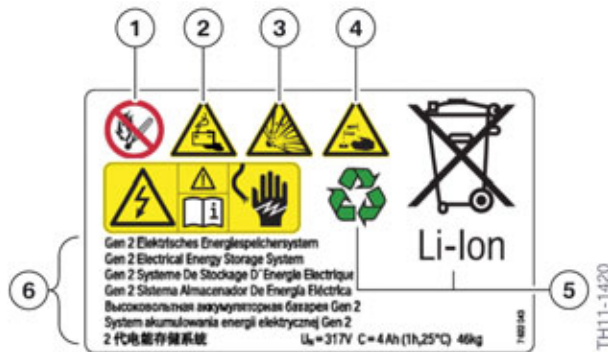
Installation location of high-voltage battery unit in the F30H from inside the luggage compartment

Index	Explanation
1	High-voltage battery unit
2	High-voltage safety connector (Service Disconnect)
3	Foam padding
4	Signal connection for battery management electronics control unit

There is a safety data sheet on the top of the housing of the high-voltage battery unit which warns against the dangers associated with working on high-voltage components.

F30H Complete Vehicle

4. High-voltage Battery Unit



Warning sticker on the high-voltage battery unit

Index	Explanation
1	Prohibition symbol: No flames, sparks or smoking
2	Warning symbol: Warning against dangers of batteries
3	Warning symbol: Warning against explosive substances / materials
4	Warning symbol: Warning against caustic substances / materials
5	Reference to disposal of high-voltage battery unit: Recycling by specialist personnel possible, not to be disposed of in domestic waste
6	Multilingual text: "Gen 2 Electrical Energy Storage System $U_N = 317\text{ V}$, $C = 4\text{ Ah}$ (1 h, 25°C), 46 kg"



Observe the safety data sheet for the high-voltage battery for all repair work on these components. Use the personal safety equipment prescribed there. If there is any uncertainty regarding a possible hazard, contact the technical support (PUMA) of the BMW Group.

In a conventional F30 a tray is located under the luggage compartment floor in the rear of the bodyshell. This can also be used by the customer as an additional storage compartment. In the F30H the high-voltage battery unit is installed in this tray. The high-voltage battery unit is bolted directly to the frame of the tray. For this purpose there are stud bolts and a threaded bushing in the bodyshell on the left and right side of the tray. The potential compensation which is required for the isolation monitoring is also established via the screw connection of the housing of the high-voltage battery unit to the bodyshell. There is no separate potential compensation line at the high-voltage battery unit of the F30H.

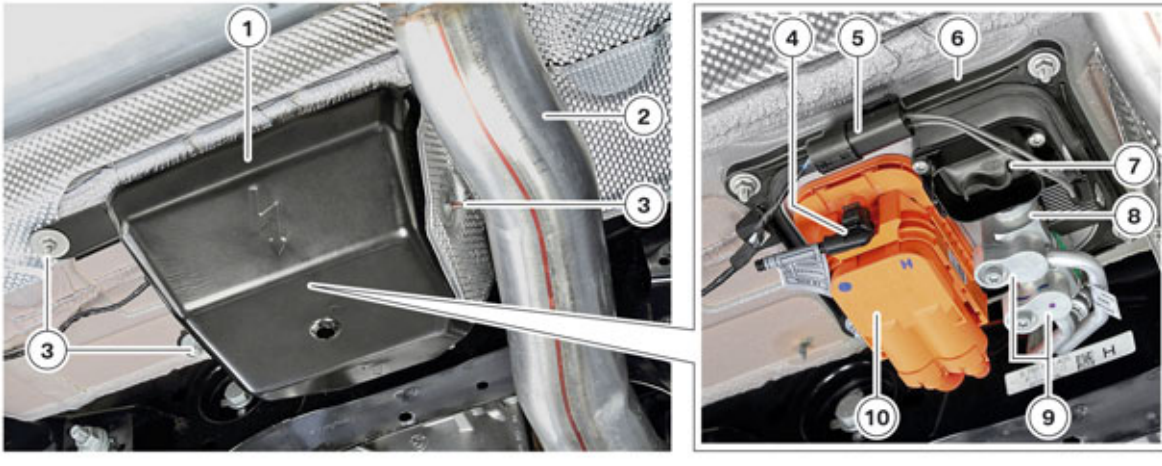


The screw connection of the high-voltage battery unit to the bodyshell also establishes the potential compensation. For this reason the screws and threaded bushings cannot be painted. The tightening torque specified for the screws/nuts must be observed.

There is a 2-pin high-voltage connection at the high-voltage battery unit, which is located on its bottom. In comparison to the F10H, the high-voltage connection and the connection for the refrigerant lines in the F30H are also accessible from the vehicle underbody.

F30H Complete Vehicle

4. High-voltage Battery Unit



TH12-0643

Lower connections of the high-voltage battery unit in the F30H

Index	Explanation
1	Stoneguard
2	Exhaust tailpipe of the exhaust system
3	Nuts for securing the stoneguard
4	Bridge for high-voltage interlock loop
5	Plug connection in the control line
6	Flexible frame (gasket)
7	Vent hole
8	Combined expansion and shutoff valve
9	Connections for refrigerant lines
10	High-voltage connector

The high-voltage connection has contact protection, a mechanical slide, as well as connection for the bridge of the high-voltage interlock loop, as is known from the F10H. The refrigerant lines and the combined expansion and shutoff valve are connected to the high-voltage battery unit by screws like in the F10H. Before the high-voltage battery unit is removed the refrigerant circuit must be evacuated. Only then can refrigerant lines and the combined expansion and shutoff valve be removed.

In the case of a cell defect excess pressure may arise in the inside of the high-voltage battery unit. Therefore there is a vent hole installed to allow the gases to escape outwards. This vent has a diaphragm which is permeable for gases, but sealed from fluids from the outside. A vent pipe like the one used in the F10H is not required in the F30H as the vent hole of the high-voltage battery unit is already located outside the vehicle.

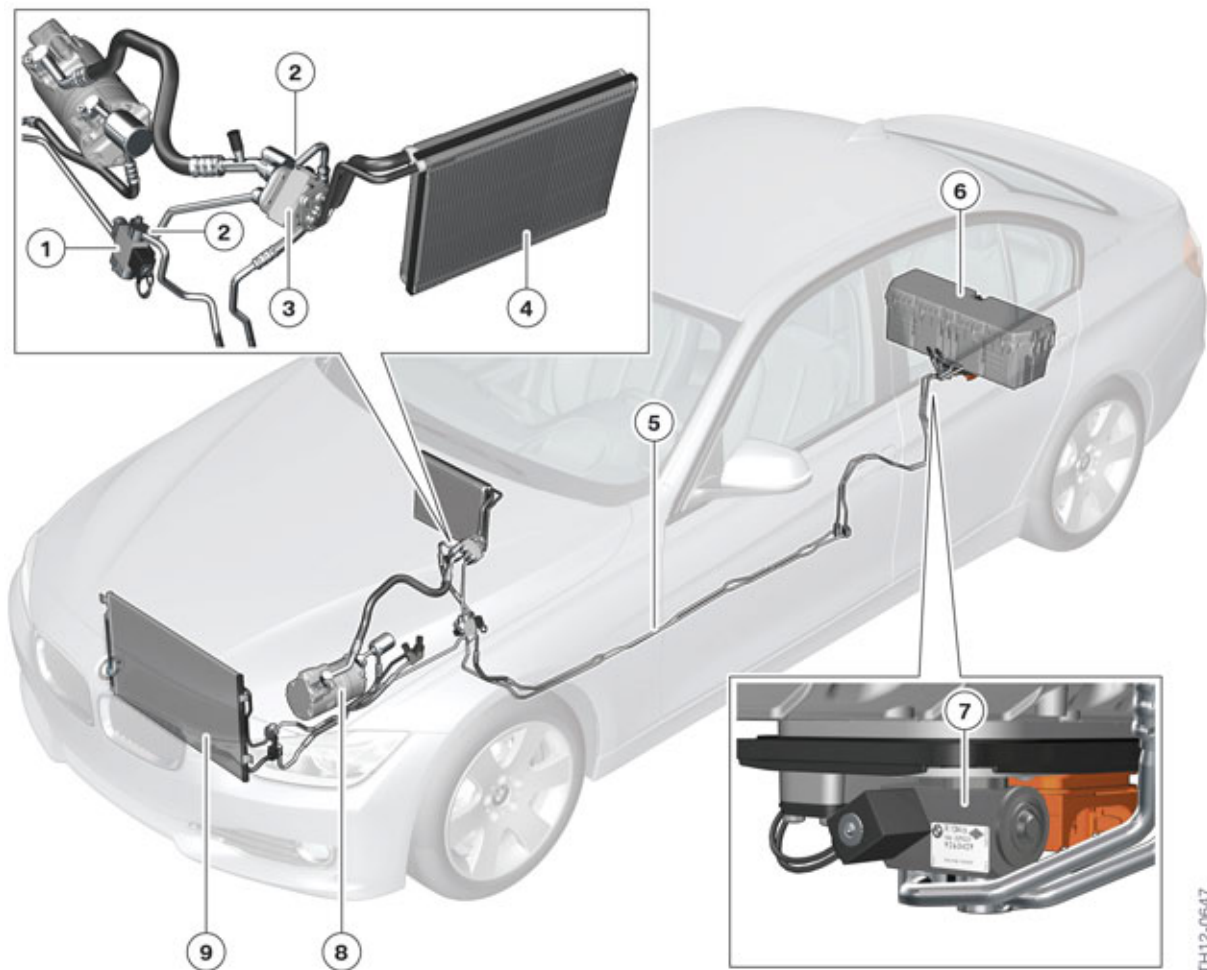
The entire lower connection field of the high-voltage battery unit or the opening in the tray of the bodyshell has a black frame made from flexible plastic. This serves as a gasket and prevents water penetrating into the luggage compartment from the outside. The connection field is also protected with a black painted panel against stone chipping or other mechanical influences. This panel is connected at the vehicle underbody with help of five nuts or together to the heat shield of the exhaust system.

F30H Complete Vehicle

4. High-voltage Battery Unit

4.3. Cooling system

To maximize the service life of the high-voltage battery and obtain the greatest possible power, it is operated in a defined temperature range. A cooling system is used which cools the high-voltage battery with refrigerant and has a similar structure in principle to the F10H. The refrigerant circuit of the air-conditioning was expanded with a parallel duct to the high-voltage battery. Differences between F30H and F10H only result from the varying vehicle geometry and the installation location of the high-voltage battery.



F30H Refrigerant circuit

TH12-0647

Index	Explanation
1	Shutoff valve for passenger compartment
2	Branching of refrigerant lines to high-voltage battery unit
3	Expansion valve for passenger compartment
4	Evaporator for passenger compartment
5	Refrigerant lines to high-voltage battery unit

F30H Complete Vehicle

4. High-voltage Battery Unit

Index	Explanation
6	High-voltage battery unit
7	Combined expansion and shutoff valve for high-voltage battery unit
8	Electric A/C compressor
9	Condenser

In the temperature range in which the high-voltage battery unit does not need to be cooled the combined expansion and shutoff valve for the high-voltage battery unit remains closed. However, if the high-voltage battery unit has to be cooled, the battery management electronics activates the combined expansion and shutoff valve at the bottom of the high-voltage battery unit. In this way refrigerant can flow to the high-voltage battery, expanding, evaporating and cooling in the process. The cooling of the high-voltage battery unit can be carried out completely independently of the cooling in the passenger compartment as these are controlled by separate expansion and shut-off valves.



Because the high-voltage battery unit is connected to the refrigerant circuit, preliminary work similar to that conducted on the air-conditioning of conventional vehicles (e.g. evacuation) will be required prior to removal and installation work. The repair instructions must always be followed to the letter.

4.4. High-voltage safety connector (Service Disconnect)

The high-voltage safety connector ("Service Disconnect") of the F10H is not a direct element of the high-voltage battery unit. Instead it is located under the luggage compartment floor behind the high-voltage battery unit in a foam padding. Its installation location is shown in a graphic in the "Installation location and interfaces" chapter.

As in the previous ActiveHybrid vehicles, the high-voltage safety connector in the F30H also fulfils two tasks :

- Disconnecting the high-voltage system from the supply
- Provide the high-voltage system with a safeguard against unintentional restarting.

The high-voltage safety connector (connected bridge) is part of the circuit of the high-voltage interlock loop. If one disconnects the connector and socket of the high-voltage safety connector, the circuit of the high-voltage interlock loop is interrupted. The voltage supply of the electromechanical switch contactors in the high-voltage battery unit is also interrupted. The high-voltage system automatically shuts down and is thus switched to a de-energized state.



The connector and socket of the high-voltage safety connector cannot be disconnected from each other completely. Both parts are mechanically secured against complete separation. To interrupt the circuit of the high-voltage interlock loop, it is sufficient to separate the two parts far enough so that the pad-lock can be used to secure against restart.

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4. High-voltage Battery Unit



Before working on the high-voltage connection of the high-voltage battery unit or another high-voltage component, it is essential to disconnect the high-voltage system from the supply, prevent against unintentional restart and verify its safe isolation from the supply.

The exact procedure to put the system to a de-energized state is described in the chapter entitled "Service instructions", subchapter "Safe working practices for working on a high-voltage system".

4.5. Functions

The functions of the high-voltage battery unit in the F30H are identical to those in the F10H. They are therefore only summarized here. All functions of the high-voltage battery unit are controlled by the battery management electronics (SME) in conjunction with other control units such as the cell supervision circuit (CSC), the electrical machine electronics (EME) or the Crash Safety Module (ACSM).

These functions are:

- **Starting the high-voltage system:**
The high-voltage system is automatically tested, the voltage is raised and finally the contacts of the electromechanical switch contactor are closed.
- **Regular shut-down of the high-voltage system:**
For example when switching off terminal 15 or at the end of the stationary cooling the high-voltage system is regularly shut down. The currents in the high-voltage electrical system are first adjusted to zero, the switch contactors of the high-voltage battery unit opened, the high-voltage circuit (link capacitors) discharged and the windings of the electrical machine short-circuited. Final tests automatically check the high-voltage system and the regular shut-down. This process can last up to several minutes.
- **Quick shut-down of the high-voltage system:**
This quick shut-down is then always carried out if, for safety reasons, the voltage in the high-voltage system has to be reduced to a safe value as quickly as possible. The following triggering criteria can cause a quick shut-down: accident, response of the overload current monitoring, short circuit, critical state of a cell, failure of the 12 V voltage supply, in response to an open in the high-voltage interlock loop.
- **Charging strategy and operational strategy:**
To maximize the service life of the high-voltage battery the full state-of-charge range is not used, only a part of it. A range of zero to 100 percent is shown to the customer in the instrument cluster (display of the on-board computer), which however corresponds to an actual state of charge of about 25 to 70 percent.
- **Monitoring functions:**
The monitoring functions guarantee the safety of the high-voltage system and the optimal operating conditions of the high-voltage battery unit. For this reason both the high-voltage interlock loop and isolation monitoring, as well as monitoring of the state of the cell (voltage, temperature) are part of these functions.

F30H Complete Vehicle

4. High-voltage Battery Unit

4.6. Service information

4.6.1. Installation and removal

Only the complete high-voltage battery unit can be replaced in a Service workshop, if this is provided for by the test schedule of the BMW diagnosis system.

The return of the faulty high-voltage battery unit can only be done via the "Warranty Parts Process" and not via the normal "Replacement Parts Process". The high-voltage battery unit of the F30H belongs to hazard class 9 (UN 3090) and cannot be transported via air freight.



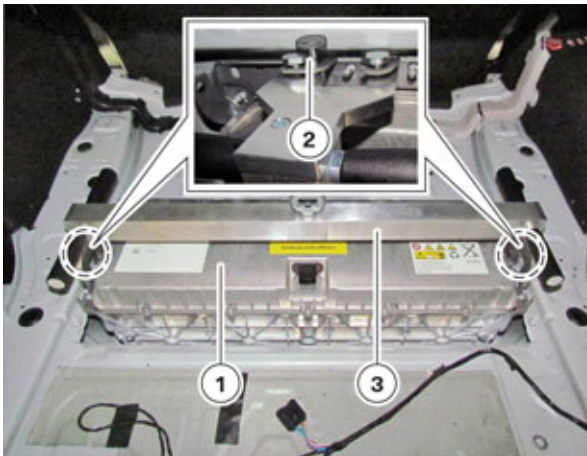
Because the high-voltage battery unit is a high-voltage component, it is essential during removal and installation to observe the electrical safety rules before starting work.

In addition, prior to removal the refrigerant must be drawn off from the refrigerant circuit and after installation re-introduced.



A precise description of the work sequence for removal and installation can be found in the repair instructions. This must be followed exactly.

The installation and removal of the high-voltage battery unit is generally done using the same set of special tools as in the F10H. However, the special tool number 0 496 722 (frame with slide rails) is not required because the high-voltage battery unit in the F30H is not installed as deep in the luggage compartment.



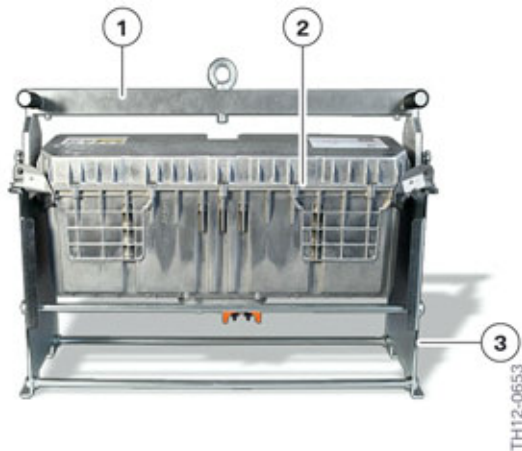
Attaching the lifting tool for removing the high-voltage battery unit

Index	Explanation
1	High-voltage battery unit
2	Retaining screw
3	Lifting tool

F30H Complete Vehicle

4. High-voltage Battery Unit

With help of the retaining screw the lifting tool is secured against unintentional removal of the high-voltage battery unit.



Removed high-voltage battery unit on set-down device

Index	Explanation
1	Lifting tool
2	High-voltage battery unit
3	Set-down device

In comparison to the F10H, the high-voltage battery unit in the F30H cannot be removed fully with help of a workshop crane. Instead it must be removed from the luggage compartment by two Service employees.



Proceed with caution when removing the high-voltage battery unit from the tray and the luggage compartment so that the connections on the bottom do not incur damage.



In the event of replacement of the high-voltage battery unit, start-up is required with help of the BMW diagnosis system.

During this start-up a type of "transport mode" of the high-voltage battery unit is reset. This "transport mode" prevents the contacts of the electromechanical switch contactors closing during transportation or storage. Only when this "transport mode" has been reset after installation is the high-voltage battery unit and thus the complete high-voltage system operational.

4.6.2. Charging the high-voltage battery unit

The high-voltage battery unit in the F30H can be charged the same way as in the F10H. The operating strategy always ensures sufficient state of charge of the high-voltage battery unit in normal customer operation. In the event of extended immobilization periods or longer workshop visits the high-voltage battery unit however should be fully charged in order to avoid damage.

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4. High-voltage Battery Unit

The following steps must be performed for this:

- Start engine
- Engage “P” and activate Automatic Hold brake
- Press and hold down brake pedal
- With help of the brake pedal keep the speed of the combustion engine at about 2000 rpm.

After a few minutes the high-voltage battery is fully charged again. The state of charge can be controlled with help of the state of charge display in the instrument cluster.

4.6.3. Safe working practices for working on a high-voltage system



Before working on high-voltage components of the F10H, it is essential to observe and implement the electrical safety rules:

- 1 The high-voltage system must be disconnected from the supply
- 2 The high-voltage system must be provided with a safeguard to prevent unintentional restarting
- 3 The safe isolation of the high-voltage system must be verified.

The following chapters provide brief descriptions on how to implement the electrical safety rules in the F30H.

Preparations

Prior to beginning any work, the vehicle must be secured against rolling away (engage the parking lock of the automatic transmission and activate the Automatic Hold brake). Disconnect any connected charging cables. Terminal 15 and terminal R must be switched off.

From an electrical safety viewpoint, work can begin on the system when it has been switched off and put to a de-energized state. However, the high-voltage system may still be active and current may flow via the contacts of the switch contactors. During shut-down this may result in a load in the contacts which can be avoided in the following manner:



Allow the vehicle or the control units to go to sleep before starting to de-energize the components. This prevents pressure being put on the contacts of the electromechanical switch contactors because the high-voltage system is regularly shut down with the go to sleep function.



Work on live high-voltage components is strictly prohibited. Prior to every operation which involves a high-voltage component, it is essential to disconnect the high-voltage system from the voltage supply and to secure it against unauthorized reconnection.

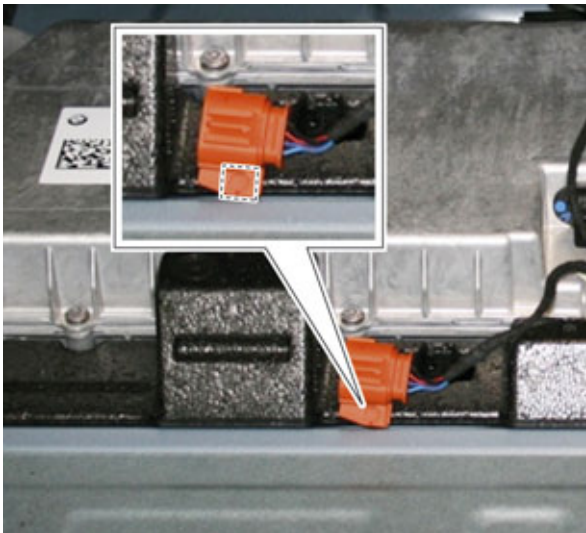
F30H Complete Vehicle

4. High-voltage Battery Unit

- 1 Switch off terminal 15.
 - 2 **De-energize** the high-voltage by opening the high-voltage safety switch (Safety Disconnect).
 - 3 **Secure** the high-voltage safety connector from being reconnected.
 - 4 Switch on terminal 15.
 - 5 **Verify** that the "High-voltage system switched-off" Check Control message is displayed in the instrument cluster.
 - 6 Switch off terminal 15 and terminal R.
-

Disconnect the high-voltage system from the supply

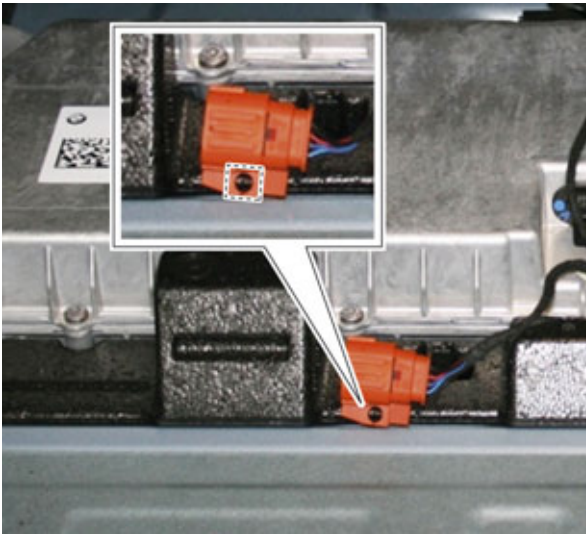
The high-voltage system in the F10H is disconnected from the supply with the high-voltage safety connector. To disconnect from the supply, the connector must be pulled from the associated socket. This interrupts the circuit of the high-voltage interlock loop. As a result of the open circuit of the high-voltage interlock loop, the voltage supply of the switch contactors is interrupted and the high-voltage system put into a de-energized state.



This image shows the high-voltage safety connector in a connected state. The circuit of the high-voltage interlock loop is not interrupted.

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4. High-voltage Battery Unit



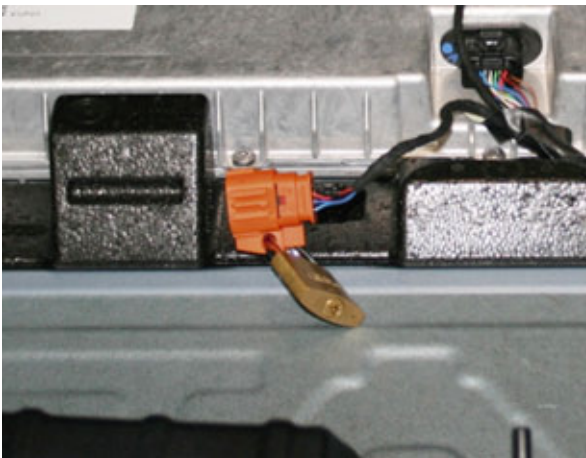
To pull apart the socket and the connector, the mechanical lock must be removed by squeezing the tab. As soon as the mechanical lock has been removed, the connector can be pulled from the socket a few milliliters.



Do not pull any further or harder if resistance can be felt. The connector and socket of the high-voltage safety connector cannot be disconnected from each other completely.

Provide the high-voltage system with a safeguard against unintentional restarting

Securing against restart is also done at the high-voltage safety connector. A commercially available pad lock is required for this purpose.



By separating the socket and connector of the high-voltage safety connector, a hole becomes free through both parts. The loop of a pad lock must be inserted in this hole.

The pad lock can now be closed. The key must be stored in a safe place during work on the high-voltage system so that an unauthorized person cannot unlock the lock. Inserting and closing the pad lock at the high-voltage safety connector is an effective way of ensuring that the high-voltage system is not switched on again without the knowledge and consent of the service technician.

Verifying safe isolation from the supply

The de-energized state is not verified using a measuring device or via the BMW diagnosis system. Instead, the high-voltage components measure the voltage themselves and transmit the measuring result via bus signal to the instrument cluster.

The instrument cluster does not generate the Check Control message to display the de-energized state unless all involved high-voltage components consistently signal the de-energized state. This Check Control symbol in red shows a crossed-out lightning bolt symbol.

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4. High-voltage Battery Unit



Display of isolation from the supply in the instrument cluster

Index	Explanation
1	Note "High-voltage system switched off"
2	Check Control symbol "High-voltage system de-energized"

In order to verify the de-energized state, the service technician must switch on terminal 15 and wait until he sees the Check Control message with the symbol shown above on the instrument cluster. Then, and only then, have you ensured that the high-voltage system is de-energized. After the de-energized state has been verified, terminal 15 and terminal R must be switched off again before you can start the actual work.



If the Check Control message is not displayed, you must **not** carry out any work on high-voltage components!



If the de-energized state cannot be determined and repairs are necessary on the high-voltage system, Technical Support (PUMA) must be contacted.

4.6.4. Procedure after an accident

The safety concept of the high-voltage system ensures that, even during or after an accident, there is no danger to the customer or the service technician. The high-voltage system is automatically deactivated in the event of an accident in such a way that no dangerous voltages are applied at those points on the high-voltage components which are accessible from the outside. Deactivation of the high-voltage system is done as follows:

In normal operation the battery management electronics (SME) is supplied via terminal 30B. The coils of the electromechanical switch contactors are also supplied. Deactivation in the event of an accident is done by an extra battery safety terminal. It contains an additional normally closed contact. This switch contact opens when the battery safety terminal is triggered at the same time as the positive battery cable is severed. Opening this switch contact causes the switch contactors in the high-voltage battery unit to open directly so that no more dangerous voltage can be fed into the high-voltage

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4. High-voltage Battery Unit

electrical system from the high-voltage battery unit. The electrical machine electronics (EME) receives a crash signal from the Crash Safety Module (ACSM). The EME then discharges the link capacitors immediately.

After an accident the safety battery terminal remains in the state described above such that the high-voltage battery unit is not operational. Thus the high-voltage system remains inactive even if terminal 15 is switched on again.



Before working on the high-voltage components or on the battery safety terminal of an F30H which has been involved in an accident with a triggered battery safety terminal, contact the Technical Support (PUMA) of the BMW Group.

4.6.5. Transport mode

To protect the high-voltage battery unit, the following functions are **not** available when the vehicle is in transport mode:

- Electrical driving
- Boost function
- Automatic engine start/stop function.

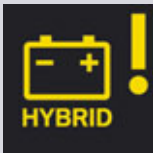
The high-voltage battery unit is always charged in transport mode if the combustion engine is running.

Display of the state of charge

Like with other vehicles, in transport mode the state of charge of the 12 V battery is displayed as a Check Control message. In the F10H, also in transport mode a Check Control message is displayed for the state of charge of the high-voltage battery unit. The state of charge of the high-voltage battery unit is displayed in three stages:

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4. High-voltage Battery Unit

State of charge	Display in the instrument cluster	Action
Battery state of charge of the high-voltage battery is OK		No further work necessary
High-voltage battery is flat		Charge high-voltage battery!
High-voltage battery is dead		Replace high-voltage battery unit!

If the high-voltage battery unit was dead, the display in the instrument cluster to the high-voltage battery unit is replaced. After resetting transport mode, there are no Check Control messages on the state of charge of the high-voltage battery unit in the instrument cluster. The state of charge can instead be called up in the display of the on-board computer.

4.7. Technical data

The following is a summary of the important characteristics of the high-voltage battery unit:

Nominal voltage	316.8 V
Battery cells	96 at 3.3 V each
Storable amount of energy	1350 Wh
Used amount of energy	675 Wh
Maximum power	43.2 kW
Storage technology	Lithium-ion
Dimensions (width, height, depth)	690 mm x 237 mm x 210 mm
Weight	about 46 kg
Cooling system	Cooling with refrigerant (R134a)

F30H Complete Vehicle

5. Hybrid Brake System

5.1. Introduction

With regard to the chassis and suspension of the F30H the biggest technical change (in comparison to the conventional F30) is in the hybrid brake system.

The following components/systems were taken from the F30 335i and implemented in the F30H:

- Twin-arm McPherson strut front suspension
- HA5 rear axle
- Electromechanical power steering (EPS) with Servotronic function / Variable sport steering with Servotronic function
- Parking brake
- Brake servo and tandem brake master cylinder
- Wheel brakes
- DSC hydraulic control unit and control unit.

The DSC was adopted for use in the F30H on account of its hardware, however the software was adapted to enable brake energy regeneration. In comparison to the conventional F30 the spring/damping system was also coordinated especially for the F30H. Despite its slightly higher vehicle weight, the F30H still offers excellent driving characteristics, such as those known from the conventional F30. The modified spring/damping coordination has an important impact for the service technicians:



When performing a wheel alignment on an F30H, the modified ride height must be considered in comparison to the conventional F30. The BMW ActiveHybrid 3 has a higher ride height than the 335i and is 5 mm higher at the rear axle.

Similar to the F10H the hybrid brake system of the F30H also enables brake energy regeneration and thus contributes to increasing efficiency. At the same time it ensures that the driver can safely and firmly decelerate the vehicle.

The hybrid brake system of the F30H is almost identical in structure to that of the F10H. The following components differentiate it from the brake system of a conventional F30:

- Electric vacuum pump
- Brake vacuum sensor
- Brake pedal travel sensor
- DSC software.

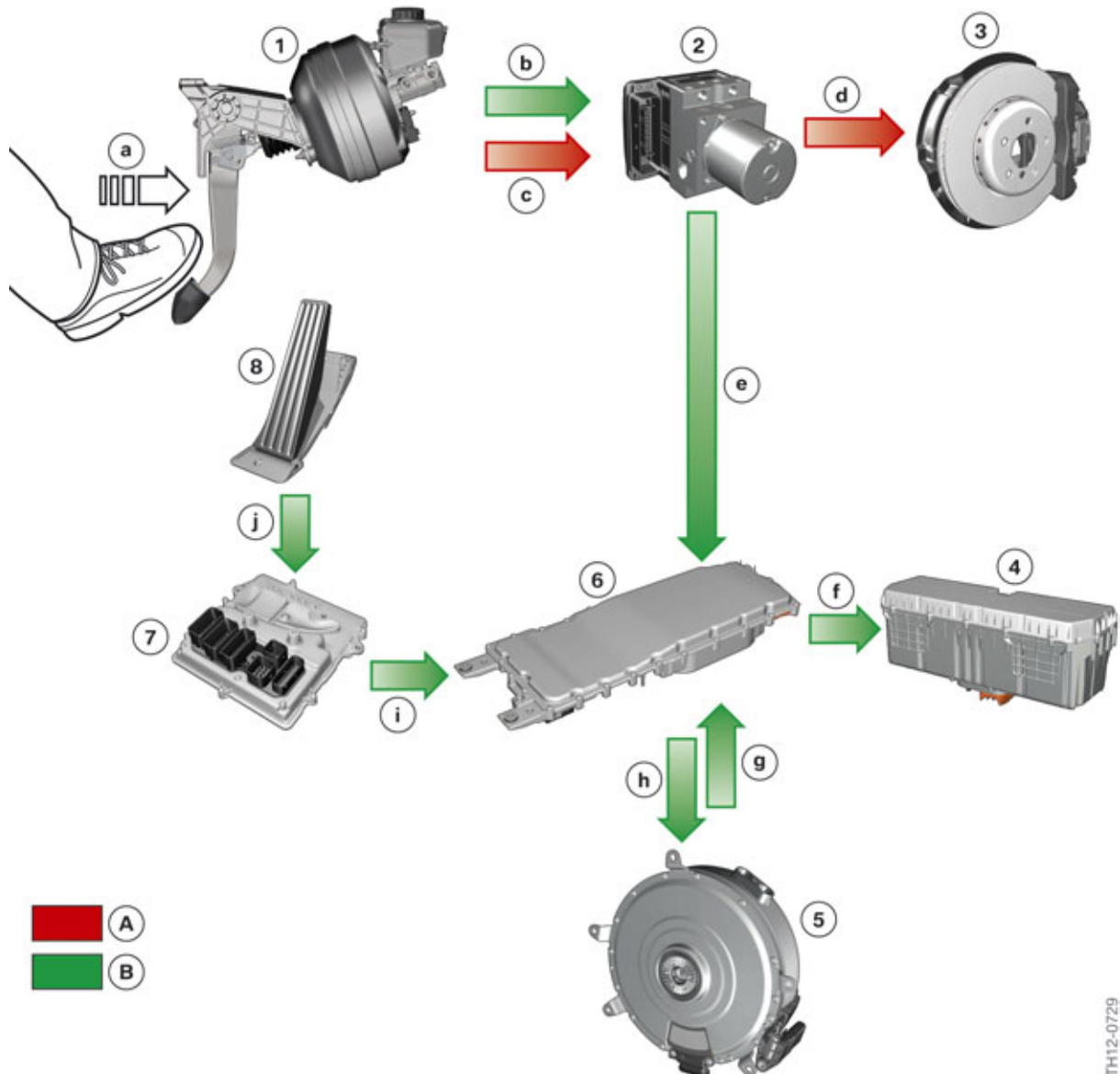
The electric vacuum pump supplies the brake servo with a vacuum in the phases in which the combustion engine is at a standstill, for instance during purely electric driving. To identify when the electric vacuum pump has to be activated, a brake vacuum sensor is required. In addition, a brake pedal travel sensor is required which indicates how far the driver has actuated the brake pedal. With this input variable the level of the brake energy regeneration can be controlled. The DSC software evaluates the signal of the brake pedal travel sensor for this purpose and makes available the result in a bus signal. The DSC software also permanently monitors the steering angle, lateral acceleration and the slip at the rear wheels. As the brake force is not applied to all four wheels but only the rear wheels

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5. Hybrid Brake System

during brake energy regeneration, this software measure is required. It is also called hybrid stability management and normally permits maximum brake energy regeneration – even before the dynamic handling characteristics reach a critical level, the hybrid stability management will reduce the level of the brake energy regeneration and thus guarantee the highest level of driving stability.

5.2. System overview



F30H system overview of hybrid brake system

TH12-0729

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5. Hybrid Brake System

Index	Explanation
A	Hydraulic braking
B	Regenerative braking
1	Brake pedal and brake booster with brake pedal travel sensor
2	Dynamic Stability Control (DSC)
3	Wheel brakes
4	High-voltage battery unit
5	Electrical machine
6	Electrical machine electronics (EME)
7	Digital Engine Electronics (DME)
8	Accelerator pedal module
a	Pressing of the brake pedal
b	Electrical signal "Brake pedal travel" from the brake pedal travel sensor to the DSC
c	Hydraulic pressure from the brake booster to the DSC
d	Hydraulic pressure from the DSC to the four wheel brakes
e	Data bus message "Target braking torque" from the DSC to the EME
f	Electrical energy to be stored in the high-voltage battery (DC voltage)
g	Electrical energy generated by the electric motor (AC voltage)
h	Phase voltages for activating the electric motor
i	Data bus message "Accelerator pedal angle" from the DME to the EME (energy recovery in coasting (overrun) mode)
j	Electrical signal "Accelerator pedal angle" from the accelerator pedal module to the DME (energy recovery in coasting (overrun) mode)

Similar to the F10H, the hybrid brake system of the F30H is a “pedal-coupled brake system” in which there is constantly a mechanical connection between the brake pedal, brake servo and tandem brake master cylinder. Accordingly, the hydraulic braking is done in principle like in a conventional vehicle. The only difference is in an enlarged response range of the brake pedal to extend the pedal travel for regenerative braking.

A deceleration process with increasing strength is as follows in the F30H:

- 1 The driver takes his foot off the accelerator pedal:
The electrical machine is now operated as an alternator. The EME activates the electrical machine in such a way that a brake force is obtained for the complete vehicle which corresponds to a conventional vehicle in overrun mode. The extent of energy regeneration is still at a low level here.
- 2 The driver presses the brake pedal (still within the response range):
The electrical machine is activated as an alternator so that the brake force increases with the brake pedal travel. The deceleration is done via the wheels at the drive axles (rear axle) and is now greater than in the coasting (overrun) mode and the brake energy regeneration is greater. The hydraulic brake system is not yet active here.

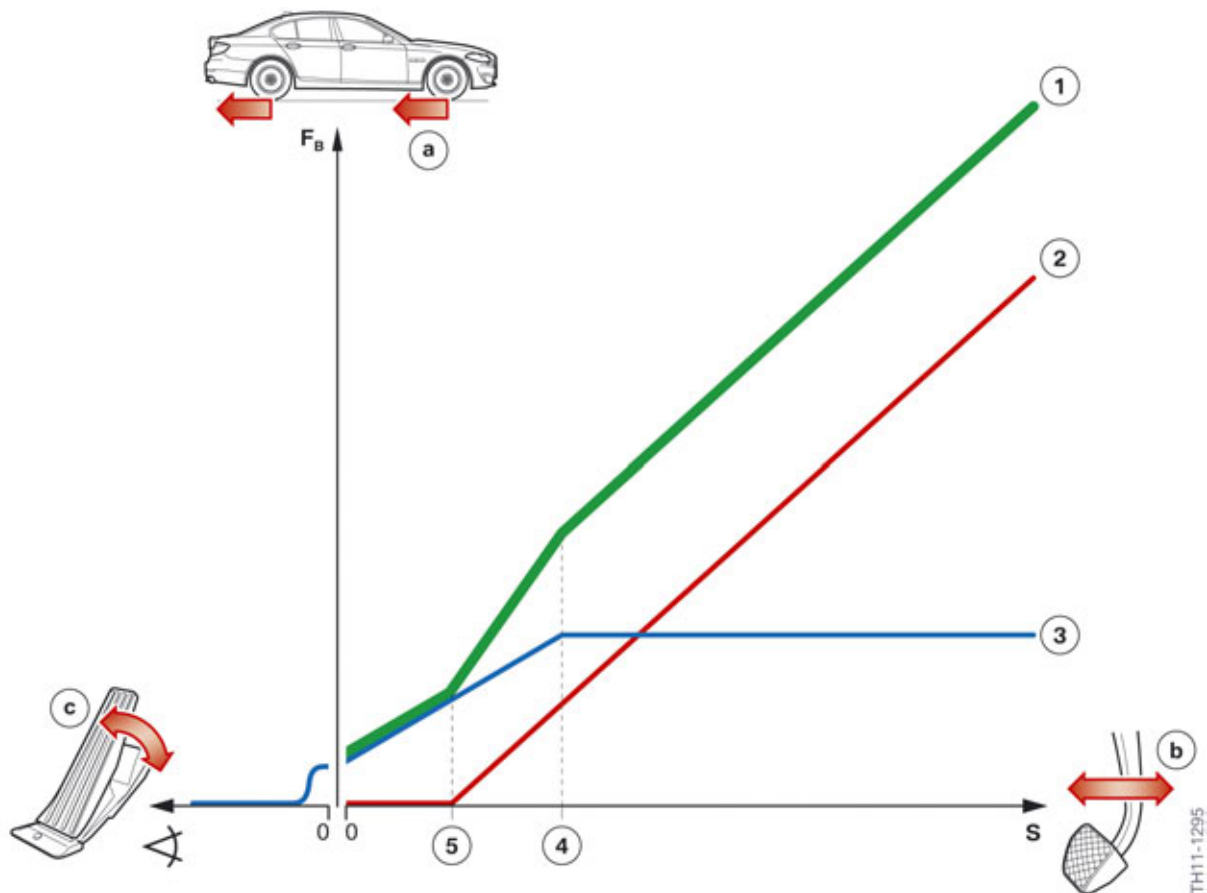
F30H Complete Vehicle

5. Hybrid Brake System

As the brake servo and tandem brake master cylinder in the F30H are identical to those in the conventional F30, the response range is also the same when the brake pedal is pressed. Tests in development have concluded that the level of the response range was sufficient for the metering of the brake energy regeneration and therefore no component change was required.

- 3 The driver presses the brake pedal (beyond the response range):
The electrical machine is activated so that the maximum possible brake energy regeneration is achieved. The brake force generated in this way is added now to the brake force generated also on mechanical/hydraulic travel.

The following graphic summarizes this process again.



F30H Diagram for distributing the brake force

Index	Explanation
a	Brake force on the wheels
b	Brake pedal travel
c	Accelerator pedal angle
1	Total brake force

F30H Complete Vehicle

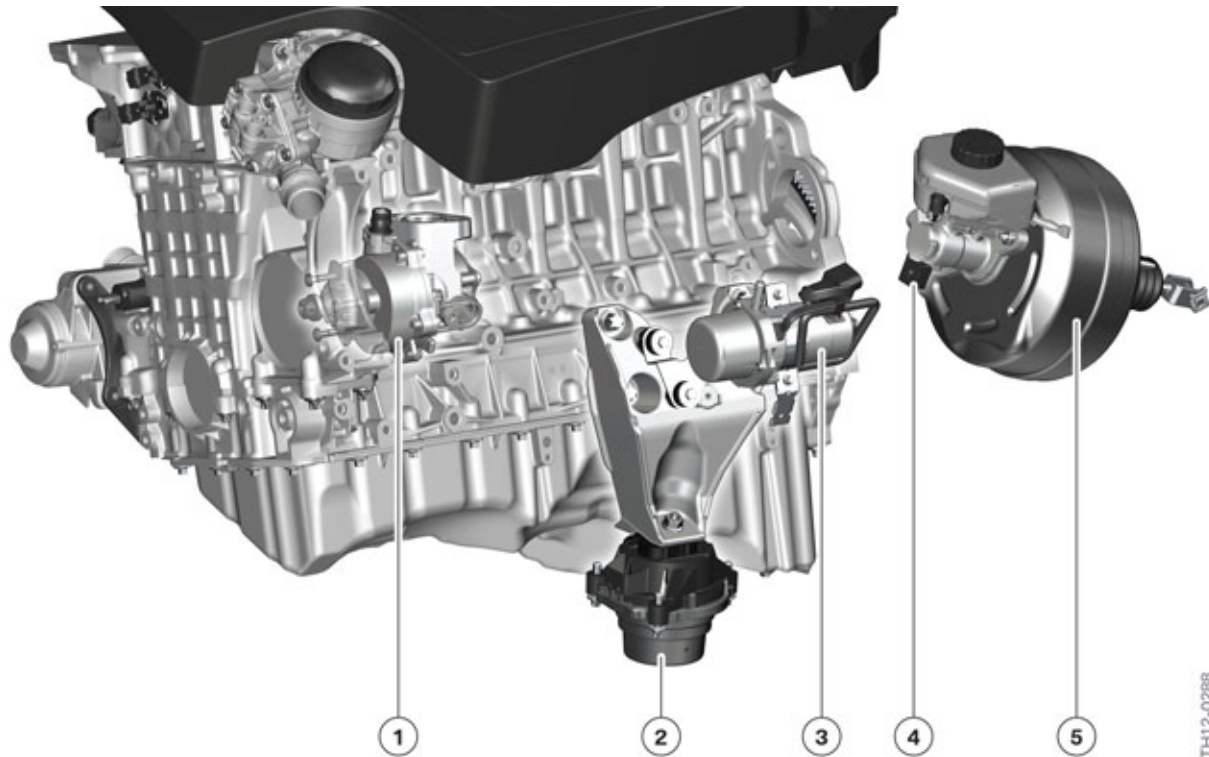
5. Hybrid Brake System

Index	Explanation
2	Hydraulically generated brake force
3	Regeneratively generated brake force
4	Brake pedal travel at which the maximum possible regenerative brake force acts
5	Brake pedal travel at which the hydraulic brake force begins (end of response travel)

5.3. System components

5.3.1. Vacuum supply

The vacuum supply is guaranteed when the combustion engine is running by the mechanically driven vacuum pump. An electric vacuum pump performs this task when the combustion engine is off.



F30H Vacuum system

TH12-0288

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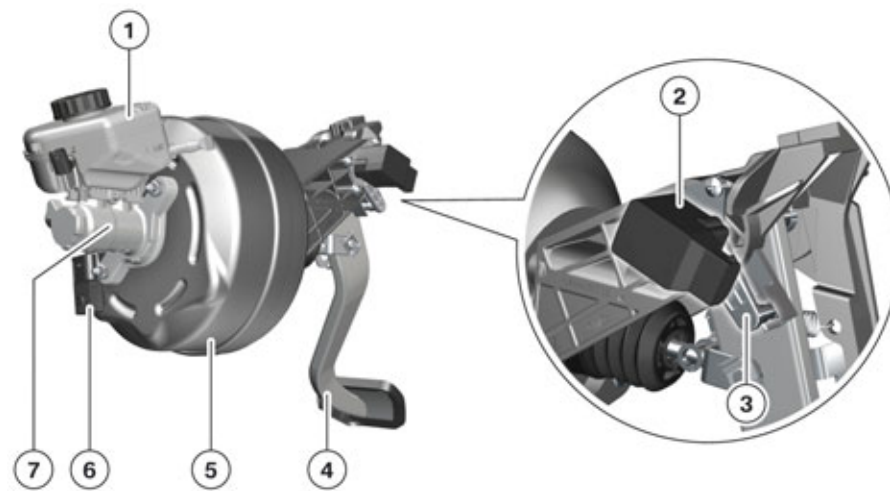
5. Hybrid Brake System

Index	Explanation
1	Mechanical vacuum pump
2	Engine mount (not controllable)
3	Electric vacuum pump
4	Brake vacuum sensor
5	Brake servo

The electric vacuum pump is activated by the electrical machine electronics. The EME contains the electrical output stages required for this and switches the electric vacuum pump on and off as required. To determine this requirements there is a brake vacuum sensor, which is installed in the brake servo. This sensor supplies an analogue output signal to the DME where it is evaluated and then made available to the EME as a bus signal.

5.3.2. Brake pedal travel sensor

In comparison to the F10H, the brake pedal travel sensor in the F30H is **not** integrated in the tandem brake master cylinder. In the F30H the same tandem brake master cylinder as in the F30 can therefore be used. The brake pedal travel sensor is located instead directly at the brake pedal. The sensor principle is similar to that of a ride height sensor. The analogue output signal is read in and evaluated by the DSC control unit. It is transmitted to the EME in the form of a bus signal, which then decides whether and how strong the electrical machine should be operated. The following graphic shows the installation location of the brake pedal travel sensor.



Installation location of the brake pedal travel sensor

F30H Complete Vehicle

5. Hybrid Brake System

Index	Explanation
1	Brake fluid expansion tank
2	Brake pedal travel sensor
3	Connecting rod between brake pedal travel sensor and brake pedal
4	Brake pedal
5	Brake servo
6	Brake vacuum sensor
7	Brake master cylinder

With help of a Service function of the diagnosis system the brake pedal travel sensor can be initialized. The sensor signal is programmed if the brake pedal is not pressed. This initialization is necessary in the following situations:



The brake pedal travel sensor must be initialized after each of the following:

- the DSC control unit was replaced
- the brake pedal travel sensor was replaced
- mechanical work was performed on pedal mechanism.

The BMW diagnosis system must be used for initialization.

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6. Low-voltage Electrical System

The low-voltage vehicle electrical system of the F30H and F10H are distinguished mainly with regard to the 12 V voltage supply. The reason for this is that the F10H is based on the conventional F10 and the F30H on the F30. Accordingly other control units and components are involved in the 12 V voltage supply. These differences are described in the following chapter. The common features with regards to the start-up system and terminal control are only briefly summarized here.

6.1. 12 V voltage supply

The 12 V vehicle electrical system of the F30H is based on the one hand on the electrical system of the F30 and on the other on the hybrid-specific part of the F10H. Also in the F30H the energy supply is no longer realized by a conventional alternator but primarily by the high-voltage electrical system. The high voltage of the high-voltage battery is converted to a lower voltage (about 14 V) using a DC/DC converter. The starter motor generator (SGR) is connected to an auxiliary battery like in the F10H. The auxiliary battery in the F30H is however a conventional lead-acid battery (**not an AGM battery**) with a capacity of 40 Ah. In the following cases the cut-off relay, through which the two 12 V vehicle electrical systems are normally separated from each other, is closed:

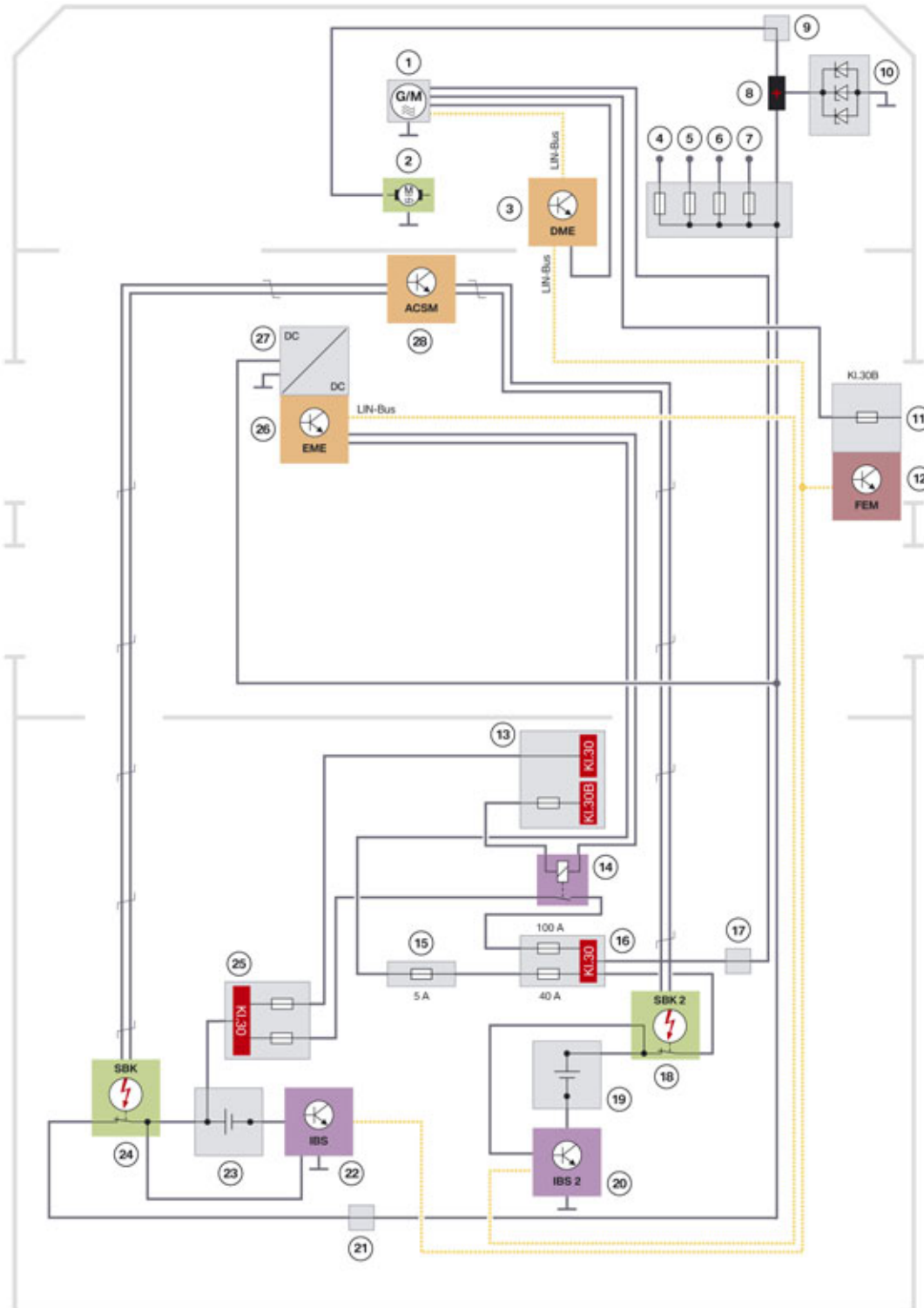
- 1 The auxiliary battery is fully charged and the starter motor generator continues to generate electrical energy or
- 2 The DC/DC converter of the EME cannot assume the energy supply due to a fault.

A second energy supply for the 12 V vehicle electrical system is available, which increases the reliability.

F30H Complete Vehicle

6. Low-voltage Electrical System

6.1.1. System overview



TH12-0310

System wiring diagram, 12 V voltage supply F30H

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6. Low-voltage Electrical System

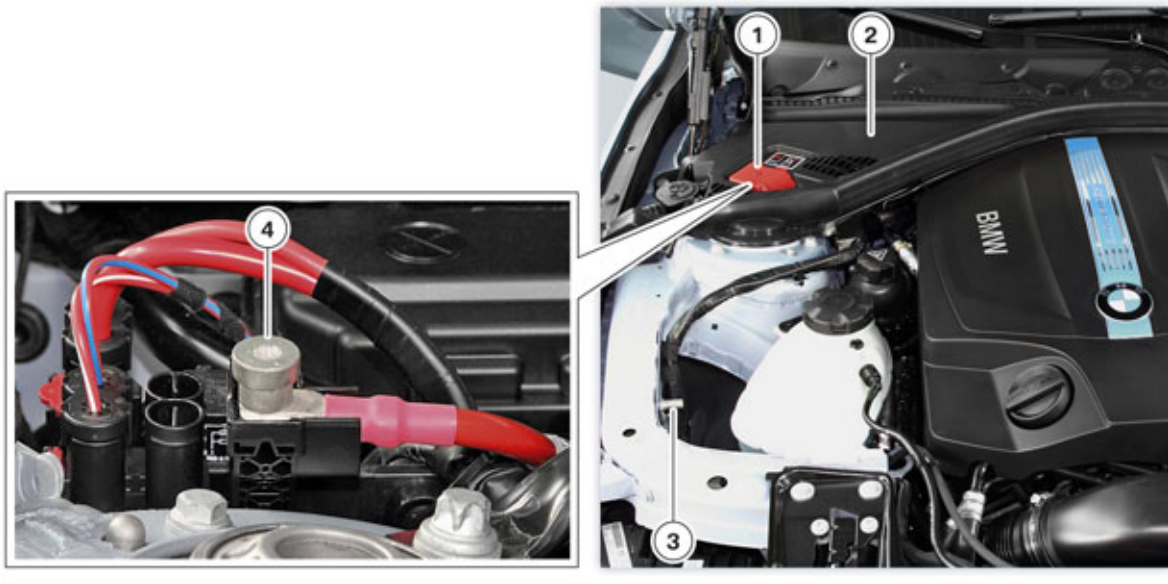
Index	Explanation
1	Starter motor generator (SGR)
2	Starter motor
3	Digital Engine Electronics (DME)
4	Supply lead (protected) from the front power distribution box to the EPS
5	Supply lead (protected) from the front power distribution box to the electric fan relay
6	Supply lead (protected) from the front power distribution box to the blower motor
7	Supply lead (protected) from the front power distribution box to the DSC hydraulic control unit
8	Jump start terminal point
9	Transfer support point (at the bulkhead)
10	Reverse polarity protection module
11	Fuse, terminal 30B in the FEM for control electronics of the starter motor generator
12	Front Electronic Module (FEM)
13	Fuses in the power distribution box of the luggage compartment
14	Cut-off relay
15	Fuse 5 A in the line from the power distribution box of the auxiliary battery to the EME
16	Power distribution box of auxiliary battery
17	Transfer support point (change of lines in the vehicle to lines at the vehicle underbody)
18	Safety battery terminal 2
19	Auxiliary battery
20	Intelligent battery sensor 2
21	Transfer support point (change of lines in the vehicle to lines at the vehicle underbody)
22	Intelligent battery sensor
23	12 V battery
24	Safety battery terminal
25	Power distribution box at the 12 V battery
26	Electrical machine electronics (EME)
27	DC/DC converter
28	Crash Safety Module

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6. Low-voltage Electrical System

6.1.2. Reverse polarity protection module, power distribution box

In this engine compartment there are some common features with regard the low-voltage system and some hybrid-specific special features. In this chapter details are provided mainly on the consumers connected at the power distribution box and the reverse polarity protection module. The following chapter deals with the start-up system. In the previous chapter all components of the low-voltage vehicle electrical system are also shown in the system wiring diagram.

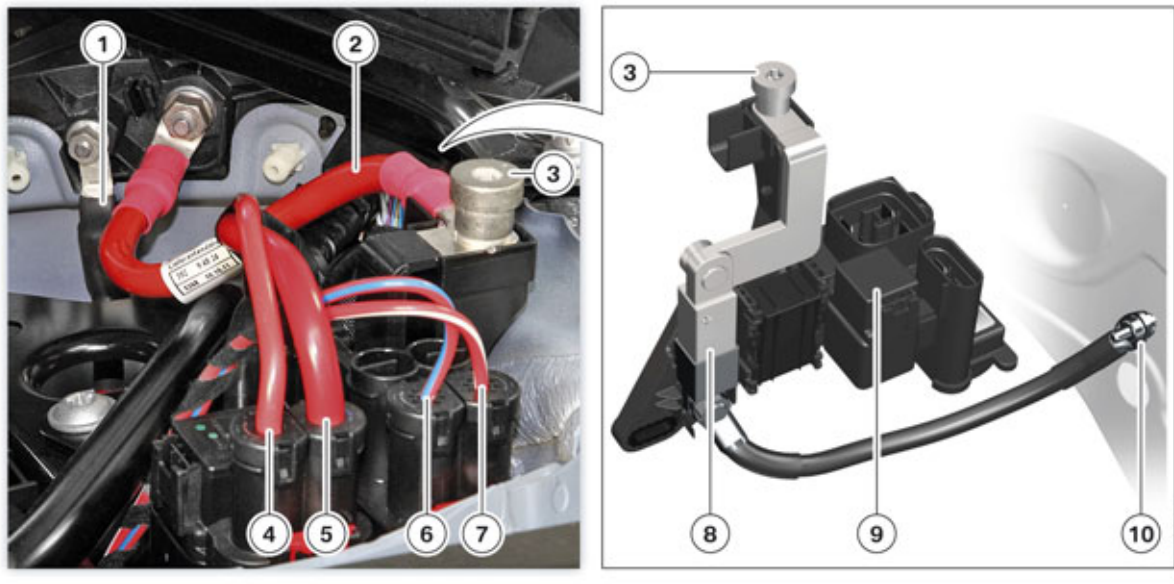


F30H Jump start terminal point

Index	Explanation
1	Cover for jump start terminal point
2	Cover for power distribution box, front
3	Ground connection for charger/jump start cable
4	Jump start terminal point

F30H Complete Vehicle

6. Low-voltage Electrical System



F30H Power distribution box, front

Index	Explanation
1	Positive battery cable/transfer support point for the power distribution module (PDM)
2	Positive battery cable from the jump start terminal point to the starter motor
3	Jump start terminal point
4	Supply lead (protected) for the electric fan relay
5	Supply lead (protected) for the electromechanical power steering (EPS)
6	Supply lead (protected) for the DSC hydraulic control unit
7	Supply lead (protected) for the blower motor, passenger compartment
8	Reverse polarity protection module
9	Electric fan relay
10	Ground connection

Like in the F10H a separate reverse polarity protection module is also used in the F30H in order to avoid subsequent damage to the vehicle electrical system and the connected electronic components during polarity reversal. For this purpose the reverse polarity protection module contains Zener diodes which are switched between the jump start terminal point and ground (see system wiring diagram 12 V voltage supply).

Reminder: Zener diodes are silicon diodes with a special property which restricts the voltage during operation in reverse direction. This property is used if an external voltage source such as a battery charger is connected to the correct (positive) polarity. In this case the Zener diodes protect the electronic components of the vehicle electrical system against overvoltage by restricting the voltage. In the opposite scenario, if an external voltage source is connected to incorrect (negative) polarity, the Zener diodes work like other diodes, i.e. a very low resistance resistor. The external voltage source is practically short-circuited and its voltage drops to almost zero volt. The electronic components in the vehicle are thus not exposed to incorrect (negative) voltage and are thus protected. If the external

F30H Complete Vehicle

6. Low-voltage Electrical System

voltage source remains connected to incorrect (negative) polarity for an extended period and a very high current is permanently flowing, then the Zener diodes are destroyed and can no longer exercise their protective effect. In this rare case the vehicle control units are also damaged.



The reverse polarity protection module can only prevent subsequent damage of a polarity reversal for a limited time. The reverse polarity protection module does not withstand a permanent polarity reversal but is damaged as a result. Subsequent damage to the electronic components of the vehicle electrical system is also possible.

6.2. Start-up system

The start-up system comprises several components:

- Starter motor generator (SGR)
- Auxiliary battery
- Power distribution box and cut-off relay in the luggage compartment.



Starter motor generator

The starter motor generator of the F30H has the same design as that of the F10H. The electrical machine is a salient-pole machine, whose rotor is equipped with permanent magnets and an excitation coil for separate excitation. The voltage at the stator can also be controlled via the excitation coil. The actual electrical machine forms one unit together with the rotor position sensor, the control module and the seven power electronics modules, and can only be replaced as a unit in the event of a defect. The control module and the DME communicate via a local interconnect network bus. The starter motor generator receives the commands for the operating mode (engine/alternator) as well as for the level of voltage generated in alternator mode.

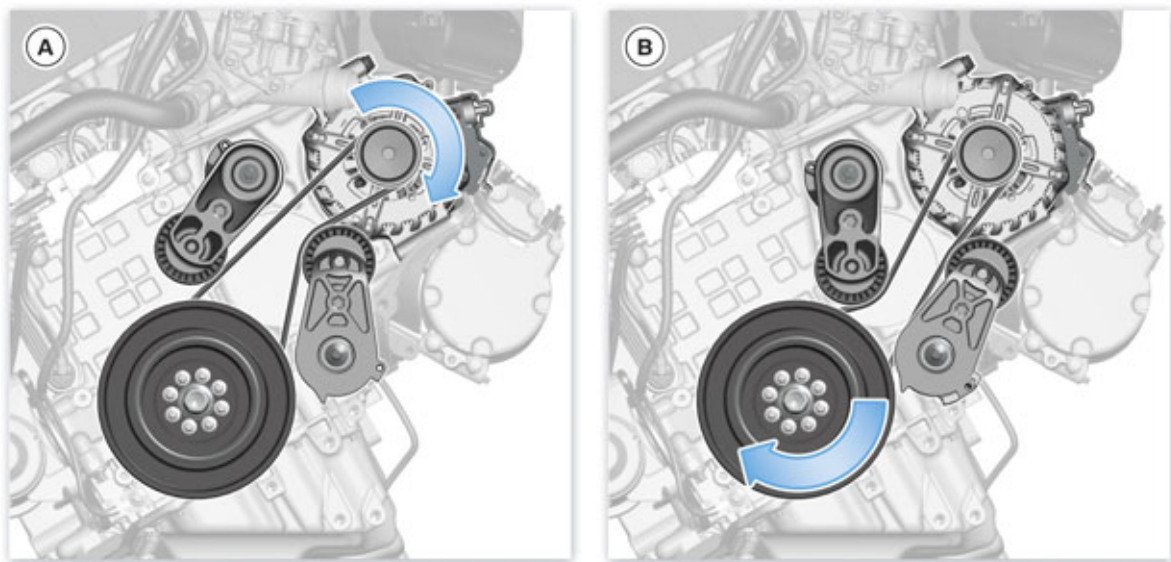
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6. Low-voltage Electrical System



To identify a fault with the starter motor generator, the auxiliary battery (state of charge or state of health), lines (resistance), screw connections of lines (contact resistance), fit and tension of drive belt, grounding cable and the local interconnect network bus connection must first be checked. The starter motor generator should only be replaced if all these checks have failed.

The starter motor generator is driven by the combustion engine via a drive belt in alternator mode or can start the combustion engine via the drive belt in starter mode. The principle of the belt drive with the two belt tensioners is also identical to that of the F10H.



Alternator mode and starter mode operation of the starter motor generator

Index	Explanation
A	Starter mode operation (starting the combustion engine)
B	Alternator mode operation (combustion engine is running and auxiliary battery is charging)

The starter motor generator with the auxiliary battery forms a separate 12 V vehicle electrical system. When the combustion engine is started (starter mode) energy is taken from the auxiliary battery; in alternator mode the auxiliary battery is charged. Due to the permanent magnets in the rotor the starter motor generator always generates electrical energy when the combustion engine is running. So as not to waste this energy when the auxiliary battery is fully charged, a large amount of the energy generated can be used to supply the conventional 12 V vehicle electrical system by closing the cut-off relay.

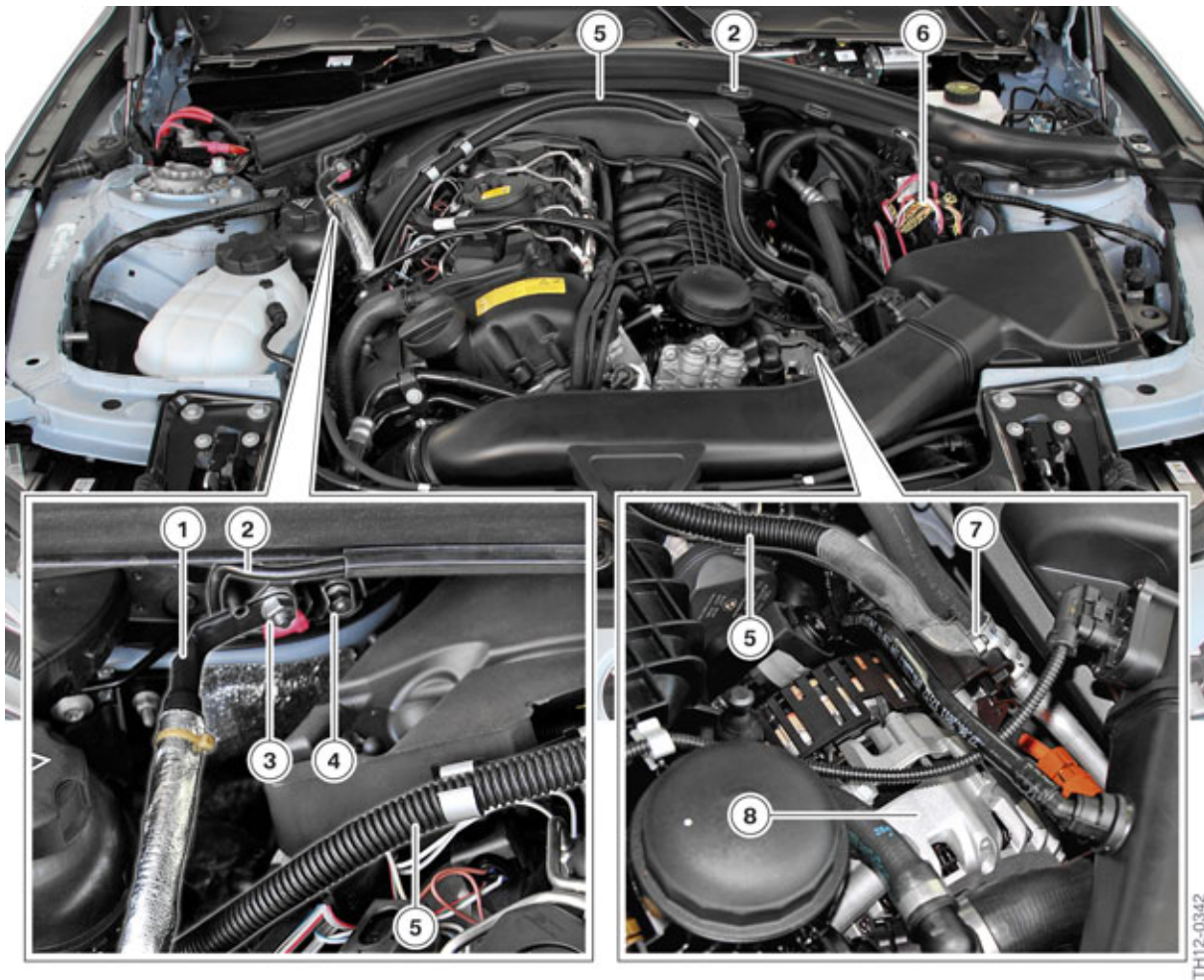
In the following two cases the combustion engine is started with help of the starter motor generator:

- The combustion engine is at operating temperature and was shut down by the automatic engine start-stop function (at a standstill or during the journey) and should be restarted now.
- When the first start (after terminal 15 is switched on) with the conventional starter motor is not possible due to a defect for example.

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6. Low-voltage Electrical System

As the installation locations in the engine compartment differ slightly between F10H and F30H, the following graphic shows the electrical wiring in conjunction with the starter motor generator. The photos were taken with the covers of the front power distribution box and acoustic cover removed.



Installation situation and electrical wiring in the engine compartment F30H

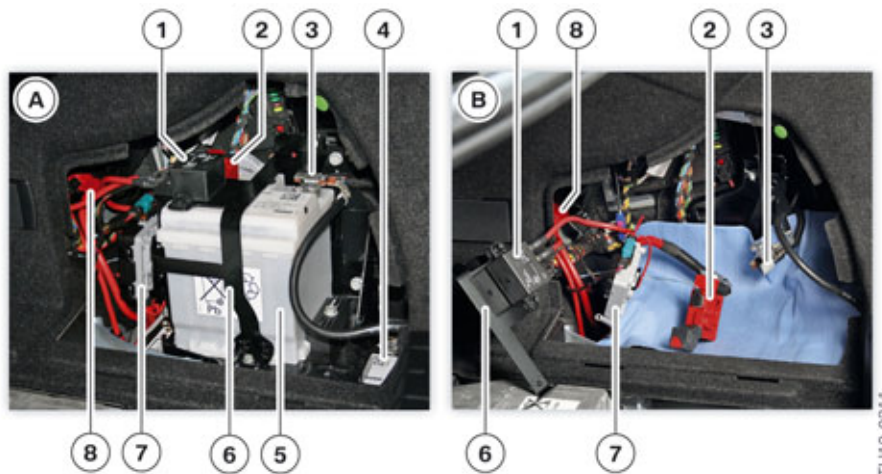
Index	Explanation
1	Line to the starter motor
2	Supply lead for the Power Distribution Module
3	Transfer support point for line to the starter motor
4	Transfer support point for line to the Power Distribution Module
5	Line to the starter motor generator
6	Power Distribution Module (PDM)
7	Connection of line at starter motor generator
8	Starter motor generator

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6. Low-voltage Electrical System

6.2.1. Auxiliary battery

The auxiliary battery of the F30H is a **conventional lead-acid battery** with a capacity of 40 Ah. It is installed together with the hybrid-specific power distribution box in the luggage compartment on the right, above the 12 V battery.



F30H auxiliary battery in the luggage compartment on the right

Index	Explanation
A	Auxiliary battery installed
B	Auxiliary battery removed
1	Cut-off relay
2	Safety battery terminal of the auxiliary battery SBK2
3	Intelligent battery sensor (auxiliary battery) IBS2
4	12 V battery
5	Auxiliary battery
6	Clamp for mechanical fixing of the auxiliary battery
7	Power distribution box of auxiliary battery
8	Battery power distribution box



A change of the auxiliary battery must always be registered.

As is already known from the F10H, the current, the voltage and the terminal temperature of the auxiliary battery are recorded by an intelligent battery sensor IBS2. The results are forwarded via local interconnect network bus to the electrical machine electronics (EME).

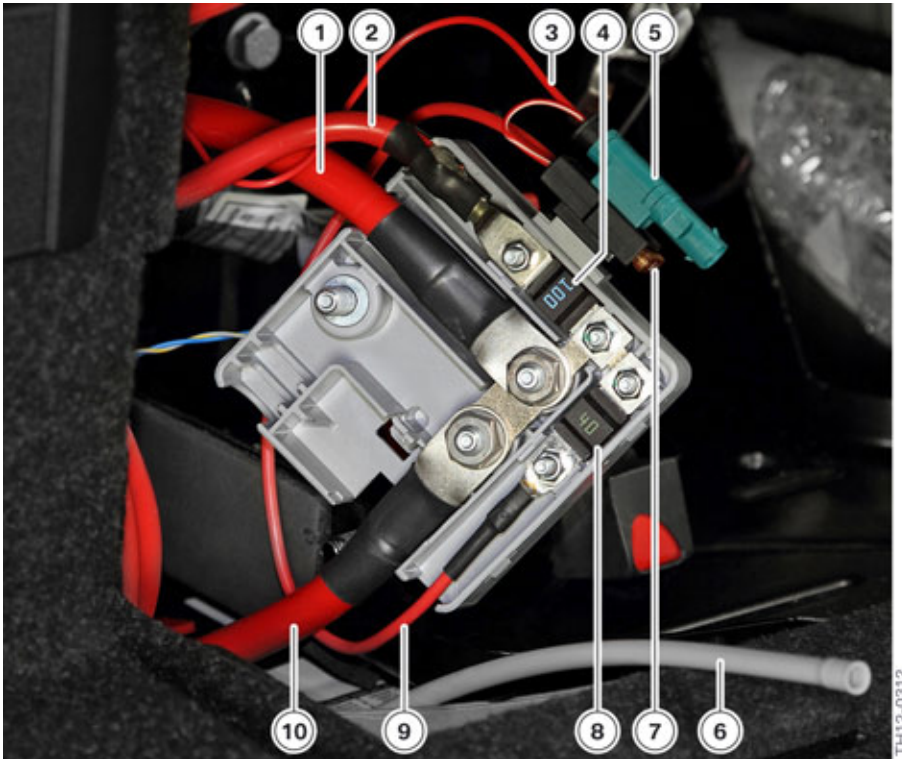
In the event of an accident of sufficient severity, a battery safety terminal (SBK2) ensures the disconnection of the positive battery cable between the auxiliary battery and starter motor generator. This battery safety terminal (SBK2) is located directly at the positive terminal of the auxiliary battery. The pyrotechnic activation of the battery safety terminal for the auxiliary battery is effected by the Crash Safety Module (ACSM).

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6. Low-voltage Electrical System

6.2.2. Power distribution box and cut-off relay

The power distribution box of the auxiliary battery and the cut-off relay are familiar components from the F10H. They only differ in the F30H by their installation location. This is located at the clamp which encloses the auxiliary battery and is shown in the graphic in the previous chapter.



Power distribution box in the luggage compartment

Index	Explanation
1	Positive battery cable between auxiliary battery and power distribution box
2	Positive battery cable between power distribution box of the auxiliary battery and cut-off relay
3	Supply line IBS2
4	100 A fuse
5	Plug connector for line to the electrical machine electronics and supply line of the IBS2
6	Ventilation line of the auxiliary battery
7	5 A fuse
8	40 A fuse
9	Line from the 40 A fuse to the 5 A fuse
10	Positive battery cable from the power distribution box to the starter motor generator

F30H Complete Vehicle

6. Low-voltage Electrical System

The intelligent battery sensor (IBS2) receives the voltage supply via a small gauge line from the battery safety terminal (SBK2) of the auxiliary battery. The positive battery cable of the auxiliary battery is also fed from the battery safety terminal (SBK2) of the auxiliary battery. Other lines also lead from this power distribution box to the starter motor generator, cut-off relay and EME. The power distribution box is the same as in the F10H and is only located at the other installation location shown.

There are two fuses in the power distribution box of the auxiliary battery. The 100 A fuse protects the positive battery cable from the power distribution box to the cut-off relay. The power distribution box is a component from the "module" which is why the 40 A fuse is always installed. The line to the EME has a smaller gauge however, and this is why a 5 A fuse is installed in series. The positive battery cable leads from the power distribution box of the auxiliary battery to the starter motor generator. This positive battery cable is made from copper and is not protected against overload current – this is one of the key reasons why a safety battery terminal is also attached at the auxiliary battery.

The cut-off relay is also a known component from the F10H and is installed (in the F30H) at the top at the clamp over the auxiliary battery. The two 12 V vehicle electrical systems are connected via this cut-off relay. The two 12 V vehicle electrical systems are connected if for example, the auxiliary battery is fully charged and the engine speed of the combustion engine increases to over 2500 rpm. The cut-off relay is opened, however, by a start-up process in order to avoid a voltage dip in the conventional 12 V vehicle electrical system. The activation of the cut-off relay is carried out by the electrical machine electronics (EME).

In the event of a malfunction with the hybrid system the starter motor generator (SGR) can assume the supply of the network consumers which is normally done by the DC/DC converter in the EME. In this case the cut-off relay is closed. Depending on the overall energy requirements in the vehicle electrical system convenience functions such as the seat heating can be deactivated.

6.3. Terminal control

The terminal control of the F30H is identical to that in the F10H. Only the display of the driving readiness in the instrument cluster is different (see section "Displays and controls").

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7. Bus Systems

The bus systems of the F30H are based on the bus systems of the F30. All the main and sub-bus systems of the F30 are also used in the F30H.

The hybrid-specific control units known from the F10H are also included.

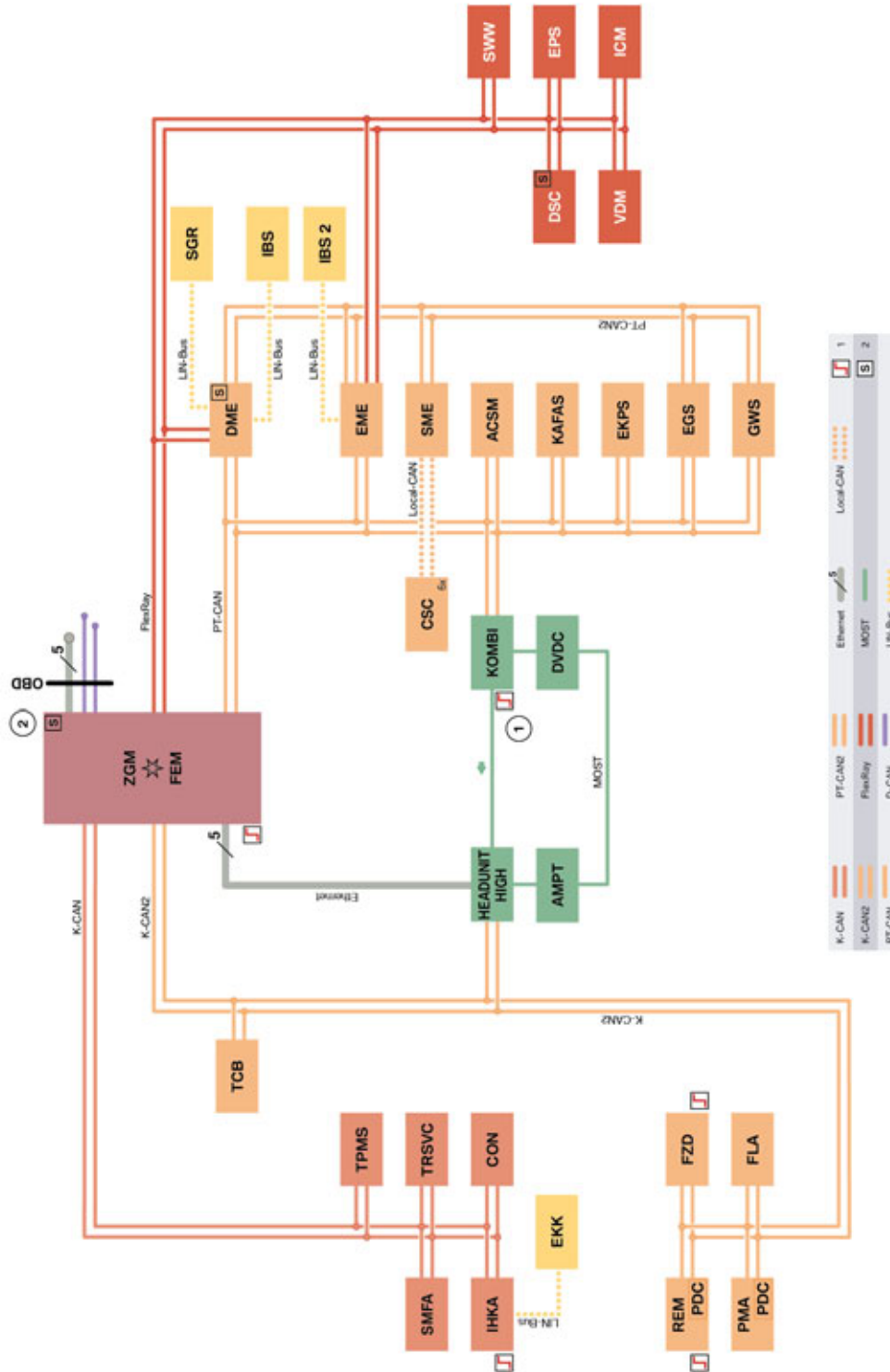
- Electric motor electronics (EME)
- Battery management electronics (SME)
- Cell supervision circuit (CSC)
- Electric A/C compressor
- Starter motor generator (SGR)
- Integrated battery sensor 2 (IBS2) for the auxiliary battery.

Due to some exceptions for optional equipment (see section "Introduction") some control units are not installed in the F30H. The resulting bus structure of the F30H is as follows.

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7. Bus Systems

7.1. Bus overview



F30H bus overview

TH12-0146_3

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7. Bus Systems

Index	Explanation
1	Control units authorized to perform wake-up function
2	Start-up nodes: Control units for start-up and synchronization of the FlexRay bus system
ACSM	Advanced Crash Safety Module
AMPT	Amplifier Top (top high fidelity amplifier)
CON	Controller
CSC	Cell supervision circuit
D-CAN	Diagnosis-on-Controller Area Network
DME	Digital Engine Electronics
DSC	Dynamic Stability Control
DVDC	DVD changer
EGS	Electronic transmission control
EKK	Electric A/C compressor
EKPS	Electronic fuel pump control
EME	Electrical machine electronics
EPS	Electric Power Steering
Ethernet	Cable-based data network technology for local data networks
FEM	Front Electronic Module
FLA	High-beam assistant
FlexRay	Fast, real-time and fault-tolerant bus system for use in automotive sector
FZD	Roof function center
GWS	Gear selector switch
HU-H	Headunit High (3rd Generation)
ICM	Integrated Chassis Management
IBS	Intelligent battery sensor
IBS2	Intelligent battery sensor 2
IHKA	Integrated automatic heating / air-conditioning system
K-CAN	Body controller area network
K-CAN2	Body controller area network 2
KAFAS	Camera-based driver assistance systems
KOMBI	Instrument panel
LIN-Bus	Local interconnect network bus
Local CAN	Local Controller Area Network
MOST	Media Oriented System Transport
OBD	Diagnostic socket

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7. Bus Systems

Index	Explanation
PDC	Park Distance Control
PMA	Parking Manoeuvring Assistant
PT-CAN	Powertrain controller area network
PT-CAN2	Powertrain controller area network 2
REM	Rear Electronic Module
SGR	Starter motor generator
SME	Battery management electronics
SMFA	Driver's seat module
SWW	Lane change warning
TCB	Telematics Communication Box
TPMS	Tire pressure monitoring system
TR SVC	Control unit for the Top, Rear and Side-view cameras
VDM	Vertical Dynamics Management
ZGM	Central gateway module

7.2. Adapted control units

The adapted control units in comparison to the basic vehicle F30 are listed in the following table together with the reasons for the adaptation.

Control unit	Reason for adaptation
ACSM	Evaluation of sensor signals from ICM (rollover), activation of second safety battery terminal at the auxiliary battery
DME	Torque coordination of electrical machine/combustion engine, communication with starter motor generator (via local interconnect network bus)
DSC	Evaluation of the brake pedal travel sensor, brake energy regeneration
EGS	Hybrid-specific shift program, activation of the additional electrical transmission oil pump
FEM	Hybrid-specific terminal control (driving readiness without running combustion engine)
ICM	For F30H always in the version with roll rate sensor and vertical acceleration sensor (identification of rollover legally required)
IHKA	Activation of the electric A/C compressor (EKK) (via local interconnect network bus)
Headunit	Hybrid-specific displays in central information display
KOMBI	Additional relevant hybrid-specific displays, as well as hybrid-specific Check Control messages

F30H Complete Vehicle

8. Displays and Controls

The hybrid-specific operating conditions, as well as other information, are displayed in the instrument cluster and if desired in the Central Information Display. The displays in the instrument cluster of the F30H differ significantly to those in the F10H. In the Central Information Display the changes are less substantial. Instead they are marked by a new visible appearance, which can be traced back to the introduction of the third generation of the HU-H.

The following hybrid-specific operating conditions and information are shown:

- "Ready to drive" mode
- Electric driving
- Boost function
- Brake energy regeneration
- State of charge of the high-voltage battery
- Information on the use of the hybrid drive and ECO PRO mode.

Some of them are shown permanently in the lower part of the rev counter. Others the driver has to call up specifically. Both the displays in the CID and in the instrument cluster are activated when terminal 15 is turned on. Their content and appearance are described in the following chapters

The hybrid-specific Check Control messages of the F30H are identical to those in the F10H. An overview can be seen in the product information bulletin "ST1203 F10H Complete Vehicle". Similar to the F10H, in the F30H no additional action is required by the driver despite the two different drive systems. The operating strategy reacts immediately to the current driver's choice and many other input variables. Based on this it uses the two drive systems particularly efficiently or dynamically to correspond to the driver's choice.

In conventional vehicles the driver can deactivate the automatic engine start-stop function with help of a separate button. In the F30H there is no such button. However engaging of the selector lever in position "DS" or "M/S" can still eliminate the stopping of the combustion engine.

8.1. Displays in the instrument cluster

8.1.1. Introduction

The display concept for the hybrid-specific operating conditions and information differ from the F10H.

The specific display concept of The F30H is described in the following points:

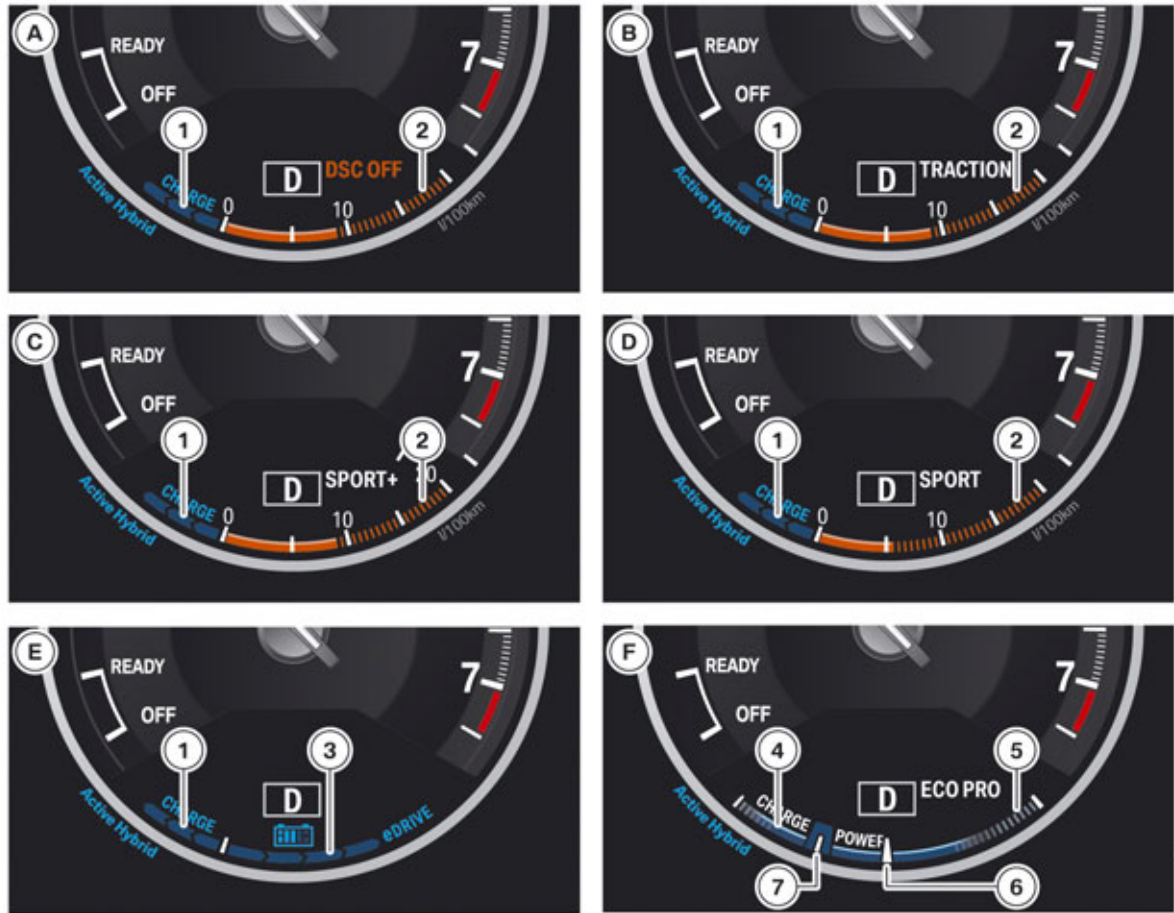
- Driving readiness: Explicit needle position of rev counter "READY"
- Electric driving and brake energy regeneration: Change of the display direction
- State of charge of the high-voltage battery: graphic/numeric
- Driving experience switch: Large dependency on selected mode.

In the following graphic it can be seen how the selected mode on the driving experience switch influences the displays in the instrument cluster. In the DSC OFF, TRACTION, SPORT+ and SPORT mode there is the familiar display of the current fuel consumption. In COMFORT mode the state of

F30H Complete Vehicle

8. Displays and Controls

charge of the high-voltage battery, as well as the electric driving power used, are shown. Finally in ECO PRO mode the driving power used or the power during brake energy regeneration is constantly shown. A white mark moves around the curved shaped scale depending on the power.



Displays in the instrument cluster of the F30H dependent on the selected mode

Index	Explanation
A	DSC OFF mode
B	TRACTION mode
C	SPORT+ mode
D	SPORT mode
E	COMFORT mode
F	ECO PRO mode
1	Three segments for display of the level of brake energy regeneration
2	Continuous display of the current fuel consumption
3	Four segments for display of the power used by the electric motor

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8. Displays and Controls

Index	Explanation
4	Continuous scale for the level of brake energy regeneration
5	Continuous scale for the driving power used
6	White mark for continuous display of the driving power or the level of brake energy regeneration
7	Neutral position of the white mark: no driving power is used or brake energy recovered

To highlight the other differences the displays of the individual operating conditions and information are shown in individual graphics in the following sub-chapters and compared with those in the F10H. The graphics shown for the F10H apply at the time of the market introduction of the F10H. They may change during the course of production as a result of technical further developments or model revisions.

8.1.2. State of charge of the high-voltage battery

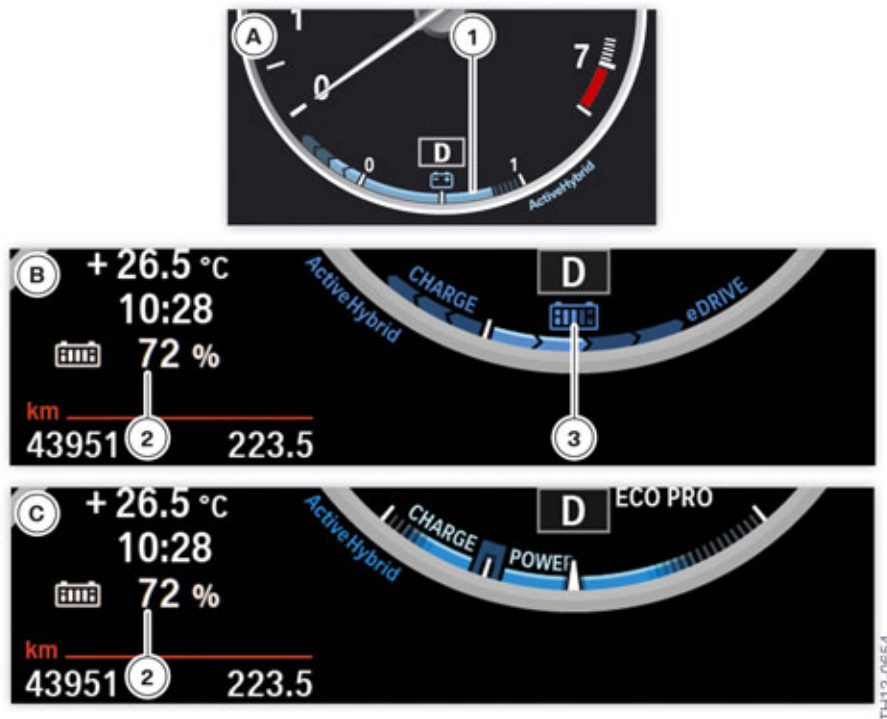
In the F10H the state of charge of the high-voltage battery is displayed on a scale from 0 to 1 when terminal 15 is switched on. The filling degree of the arc provides information on the customer-relevant state of charge of the high-voltage battery. The customer-relevant state of charge between 0 and 1 corresponds to a real state of charge between approximately 25% and 70%.

In the F30H the customer-relevant state of charge of the high-voltage battery can also be shown in the instrument cluster. If the customer selects COMFORT mode with the driving experience switch, then a small battery symbol is permanently shown under the display of the drive position. The number of blue bars in this battery symbol shows the approximate state of charge. This display is **not** available in the other modes of the driving experience switch.

Beside the graphic display there is a numerical display of the state of charge: The customer can call up the state of charge as a function of the on-board computer and have it displayed permanently if desired. This display once again shows the customer-relevant state of charge of the high-voltage battery, which is from 0% to 100%.

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8. Displays and Controls



Display of the state of charge in the instrument cluster

Index	Explanation
A	F10H
B	F30H, COMFORT mode
C	F30H, ECO PRO mode
1	Continuous display in the form of an arc
2	Numerical display as an on-board computer function
3	Display in the form of bars

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8. Displays and Controls

8.1.3. "Ready to drive" mode

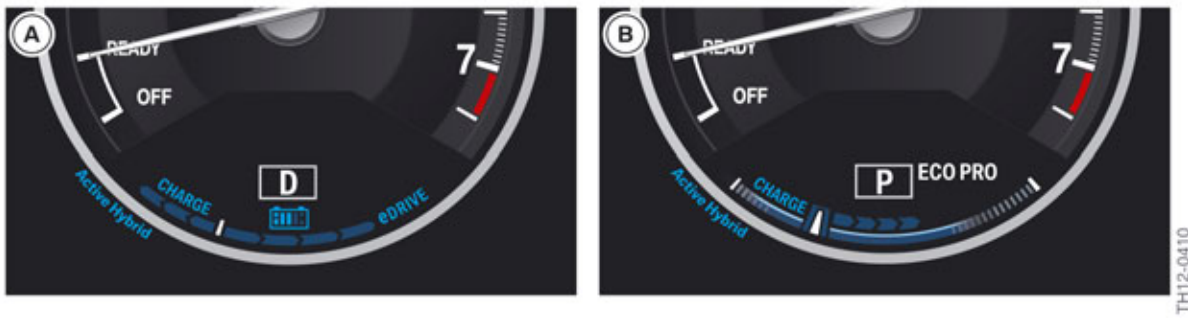
The "Ready to drive" state is captured when the drivetrain is at operating temperature (engine and transmission) and at a standstill. The combustion engine is thus switched off. When drive position "D" or "R" is engaged the vehicle can be set in motion by releasing the brake pedal and pressing the accelerator. This is realized depending on the state of charge and temperature of the high-voltage battery, as well as the accelerator pedal angle with help of the electric motor or with help of the combustion engine.

In the F10H the "Ready to drive" state is recognizable in the instrument cluster by the following: the rev counter is at "zero" and under the rev counter the blue writing "READY" is shown.



F10H "Ready to drive" display

In the F30H the rev counter includes the intended position "READY" for the "Ready to drive" state. No additional writing "READY" must therefore be shown.



F30H "Ready to drive" display

Index	Explanation
A	COMFORT mode
B	ECO PRO mode

In COMFORT mode both the segments for the driving power used and the level of the brake energy regeneration are not highlighted. The white mark on the scale in ECO PRO mode is in the neutral position. In both cases it signals that no driving power is used and no brake energy is recovered.

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8. Displays and Controls

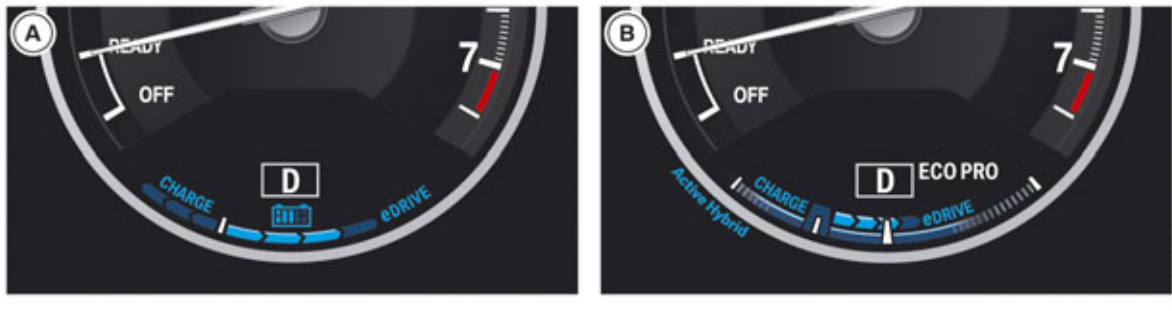
8.1.4. Electric driving

The driving power used during the purely electric driving is displayed in the form of four segments both in the F10H and in the F30H. The more segments that are highlighted the more driving power that is used. In the F10H these segments point in the direction of the rev counter, i.e. clockwise.



F10H Electric driving display

In contrast, in the F30H the segments are in the reverse direction like in the F10H, i.e. counter-clockwise. The rev counter remains in the “READY” position.



F30H Electric driving display

Index	Explanation
A	COMFORT mode
B	ECO PRO mode

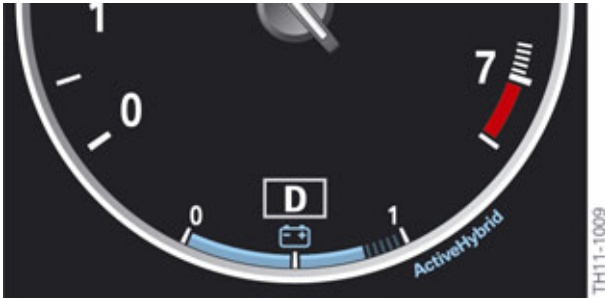
Both in COMFORT mode and in ECO PRO mode the customer can read off the driving power used using the number of highlighted segments. In ECO PRO mode the white mark also moves over the curved shaped power scale and permanently shows the driving power used. Like in the F10H the maximum usable power is reached when four segments are highlighted. If the driver further increases the accelerator pedal angle, the combustion engine is started.

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8. Displays and Controls

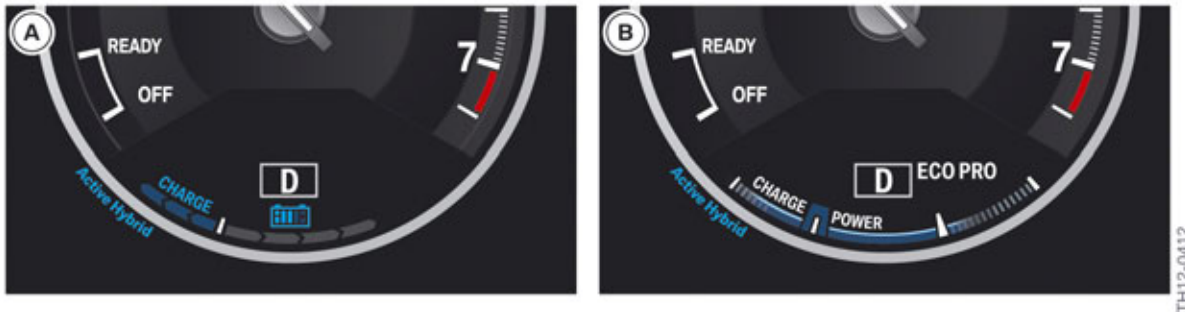
8.1.5. Operation using combustion engine

The rev counter in both vehicles typically displays the current engine speed. In the F10H the segments for the electric motor are not even visible; only the state of charge of the high-voltage battery is shown permanently.



F10H Operation using combustion engine display

In COMFORT mode of the F30H the segments for the electric driving power used and the level of the brake energy regeneration are permanently visible, but are not highlighted in combustion engine operation. In ECO PRO mode of the F30H the segments for the electric motor are no longer shown. Instead the curved shaped power scale is shown, which now has the writing "POWER". The position of the white mark now no longer shows how much electric driving power is used, but how much driving power is requested by the combustion engine.



F30H Operation using combustion engine display

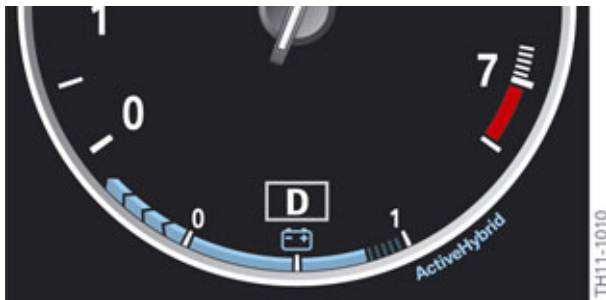
Index	Explanation
A	COMFORT mode
B	ECO PRO mode

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8. Displays and Controls

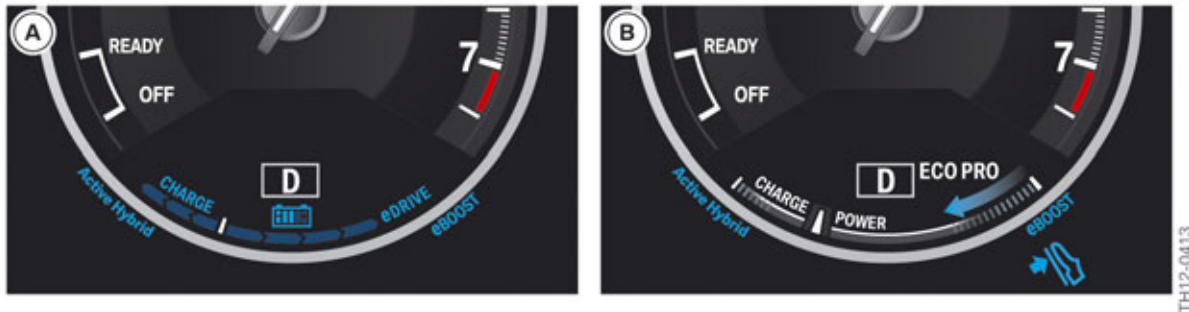
8.1.6. Boost function

With active Boost function both drive types are active at the same time and ensure maximum acceleration. In the F10H this is shown by the position of the rev counter at a high engine speed and a simultaneous display of all four segments for the electric motor.



F10H Boost function

If the driver in the F30H uses the Boost function when COMFORT mode is activated then the segments for the electric motor are **not** highlighted. Instead the writing "eBOOST" appears on the right end of the scale. In ECO PRO mode this writing also appears and the maximum driving power is also implemented. The driver no longer uses the vehicle efficiently. In ECO PRO mode the driver therefore receives the message to reduce the accelerator pedal angle. At the same time the power scale changes from blue (efficient) to grey (inefficient).



F30H Boost function display

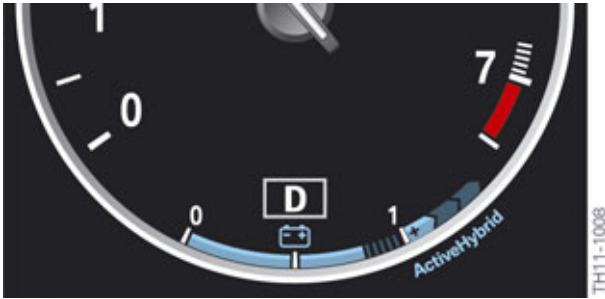
Index	Explanation
A	COMFORT mode
B	ECO PRO mode

F30H Complete Vehicle

8. Displays and Controls

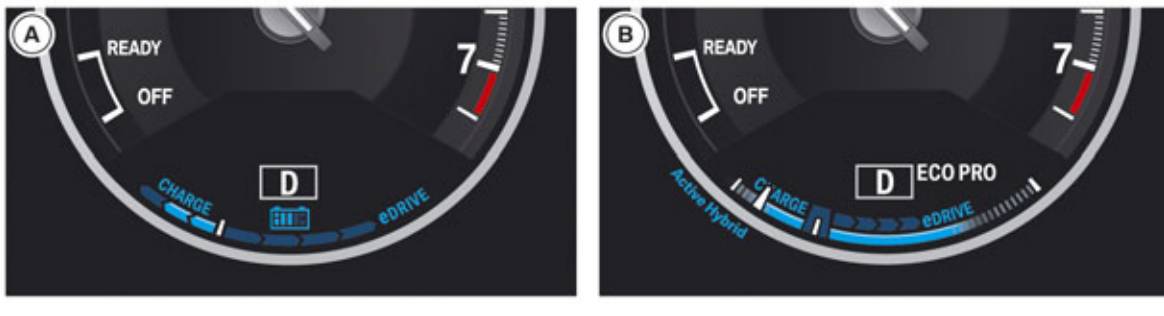
8.1.7. Brake energy regeneration

The instrument cluster in the F10H shows the driver the use of the brake energy regeneration by up to three highlighted segments. The first of the segments which has a plus symbol appears as soon as the driver takes the foot off the accelerator pedal. Upon actuation of the brake pedal the other two segments are highlighted. These segments point in a counterclockwise direction in the F10H.



F10H Brake energy regeneration display

In the F30H the display for the level of the brake energy regeneration points in the opposite direction, i.e. counterclockwise. This applies to all modes of the driving experience switch. In all modes, except ECO PRO mode, there are up to three highlighted segments: the first segment is highlighted, as soon as the driver releases the accelerator pedal, the second and third segment are highlighted when the brake pedal is pressed. In ECO PRO mode the display is not done by highlighted segments, but by the white mark which now moves over the neutral position to the left.



F30H Brake energy regeneration display

Index	Explanation
A	COMFORT mode
B	ECO PRO mode

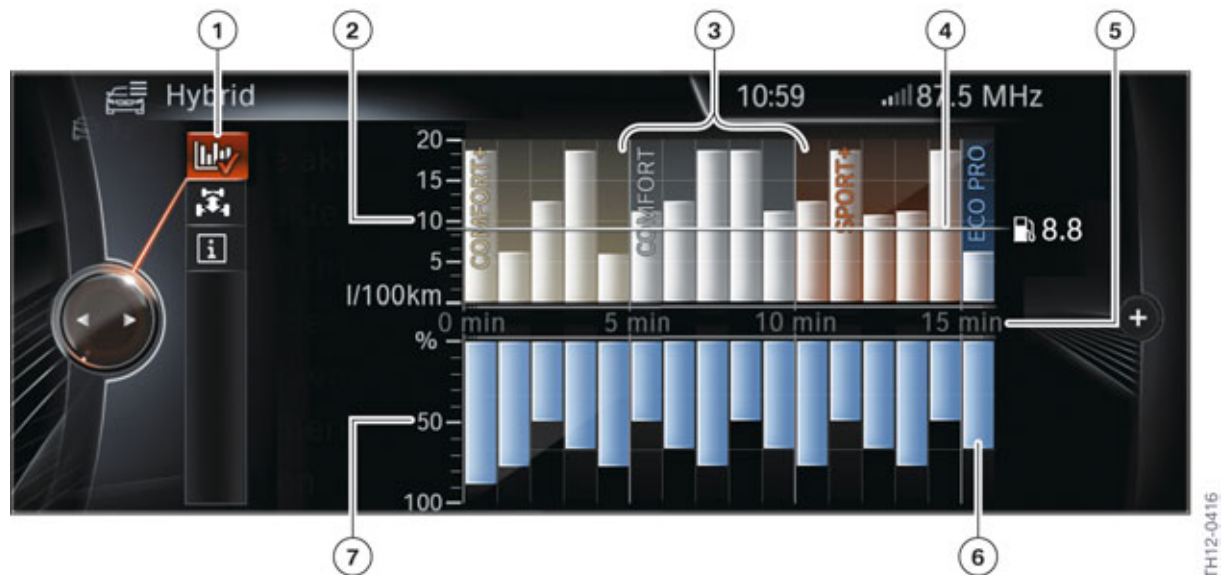
F30H Complete Vehicle

8. Displays and Controls

8.2. Displays in central information display

The hybrid-specific displays in the central information display CID are also called up in the F30H via the menu "Vehicle info > Hybrid". The content shown is basically identical to that in the F10H. The changes are suitable to the display style only in the detail and in the graphic preparation, which is defined by the third generation HU-H. The following subchapters show the content of these displays using an example: statistics for hybrid usage, energy/power flows and information in ECO PRO mode.

8.2.1. Statistics for hybrid usage



Display for utilization of the hybrid system

Index	Explanation
1	Selection of the display for hybrid use
2	Consumption scale of the combustion engine
3	Period of time in which the COMFORT mode was used
4	Average consumption of the combustion engine
5	Time axis (16 minutes)
6	Bar represents one minute
7	Percentage scale for usage of the electric motor

The utilization of the hybrid system in the last 16 minutes of driving can be shown in the CID. One bar stands for a period of one minute. The time is also counted during engine stop phases. The higher the bar, the higher the fuel consumption or the use of the electrical machine.

The grey bars at the top show the fuel consumption of the combustion engine. A line and a value to the right beside the diagram show the average consumption.

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8. Displays and Controls

In comparison to the F10H, the phases are differentiated in color dependent on the selected mode on the driving experience switch. This way the driver can identify the influence of the mode on the fuel consumption.

The blue bars show the percentages of how much the electric motor was used. The electrical machine can thus be operated as an alternator (during brake energy regeneration) or as an engine (during electric driving). Also phases of standstill, in which the combustion engine was switched off and the vehicle electrical system was supplied by the high-voltage battery, are included in the blue bars. The higher the bar, the greater the use of the electric motor and thus higher fuel economy.

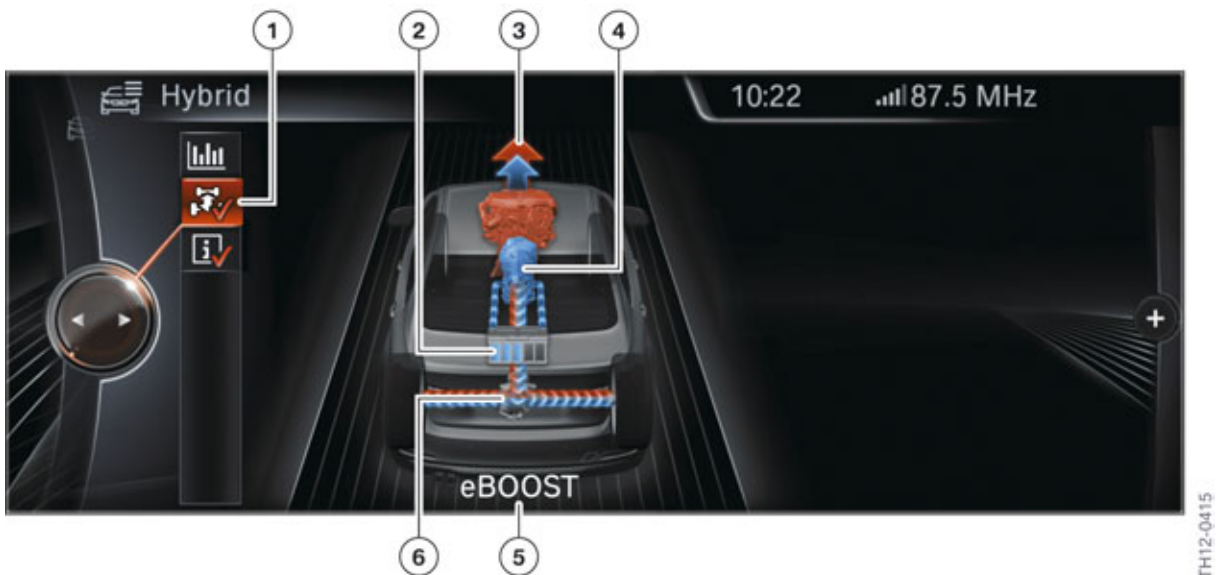
The two red marks on the vertical axis of the display indicate the bars from the last minute.

8.2.2. Energy/Power flows

The display of the energy/power flows in the CID of the F30H is effected according to the same principle in the F10H:

- Blue: Electrical energy
- Red: Energy of the combustion engine
- Arrow: Direction of the energy / power flow

A driving situation is now shown by way of an example. The other driving situations can be deduced from this.



Energy/Power flows display in the central information display during heavy acceleration

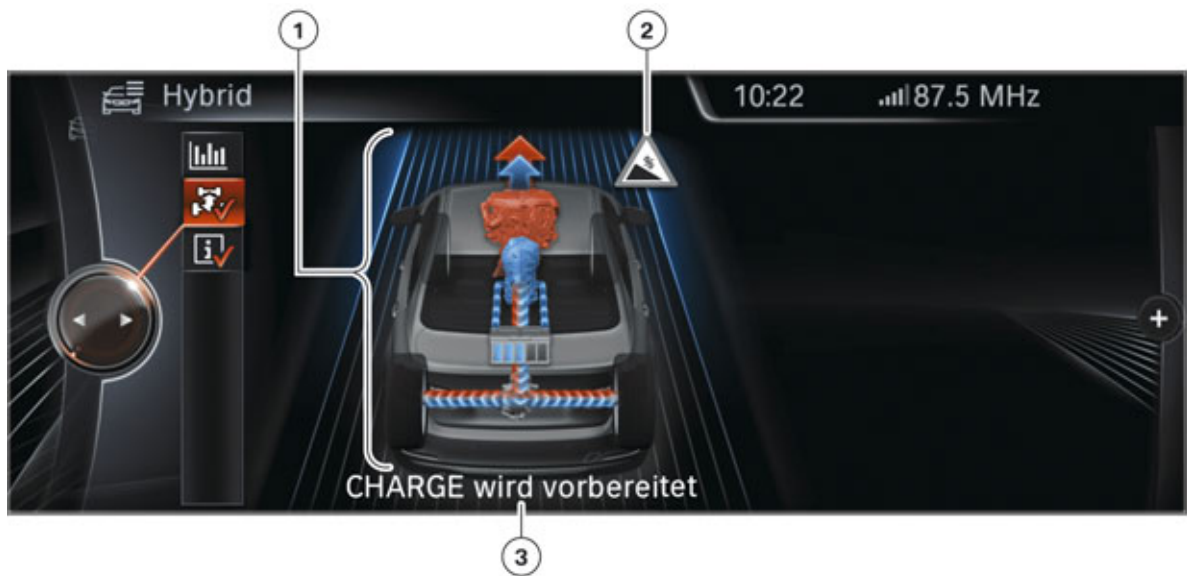
F30H Complete Vehicle

8. Displays and Controls

Index	Explanation
1	Selection of display for energy / power flows
2	State of charge of the high-voltage battery
3	Drive arrow for combustion engine (red) and drive arrow for electric motor (blue)
4	Automatic transmission with electric motor
5	Writing for current driving situation "eBOOST"
6	Power flow to the rear axle

Apart from the new visible design, no change for the F10H was made to this display.

In this display, which can also be split in the right part of the split screen, the driver also receives information if the hybrid system optimizes the operating strategy with help of the stretch of road ahead. A prerequisite for this is that the vehicle is equipped with a navigation system and the route guidance is activated. The electrical machine electronics evaluates the bus telegrams from the navigation system and takes into consideration the distance to the destination. Also the electric driving shortly before reaching the destination can be enabled by prior charging of the high-voltage battery. If such preparation of the hybrid system is under way, symbols and writing are also shown in the display of the energy/power flows. You can see an example in the following graphic.



Display of energy/power flows in the CID for downhill gradient ahead

Index	Explanation
1	Energy/Power flows diagram
2	Symbol to clarify what situation/road type is ahead (here: slope)
3	Writing "CHARGE is prepared"

While the hybrid system is prepared for the situation ahead, the symbol (item number 2) receives a green border. If the situation occurs, the same symbol is shown but with a blue border.

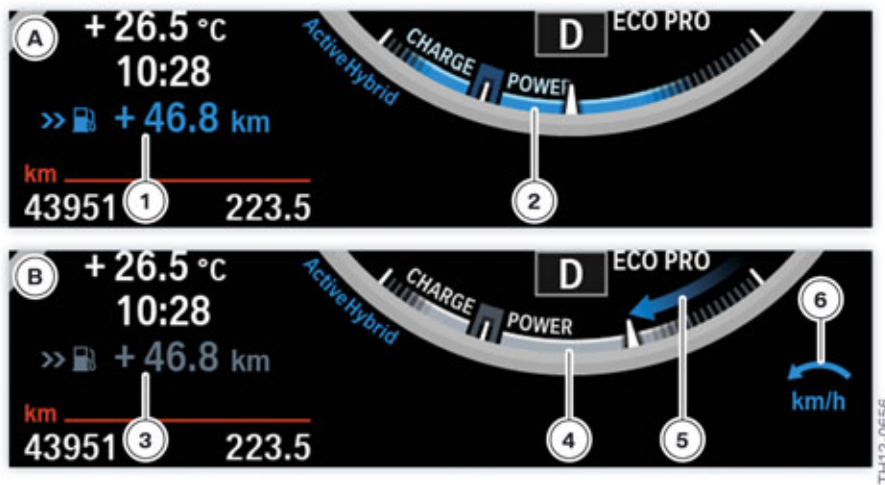
F30H Complete Vehicle

8. Displays and Controls

8.3. ECO PRO mode

Like in the F10H and in conventional F30 models, the driver can select ECO PRO mode also in F30H. ECO PRO mode supports the driver during efficient driving, optimizes the operating strategy for even more economical fuel consumption and increases the range of the vehicle.

ECO PRO mode is activated using the driving experience switch. In the instrument cluster the writing "ECO PRO" is shown beside the drive position, as well as switched to the curve-shaped scale for the driving power used. The driver also receives the so-called bonus range displayed as an on-board computer function in blue writing. The bonus range is the additional range which was achieved by using ECO PRO mode.



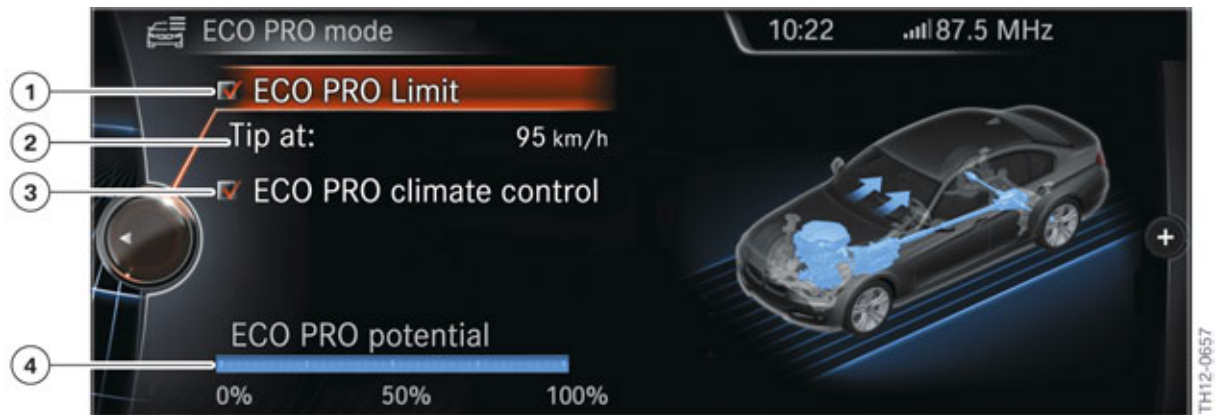
ECO PRO mode displays

Index	Explanation
A	ECO PRO mode, efficient driving style
B	ECO PRO mode, driving style outside efficient range
1	Bonus range in blue writing
2	Power scale in blue
3	Bonus range in grey writing
4	Power scale in grey
5	Arrow requests less power to be requested
6	Reason for the request to call up less power: ECO PRO speed limit exceeded

ECO PRO displays in the instrument cluster are only shown in blue if the driver uses the vehicle in the efficient range. If the driver leaves the efficient range, both the color of the bonus range and the power scale change from blue to grey. In the graphic above the reason the efficient range was left is because the ECO PRO speed limit set by the driver was exceeded. But also if the accelerator pedal is pressed too firmly, the driver receives a corresponding display which suggests to him to request less power. The display in the instrument cluster thus supports the driver in the efficient operation of the vehicle.

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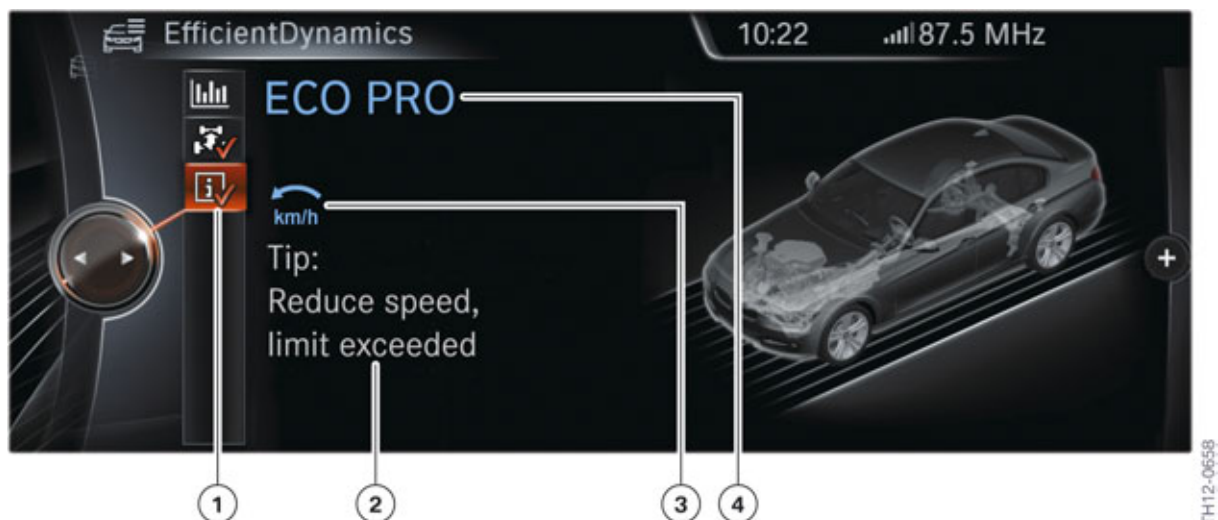
8. Displays and Controls



Configure ECO PRO mode

Index	Explanation
1	Activation/Deactivation of information when ECO PRO speed limit exceeded
2	Selection of speed for issue of ECO PRO tips
3	Activation/Deactivation of restricted climate control in ECO PRO mode
4	How much percent of the possible saving potential can be achieved with the current configuration is displayed

After activation of ECO PRO mode the driver can configure this via a menu that is opened automatically. The driver can choose to set a speed limit and activate restricted climate control. The fuel consumption is either reduced directly because the electric A/C compressor is operated at lower power for example. Or the fuel consumption is reduced indirectly because the driver receives a message when the set speed limit is exceeded. These types of messages are not only shown in the instrument cluster, but also on the central information display.



Example of ECO PRO message

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8. Displays and Controls

Index	Explanation
1	The issue of ECO PRO information is activated
2	Message in text form, e.g. "Reduce speed, limit exceeded"
3	Information in graphic form
4	Name of the menu (ECO PRO information)

In conclusion, the technical measures of ECO PRO mode are summarized which support the driver in adopting an efficient driving style:

- Alteration of the accelerator pedal characteristic and the shift program in automatic transmissions
- Message to the driver when efficient range is left
- Reduction of power of electric Comfort consumer (e.g. seat heating)
- Power reduction of heating/air-conditioning
- Rolling without energy consumption ("coasting") up to a driving speed of 160 km/h is possible.

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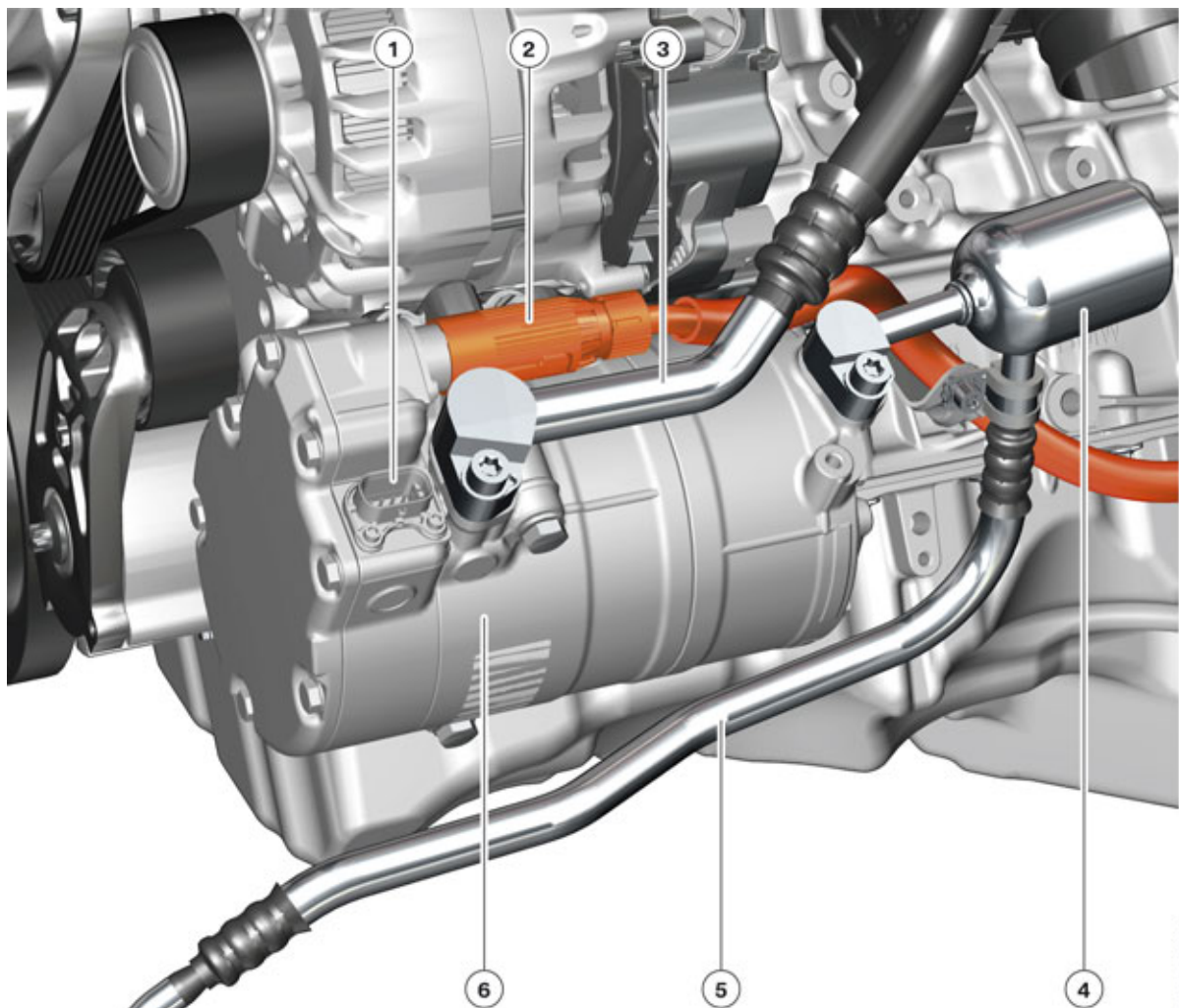
9. Climate Control

The F30H climate control system (already known from the F10H) is used to cool the passenger compartment and the high-voltage battery. The components in the refrigerant circuit, as well as the functions, are in principle identical.

The functions of the climate control system are:

- Cooling/heating of the passenger compartment during driving
- Stationary cooling and conservation cooling when vehicle is stopped
- Cooling of the high-voltage battery.

The F30H electric A/C compressor (EKK) has the same design as that in the F10H and therefore also uses a spiral compressor for compressing the refrigerant. It is supplied with high-voltage DC via the EME and absorbs a maximum electrical power of 4.5 kW. As it can be operated independently of the running combustion engine, it enables the cooling function also during electric driving and when the vehicle is stopped. As in the F10H, the electric A/C compressor in the F30H is also equipped with a silencer in the pressure line to guarantee quiet operation even when the combustion engine is not running.



Electric A/C compressor in the F30H

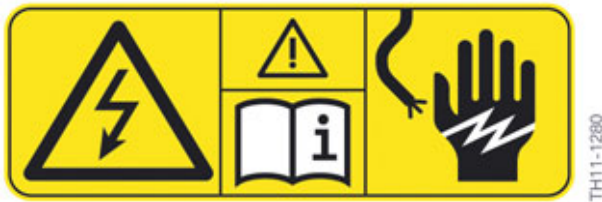
TH12-0728

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9. Climate Control

Index	Explanation
1	Signal connector
2	High-voltage connector for compressor
3	Intake pipe (gaseous refrigerant with low temperature and low pressure)
4	Silencer
5	Pressure line (gaseous refrigerant with high temperature and high pressure)
6	Electrical A/C compressor (EKK)

The electric A/C compressor is a high-voltage component!



High-voltage component warning sticker



Only service technicians who satisfy all the hybrid training prerequisites are permitted to work on the designated high-voltage components. Technicians must have all suitable qualifications, comply with the electrical safety rules and must follow procedure and the repair instructions to the letter.



The electric A/C compressor (EKK) is NOT part of the High-voltage interlock loop!

By removing the high-voltage connector at the EKK the high-voltage system is not automatically shut down!

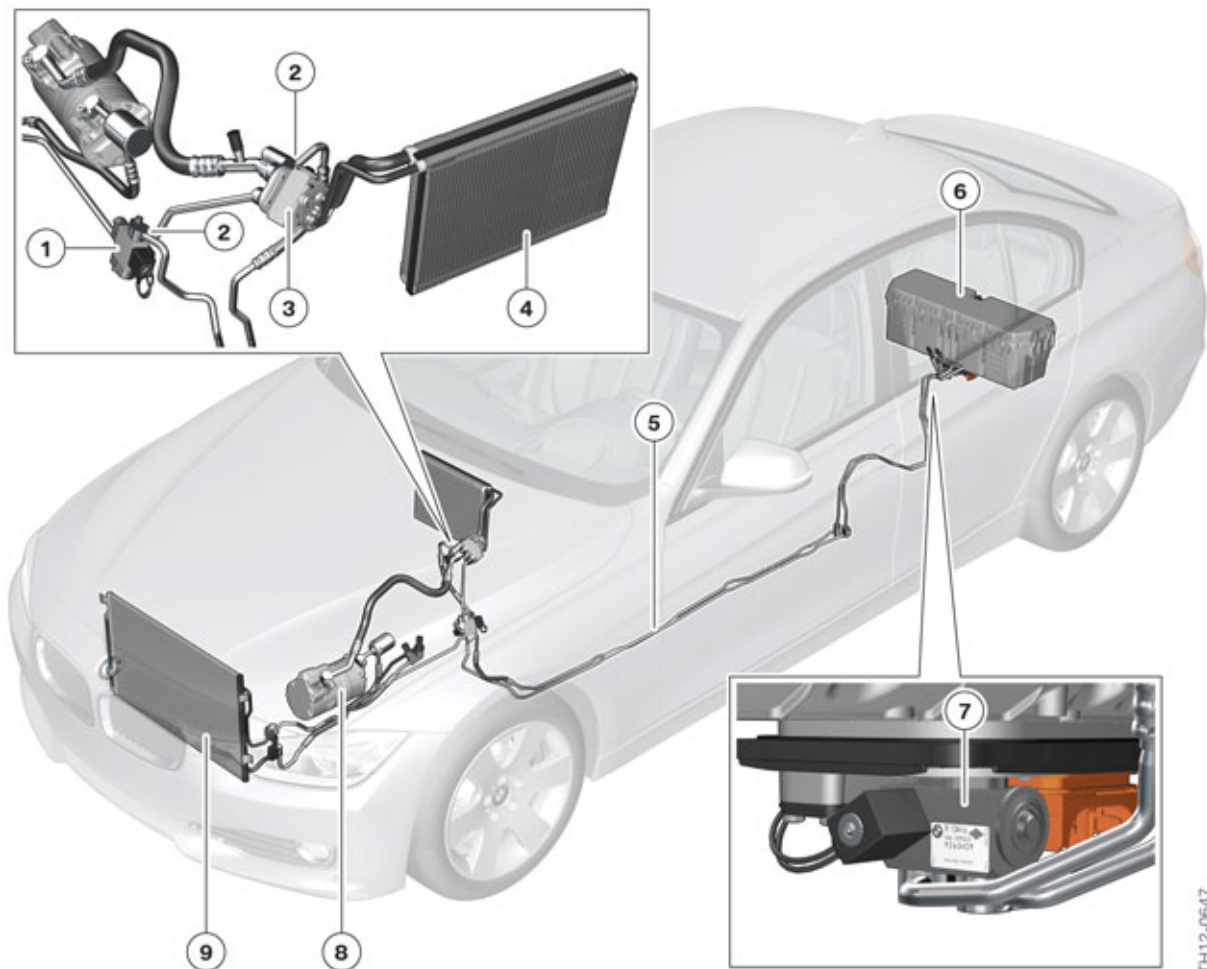
Before removing the high-voltage connector at the electric A/C compressor (EKK) the high-voltage system must be de-energized.

PAG oil also used in the refrigerant circuit of the F30H (as in F10H) for lubricating the AC components.

In addition to the electric A/C compressor the other components in the refrigerant circuit of the F30H such as the expansion and shutoff valve are also known from the F10H. however Due to the different vehicle body in the F30H they are installed at different locations and therefore the refrigerant lines are routed differently. The following graphic provides an overview of the refrigerant circuit and the installation locations of the components.

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F30H Refrigerant circuit

Index	Explanation
1	Shutoff valve for passenger compartment
2	Branching of refrigerant lines to high-voltage battery unit
3	Expansion valve for passenger compartment
4	Evaporator for climate control of passenger compartment
5	Refrigerant lines to high-voltage battery unit
6	High-voltage battery unit
7	Combined expansion and shutoff valve for high-voltage battery unit
8	Electric A/C compressor (EKK)
9	Condenser

In contrast to the F10H, the valves used for cooling the passenger compartment are not a combined expansion and shutoff valve. Instead the shutoff valve and the expansion valve are separated due to the available space. Both the shutoff valve for passenger compartment and for the combined expansion and shutoff valve for high-voltage battery cooling are **closed** when de-energized and are powered

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9. Climate Control

open. The shutoff valve for cooling the passenger compartment is activated (powered open) by the EME. The combined expansion and shutoff valve for the high-voltage battery is in contrast activated (powered open) by the battery management electronics (SME).

The diagnosis system offers a Service function with which the two refrigerant circuit valves can be activated. This Service function can be used if the refrigerant circuit is to be evacuated. If an A/C service station is used for the evacuation (which generates a vacuum in both lines) then the valves do not necessarily have to be activated. Through the vacuum on both lines the refrigerant is reliably drawn off also when the valves are closed. If in contrast an A/C service station is used (which only generates a vacuum in one line) then this Service function must be used to properly draw off the refrigerant completely. **To ensure that the evacuation is always performed properly it is therefore recommended to apply the Service function for activating the valves.**



Before the evacuation of the refrigerant a Service function in the diagnosis system must be used to activate and open the two valves in the refrigerant circuit. This ensures that the refrigerant is completely and properly evacuated.

The stationary cooling function can be activated by the customer via the vehicle remote key (fourth button). So that the stationary cooling can be performed, sufficient state of charge of the high-voltage battery is necessary and an ambient temperature which must be greater than approx. 15 °C (59 °F). The stationary cooling is automatically stopped when one enters the vehicle again and terminal 15 is switched on.

The conservation cooling function can be activated with terminal 15 switched off by pressing the button for increasing the air volume. The ventilation and electric A/C compressor are thus activated for a maximum of roughly 6 minutes so that the interior temperature is kept at a pleasant level.



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